

TEJAS
MARK 3

JEE ADVANCED

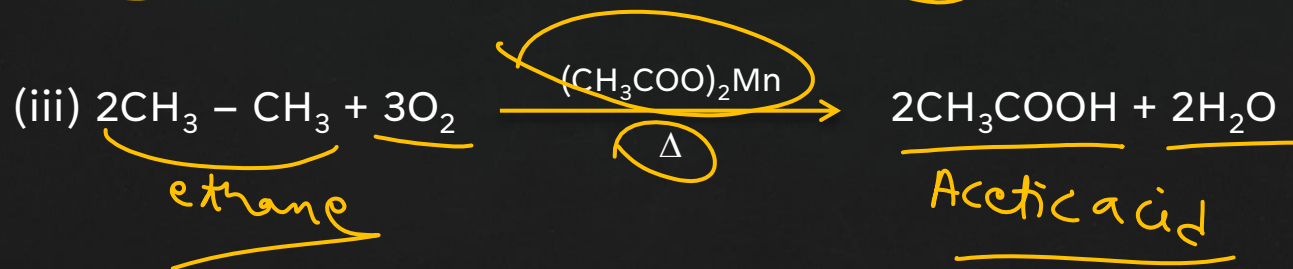
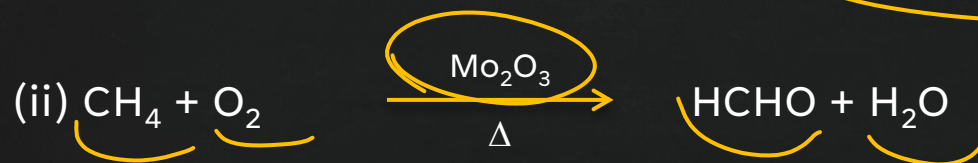
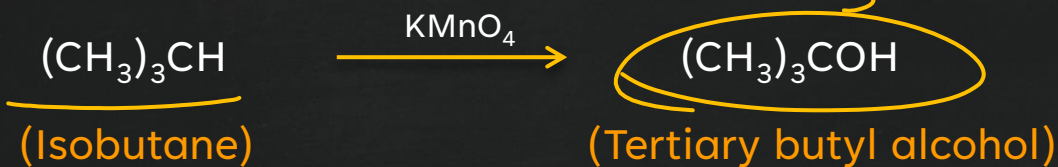
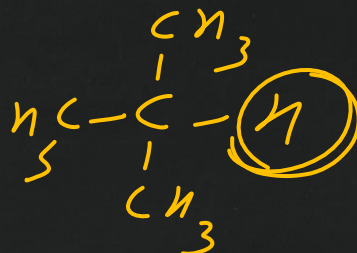
≡ CHEMISTRY ≡

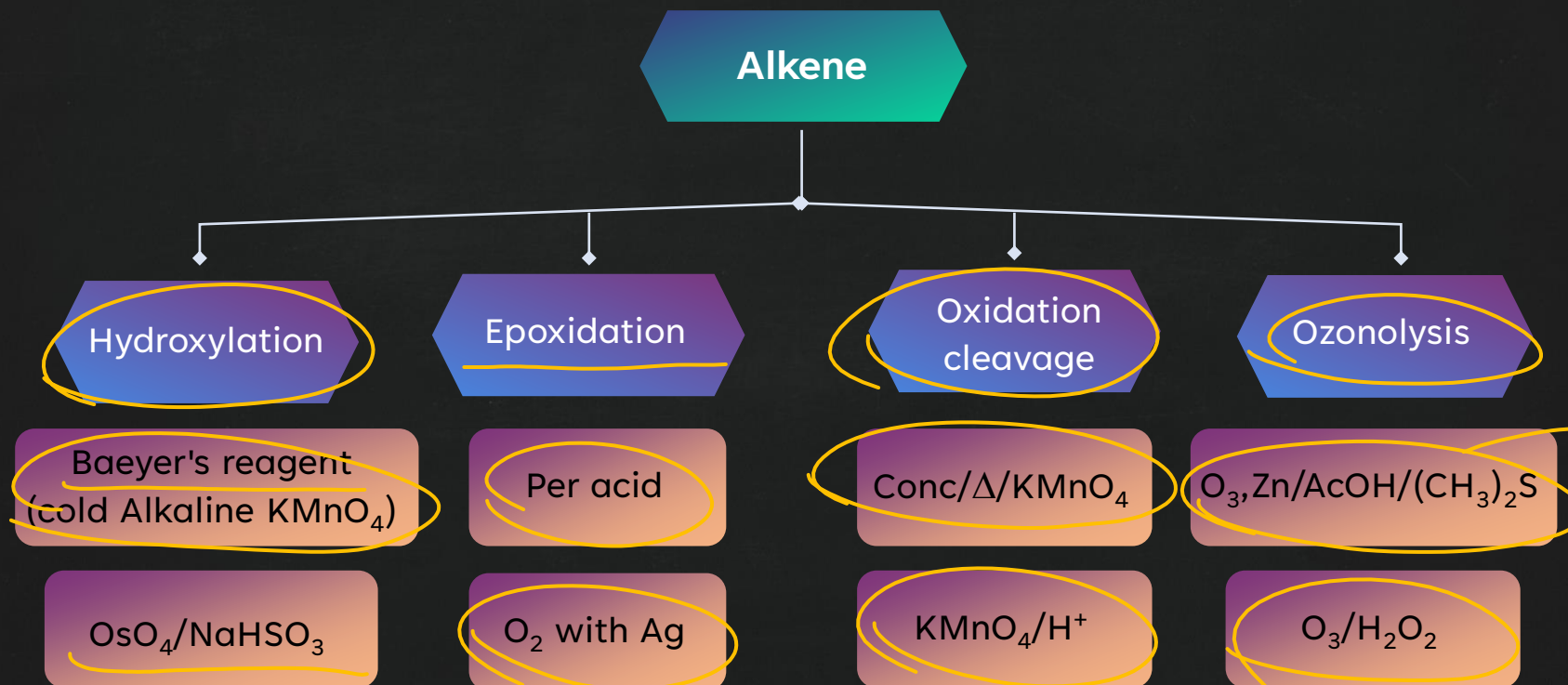
ONE SHOT 

REAGENTS

OXIDISING REAGENTS

Oxidation of alkanes

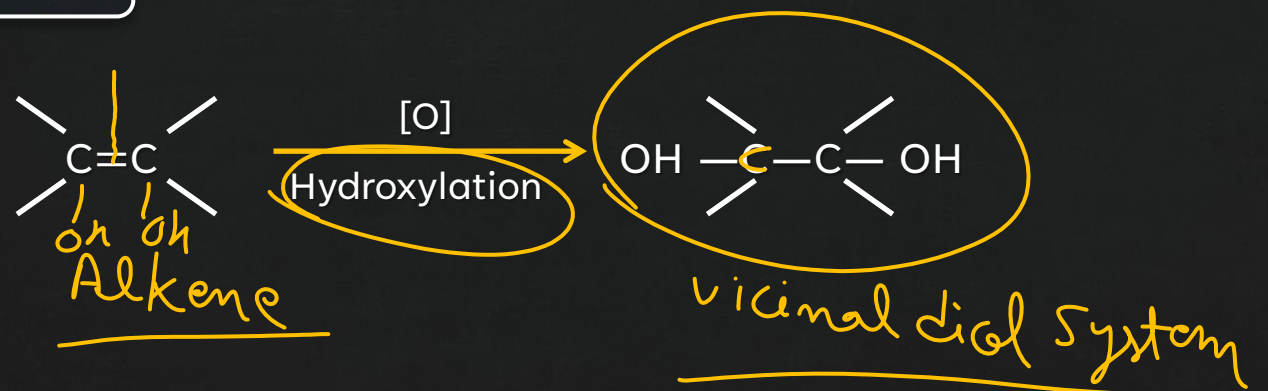




Hydroxylation

 Oxidation of carbon-carbon double bond to $\begin{array}{cc} | & | \\ -C & -C- \\ | & | \\ OH & OH \end{array}$ is known as hydroxylation.

Alkene

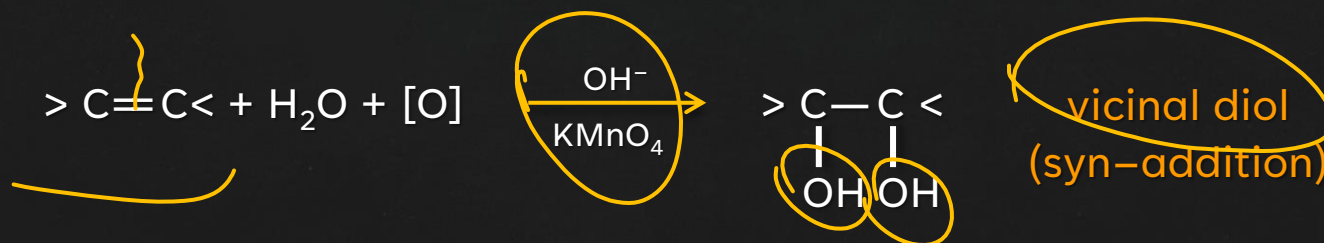


Reagent - (i) cold dilute alkaline 1% cold KMnO_4 (Baeyer's reagent),
 (ii) $[\text{OsO}_4/\text{NaHSO}_3]$ (iii) $\text{OsO}_4/\text{H}_2\text{O}_2$

Function - Hydroxylation of alkene and alkyne

Stereochemistry - Syn Dihydroxylation.

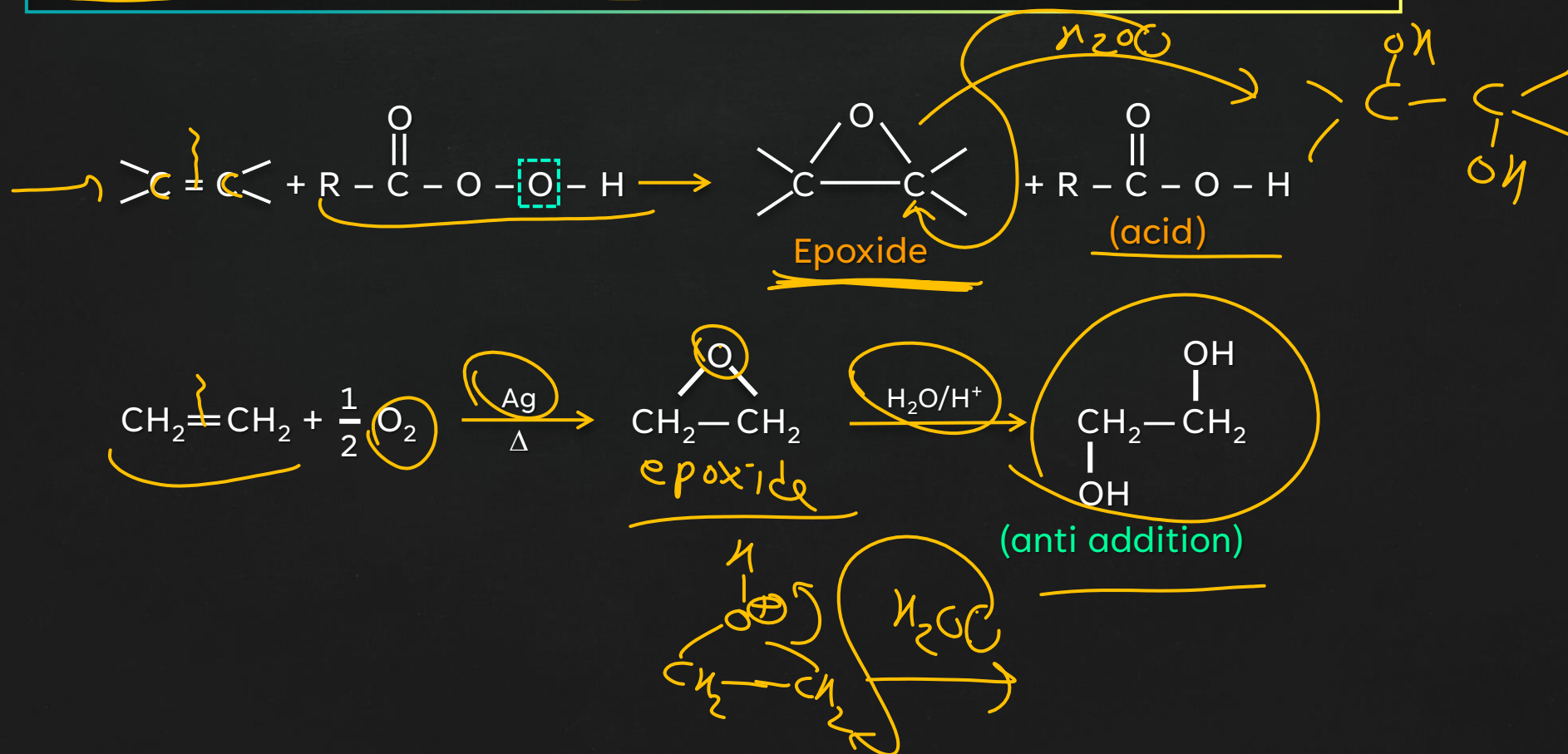
Reactant	Alkene	
Product	Vicinal diol	



Reagent - (i) Peroxyacetic acid + H_2O (ii) O_2 , Ag

Function - Epoxidation followed by Hydroxylation *Hydrolysis*

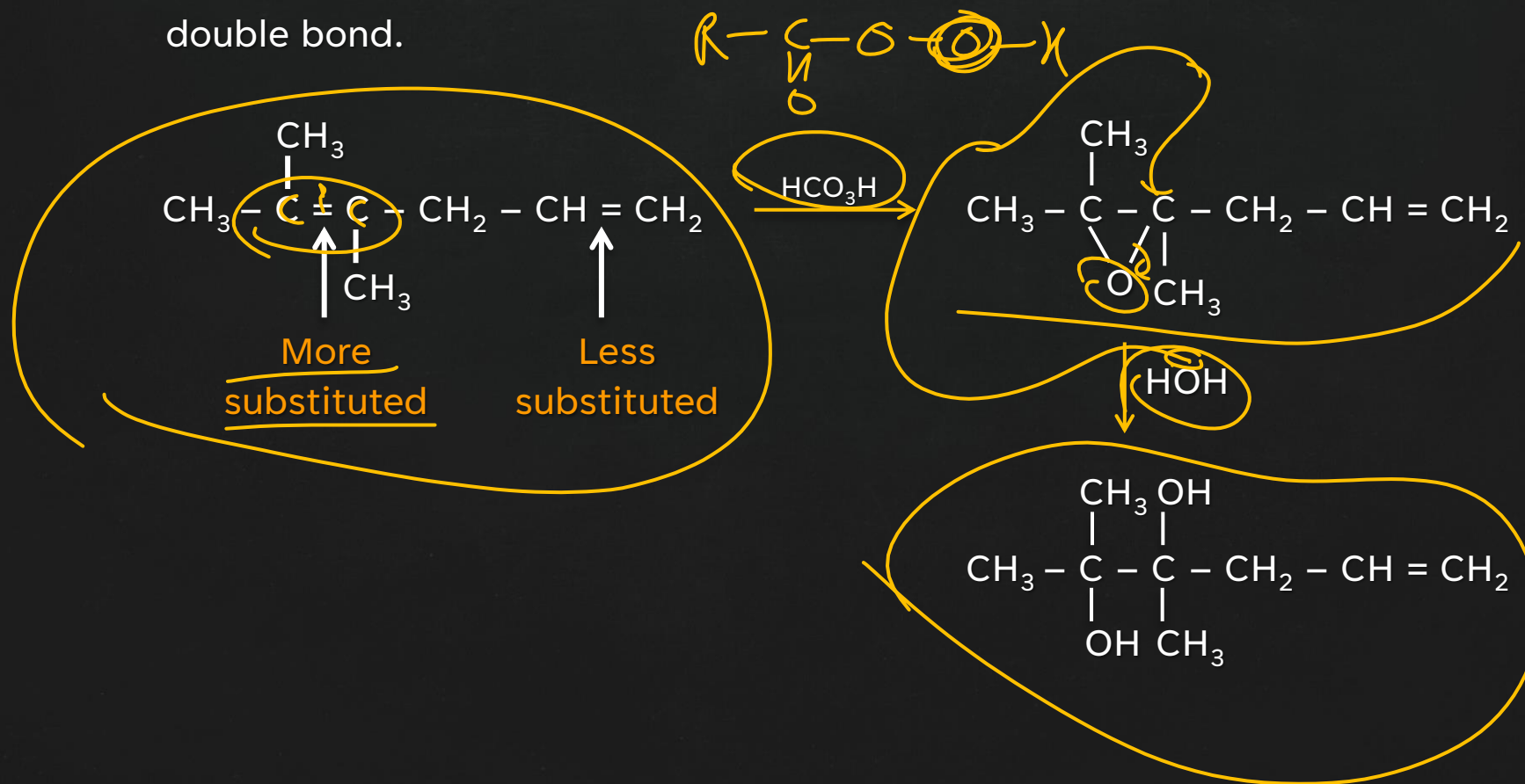
Stereochemistry - Anti Dihydroxylation



NOTE

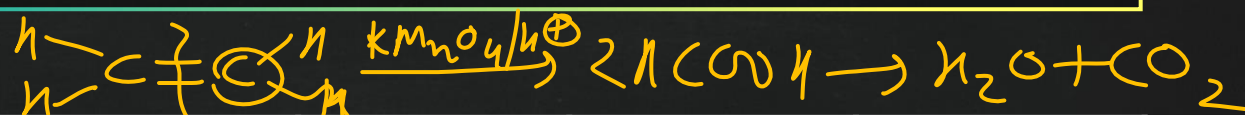


The more **highly substituted olefinic** bond is more nucleophilic it is and therefore reacts faster with the peroxyacid than the less substituted double bond.

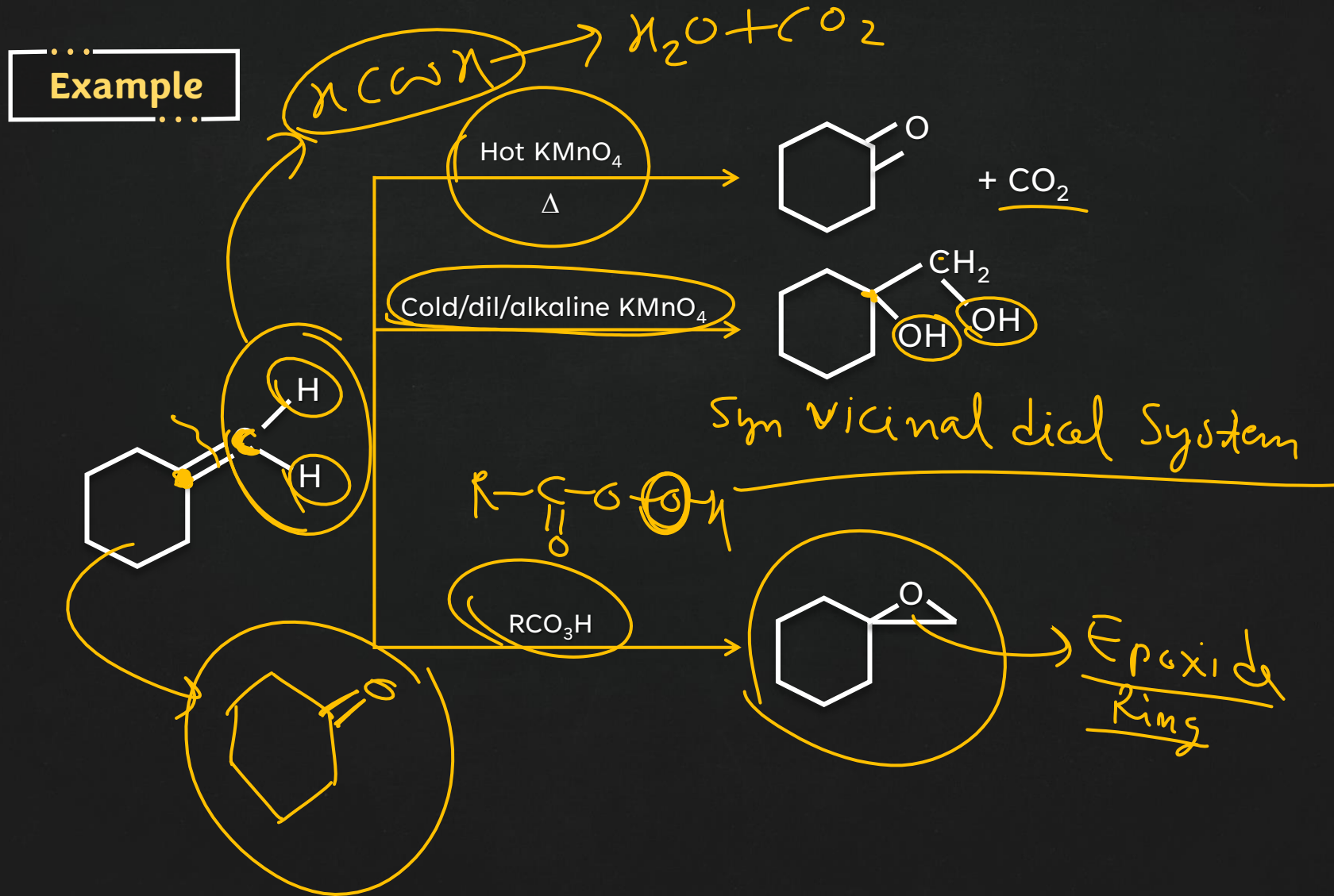


Reagent - (i) Hot alkaline KMnO_4 , (ii) Hot acidic KMnO_4
 (iii) Hot acidic $\text{K}_2\text{Cr}_2\text{O}_7$

Function - Oxidative Cleavage of Alkene & Alkyne



Reagent	KMnO_4/H^+	KMnO_4/H^+	KMnO_4/H^+	KMnO_4/H^+	KMnO_4/H^+
Reactant	$=\text{CH}_2$ group	$=\text{CH R}$ group	$=\text{CR}_1\text{R}_2$ group	$\equiv\text{CH}$ group	$\equiv\text{CR}$ group
Product	CO_2	RCOOH	$\text{O}=\text{CR}_1\text{R}_2$ group	HCOOH CO_2	RCOOH
	<u>Carbon dioxide</u>	<u>Carboxylic acid</u>	<u>Ketone</u>	Carbon dioxide	<u>Carboxylic acid</u>

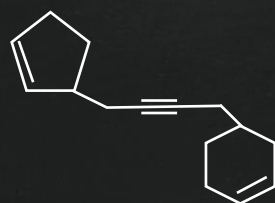




Example

[JEE-Advance 2019]

Total number of hydroxyl groups present in a molecule of the major product P is



(i) H_2 , Pd-BaSO₄, quinoline

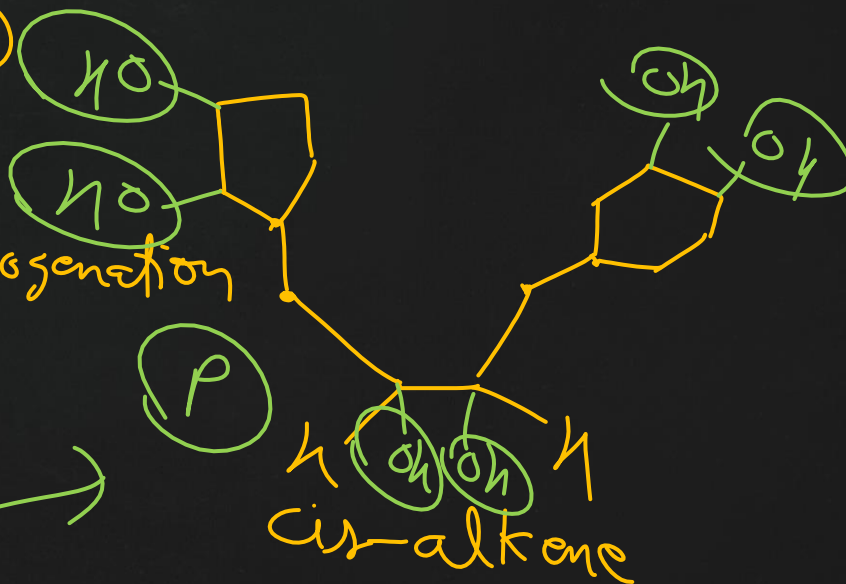
(ii) dil. KMnO₄ (excess), 273K

P

Solution



(i) Poisoned Hydrogenation



cis-alkene

Baeyer-Villiger reagent

Ans. (6.00)

6

Ozonolysis

Reductive ozonolysis

- 1. $\text{O}_3/\text{Zn}, \text{H}_2\text{O}$
- 2. $\text{O}_3/\text{H}_2/\text{Pd}$
- 3. $\text{O}_3/\text{Me}_2\text{S}$
- 4. $\text{O}_3/\text{Ph}_3\text{P}$



It gives aldehyde and ketone

Oxidative ozonolysis

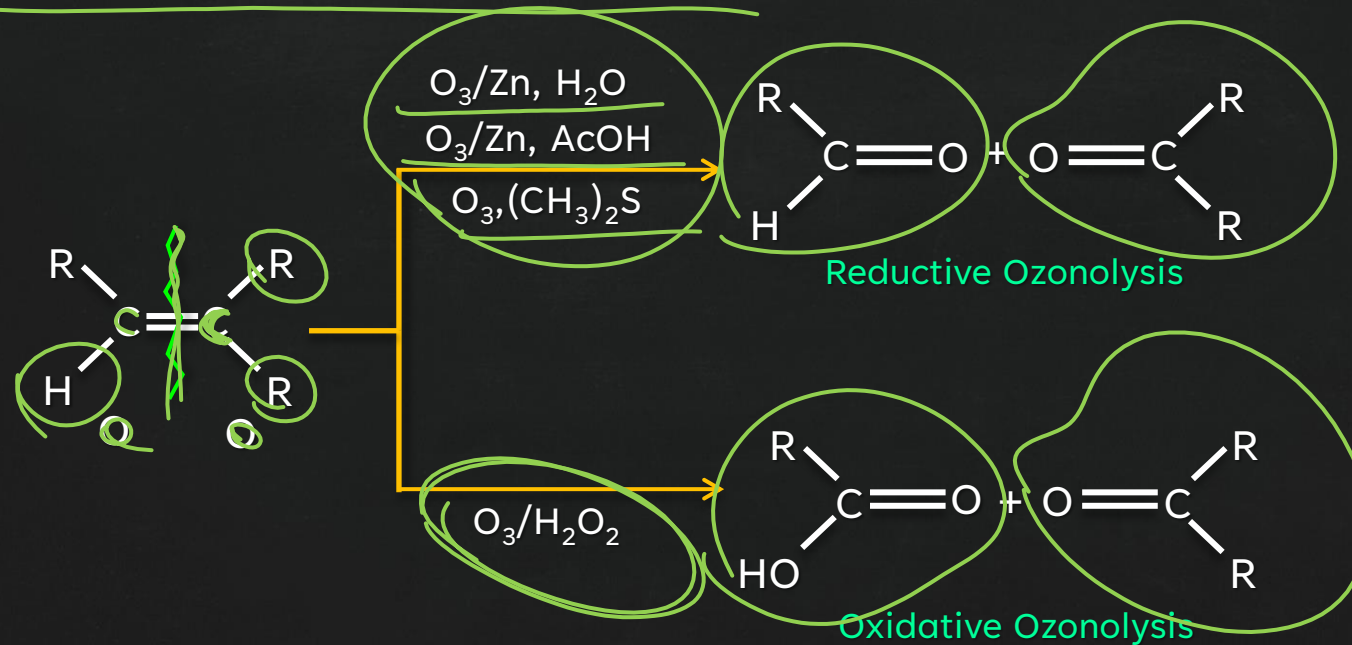
- 1. $\text{O}_3/\text{H}_2\text{O}$
- 2. $\text{O}_3/\text{Ag}_2\text{O}$
- 3. $\text{O}_3/\text{H}_2\text{O}_2$
- 4. $\text{O}_3/\text{Per-acid}$

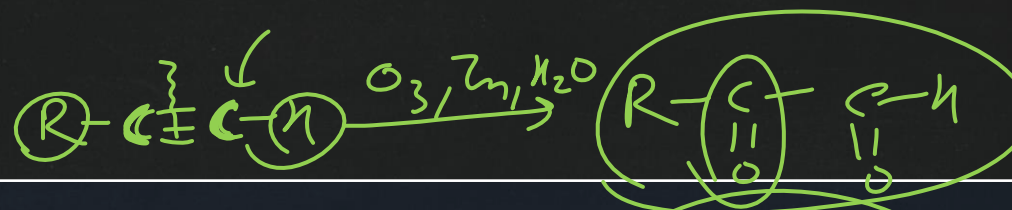


It give acids and ketones

Ozonolysis

A test for unsaturation in a molecule





Reductive Ozonolysis Products ($\text{O}_3/\text{Zn}, \text{H}_2\text{O}$)

$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$	$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$	$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$	$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$	$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$	$\text{O}_3/\text{Zn}, \text{H}_2\text{O}$
Reactant	$=\text{CH}_2$ group	$=\text{CH}^{\text{R}}$ group	$=\text{CR}_1\text{R}_2$ group	$\equiv\text{CH}$ group	$\equiv\text{CR}$ group
Product	HCHO	$\text{O}=\text{CHR}$ group	$\text{O}=\text{CR}_1\text{R}_2$ group	Keto aldehyde	diketone
	<u>Formaldehyde</u>	<u>Aldehyde</u>	<u>Ketone</u>	Keto aldehyde	diketone

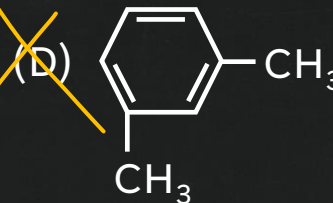
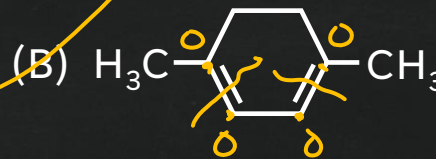
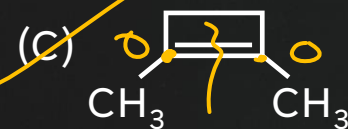


Oxidative Ozonolysis Products

O_3/H_2O_2	O_3/H_2O_2	O_3/H_2O_2	O_3/H_2O_2	O_3/H_2O_2	O_3/H_2O_2
Reactant	$=CH_2$ group	$=CHR$ group	$=CR_1R_2$ group	$\equiv CH$ group	$\equiv CR$ group
Product	CO_2	$RCOOH$	$O=CR_1R_2$ group	$HCOOH$ $\rightarrow CO_2 + H_2O$	$RCOOH$
	Carbon dioxide	<u>Carboxylic acid</u>	<u>Ketone</u>	Carbon dioxide	<u>Carboxylic acid</u>

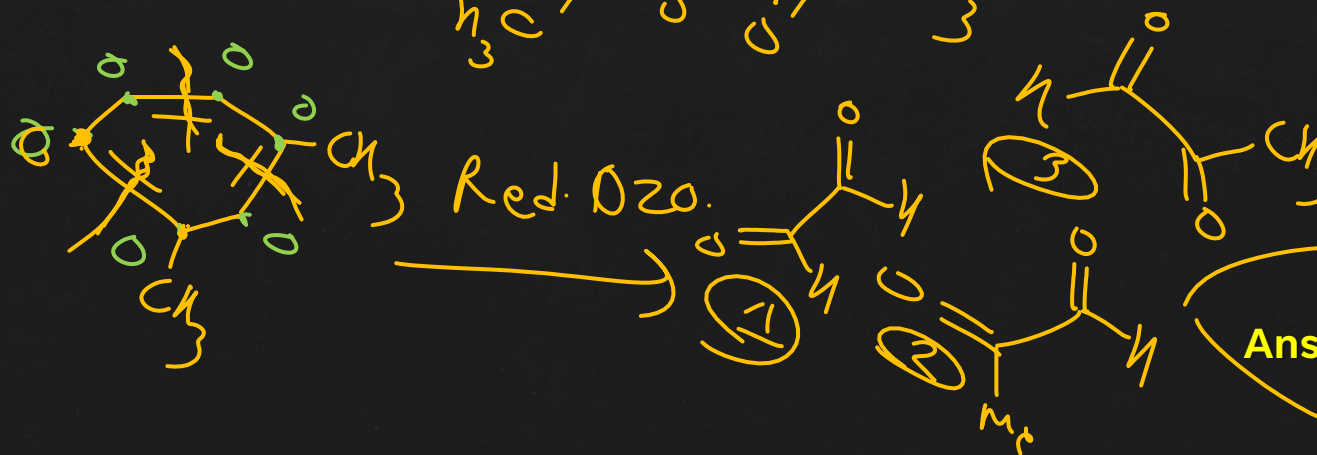
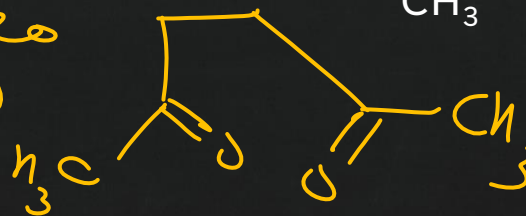
An organic compound on reaction with O_3 followed by Zn and H_2O gives

(A) $\text{CH}_3-\text{C}(\text{CH}_2)_2-\text{C}(\text{CH}_3)=\text{CH}_2$



Solution

Red 020



Ans. (A,B,C)



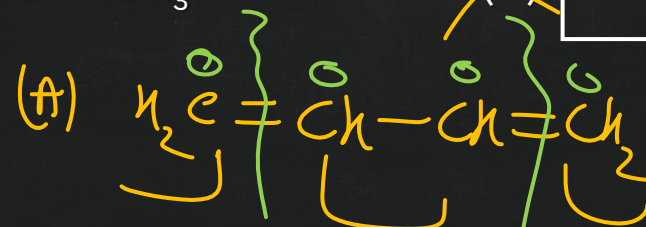
Example

An organic compound C_4H_6 on ozonolysis give $HCHO$, CO_2 , CH_3CHO .

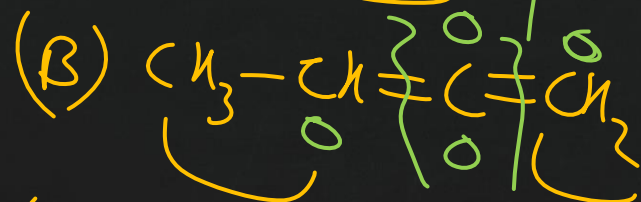
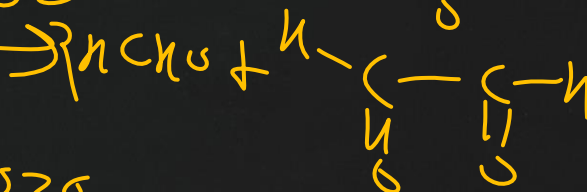
Compound will be :



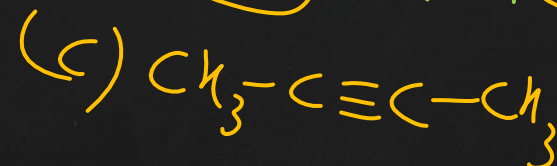
Solution



Red. Ozo.



Red. ozo.



Red. Ozo.

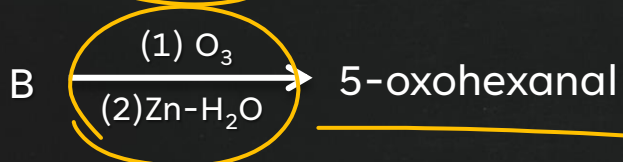
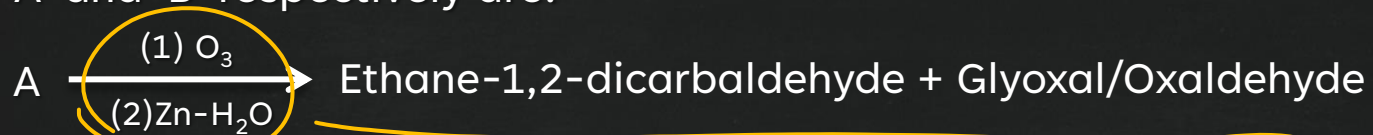


Ans. (B)



Example

'A' and 'B' respectively are:



(1) 1-methylcyclohex-1, 3-diene & cyclopentene

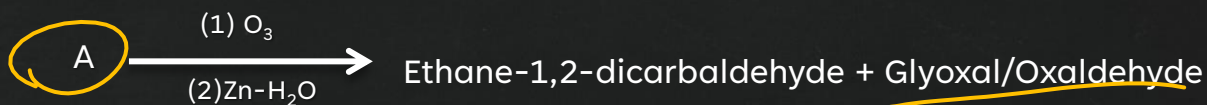
(2) Cyclohex-1, 3-diene & cyclopentene

(3) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene

(4) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene

Solution

'A' and 'B' respectively are:

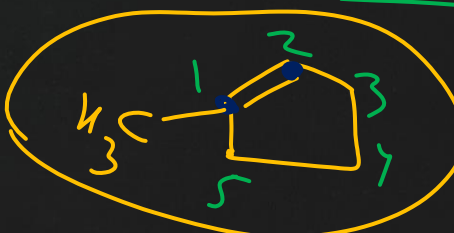
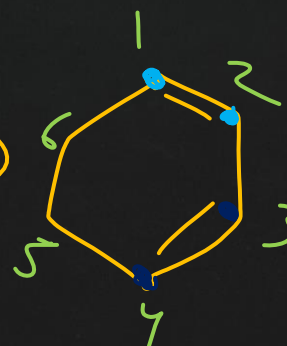
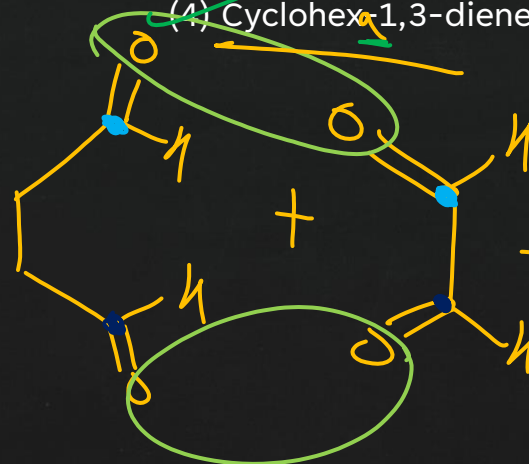


(1) 1-methylcyclohex-1, 3-diene & cyclopentene

(2) Cyclohex-1, 3-diene & cyclopentene

(3) 1-methylcyclohex-1,4-diene & 1-methylcyclopent-1-ene

(4) Cyclohex-1,3-diene & 1-methylcyclopent-1-ene



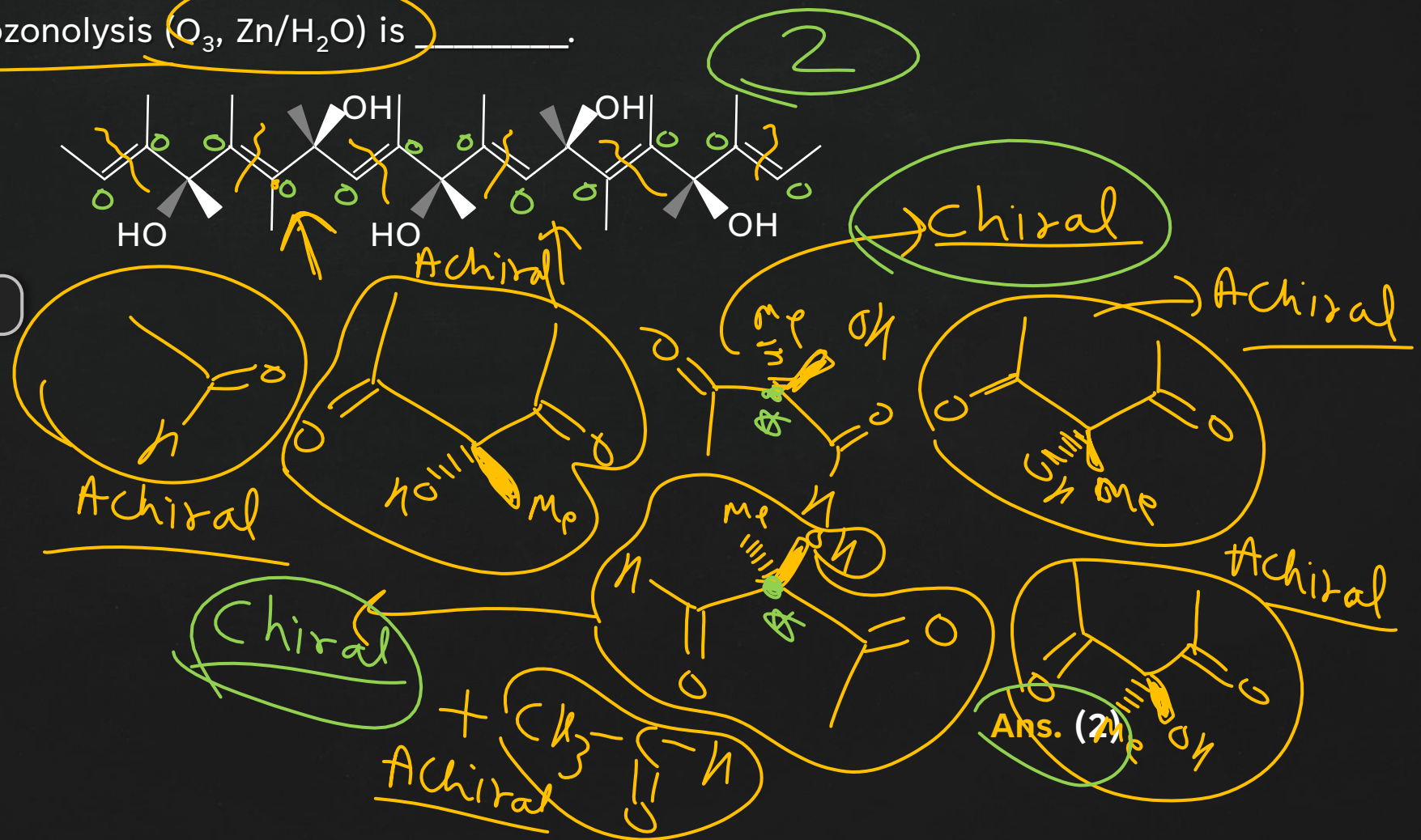
Cyclohexa-1,3-diene

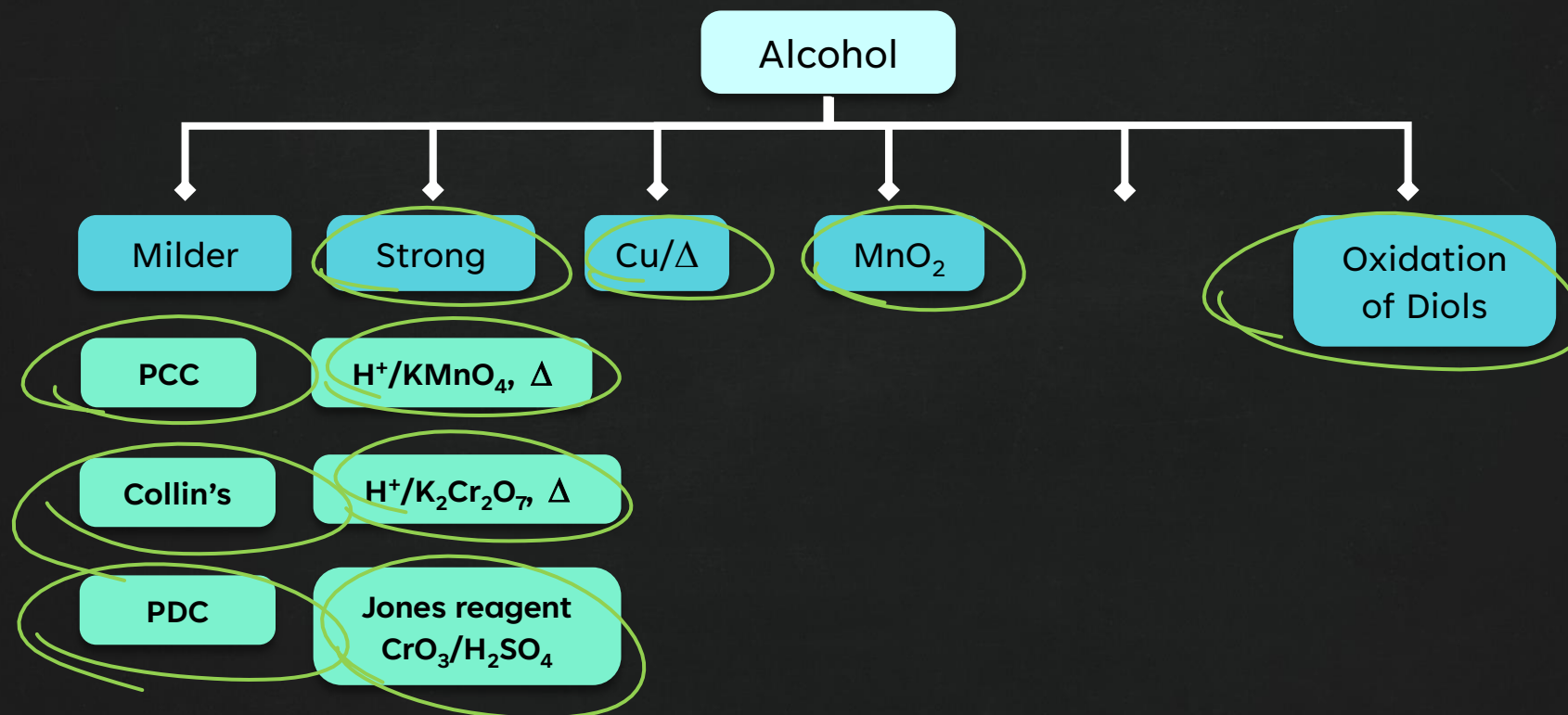
Ans. (4)

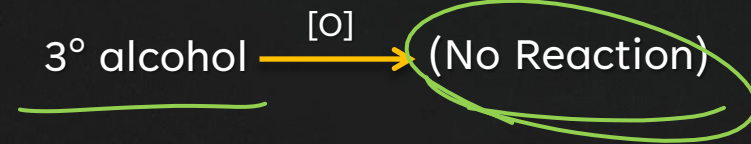
[JEE-Advance 2022]

The total number of chiral molecules formed from one molecule of P on complete ozonolysis (O_3 , Zn/H_2O) is _____. 2

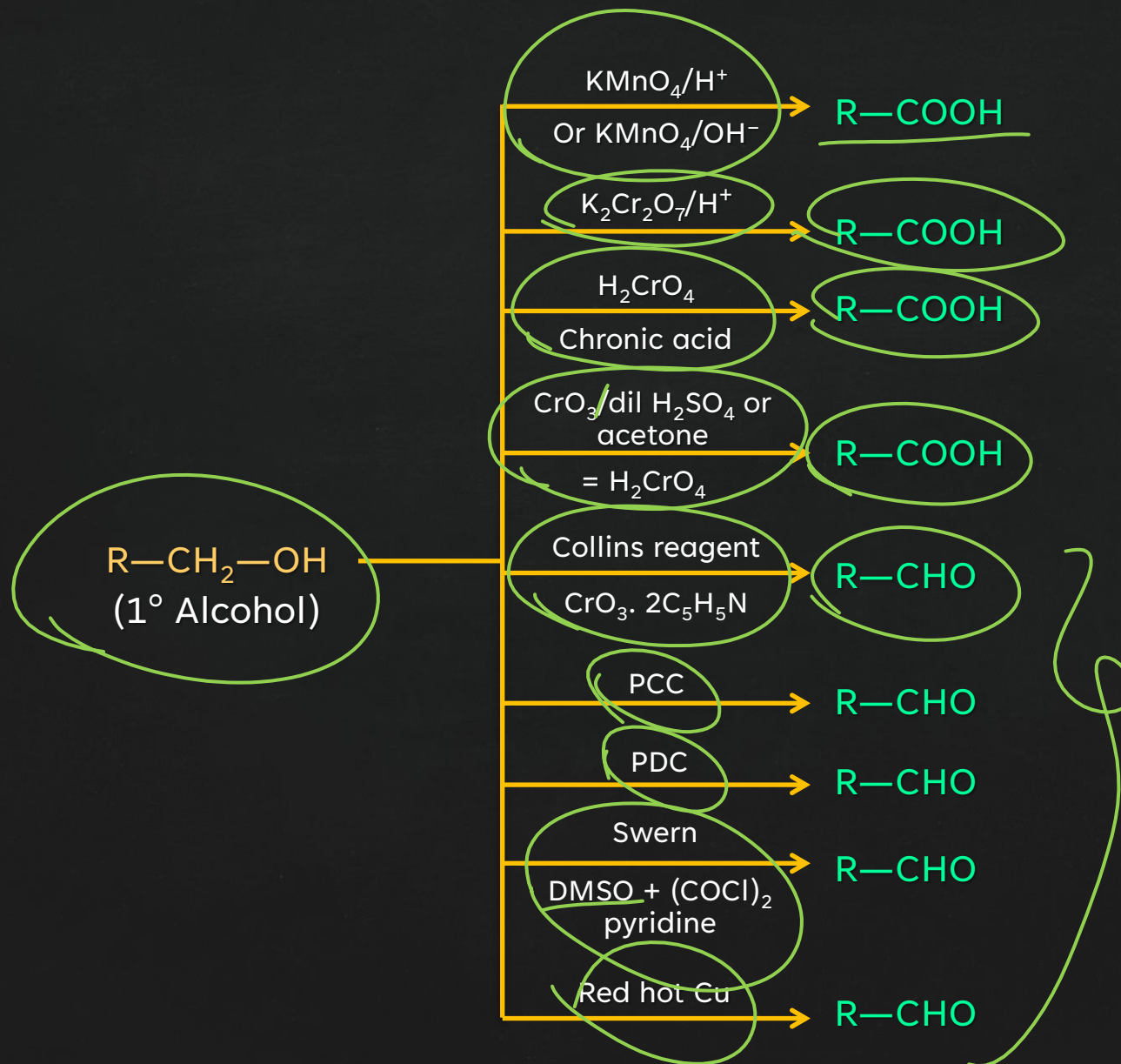
Solution

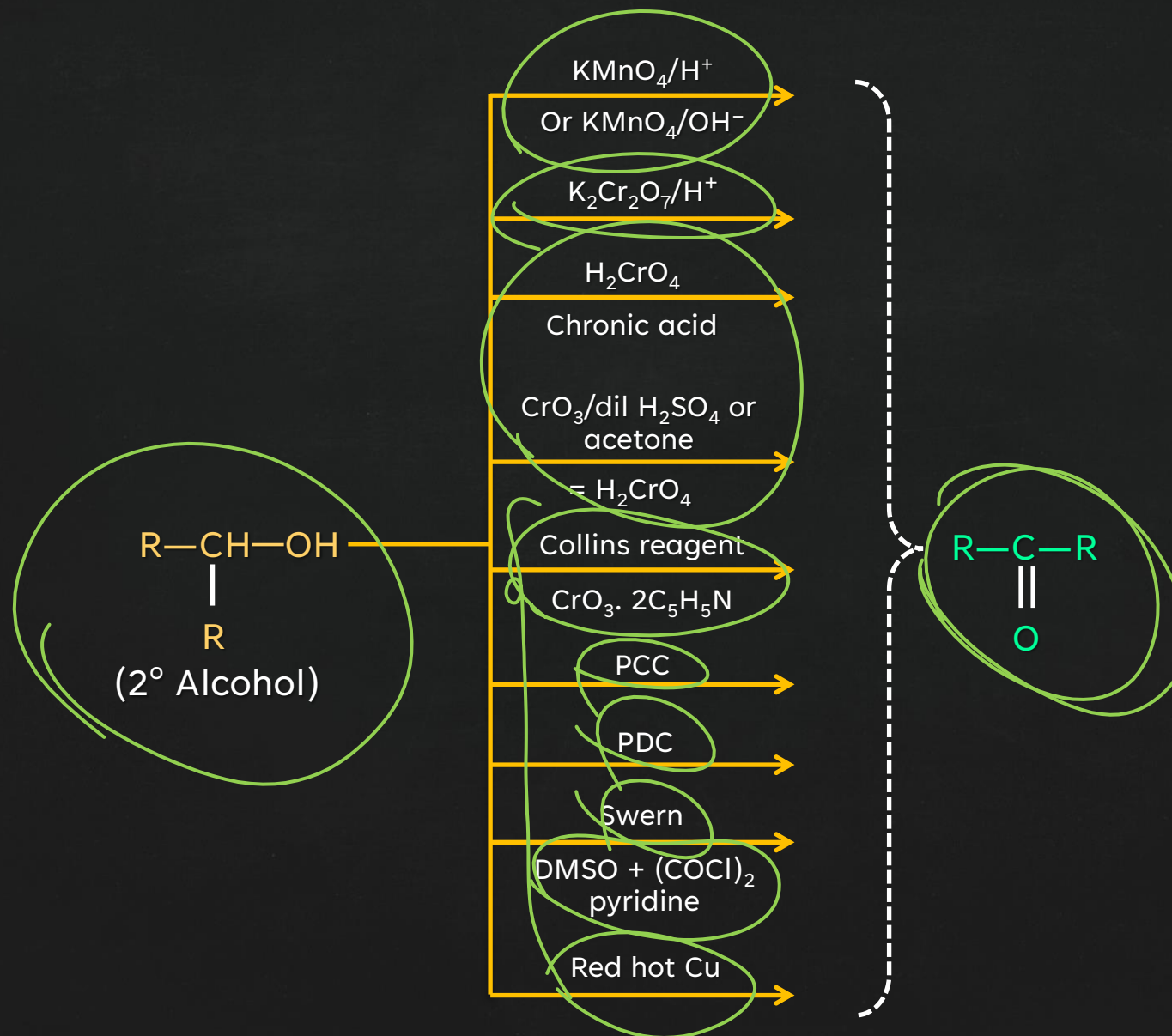


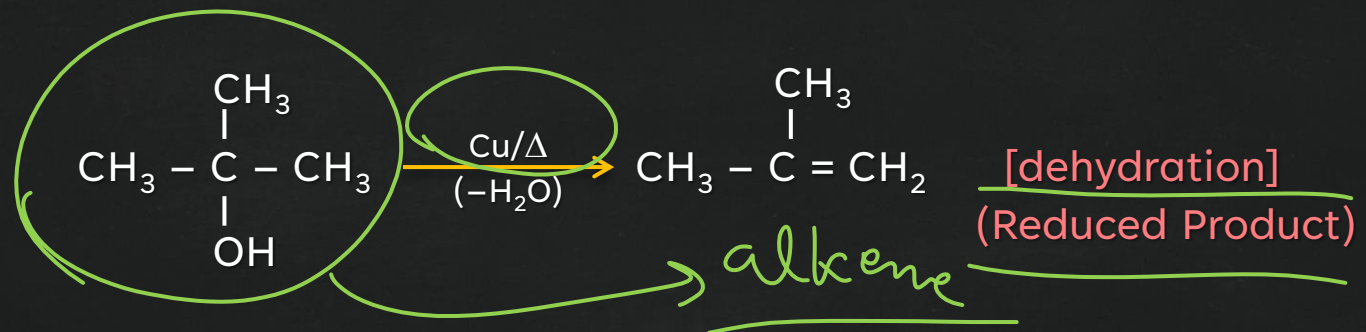




Strong Oxidising Agent
Carboxylic acid



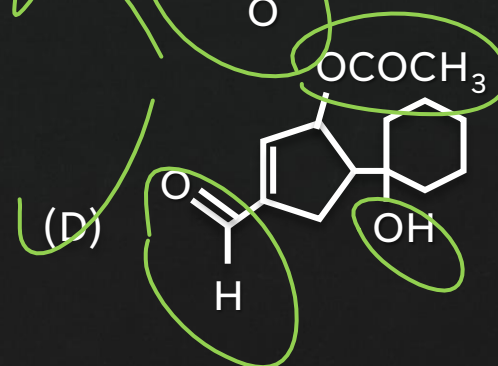
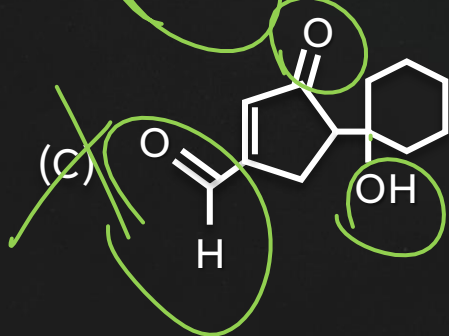
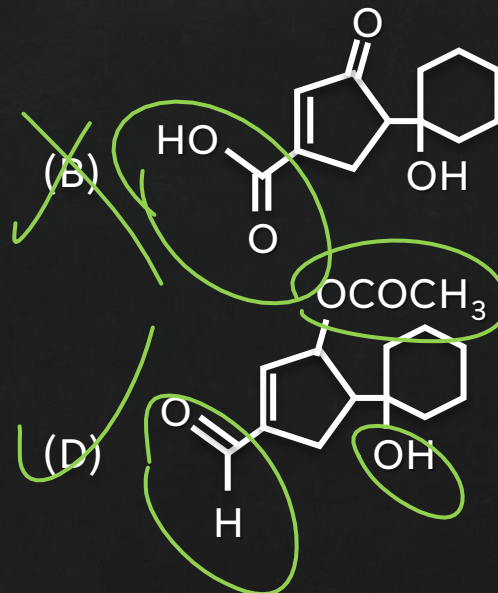
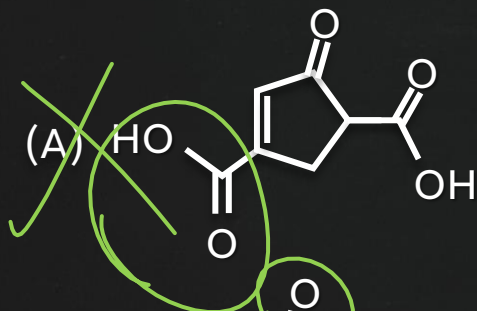
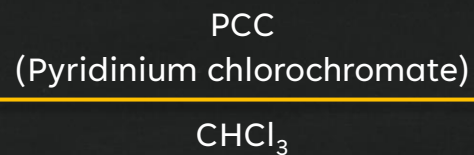
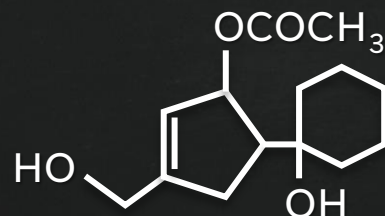






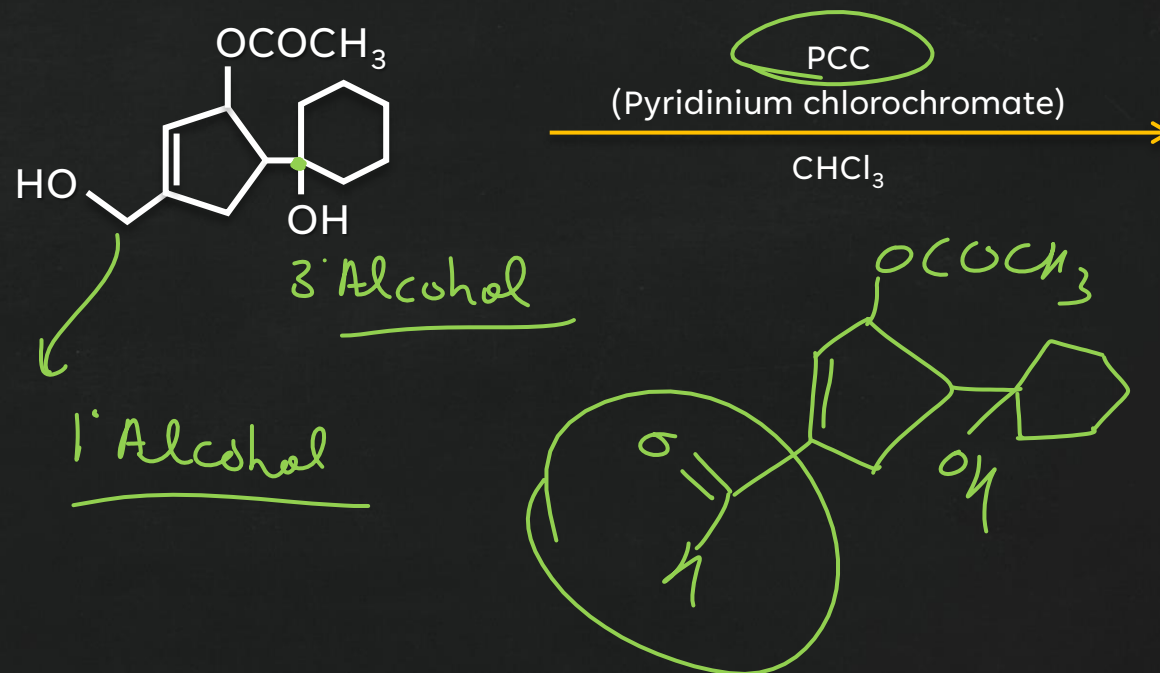
Example

The major product formed in the following reaction is :-



Solution

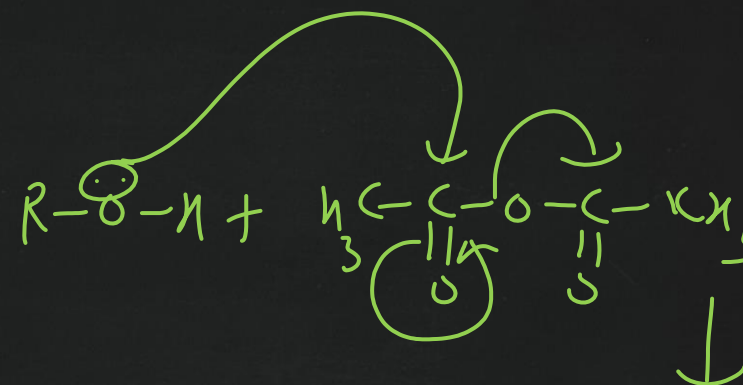
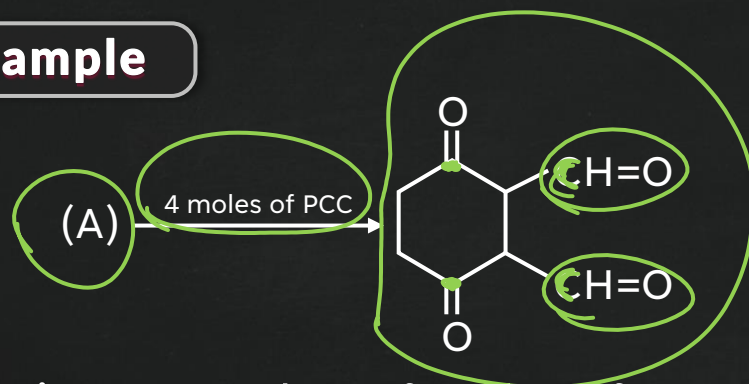
The major product formed in the following reaction is :-



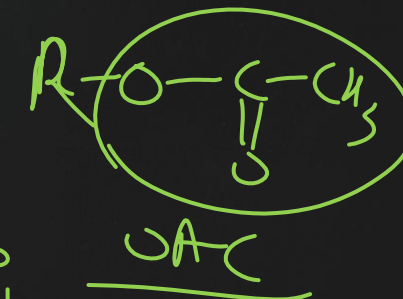
Ans. (D)



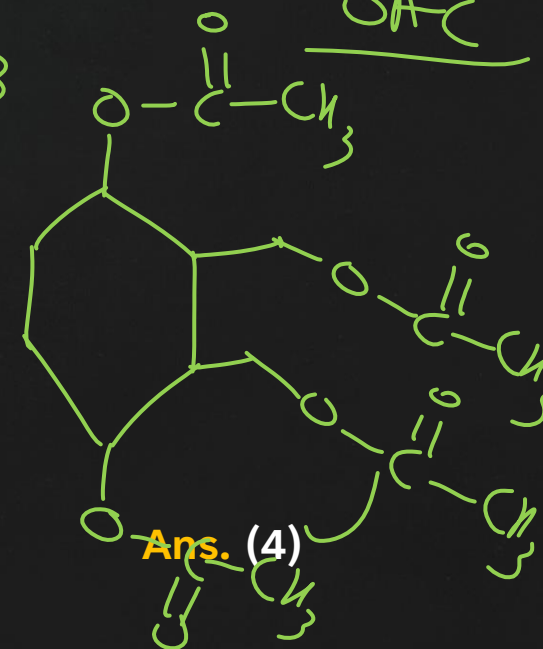
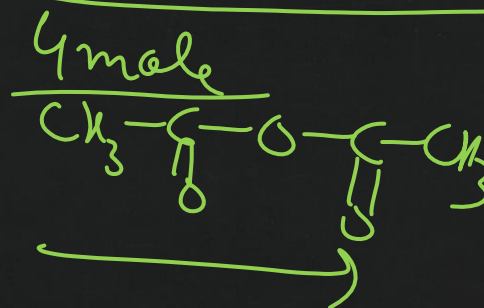
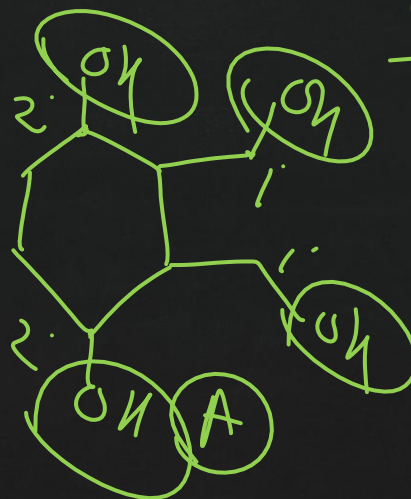
Example



Maximum number of moles of Ac_2O consumed by reactant (A) is :



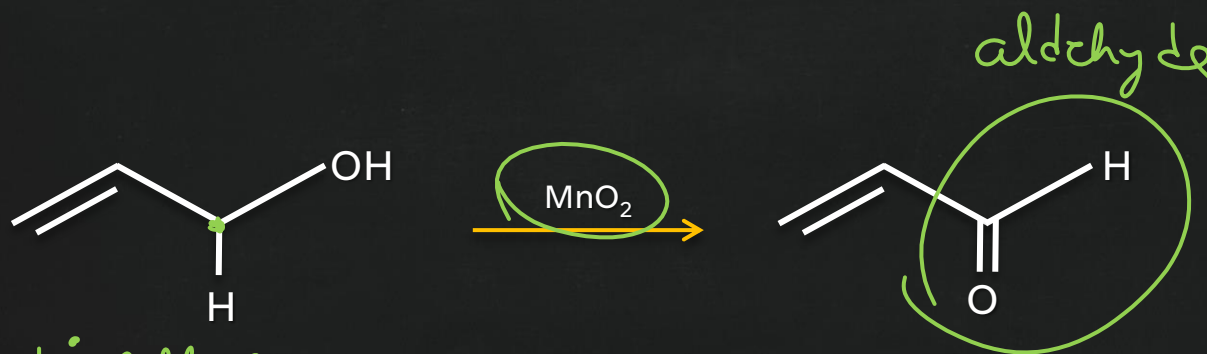
Solution



Ans. (4)

Reagent - Oxidation by MnO_2

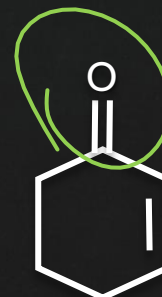
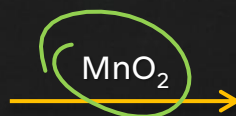
Function - Selectively oxidises allylic or benzylic 1° and 2° alcohol to give aldehyde and ketones, respectively.



1° Allylic Alcohol

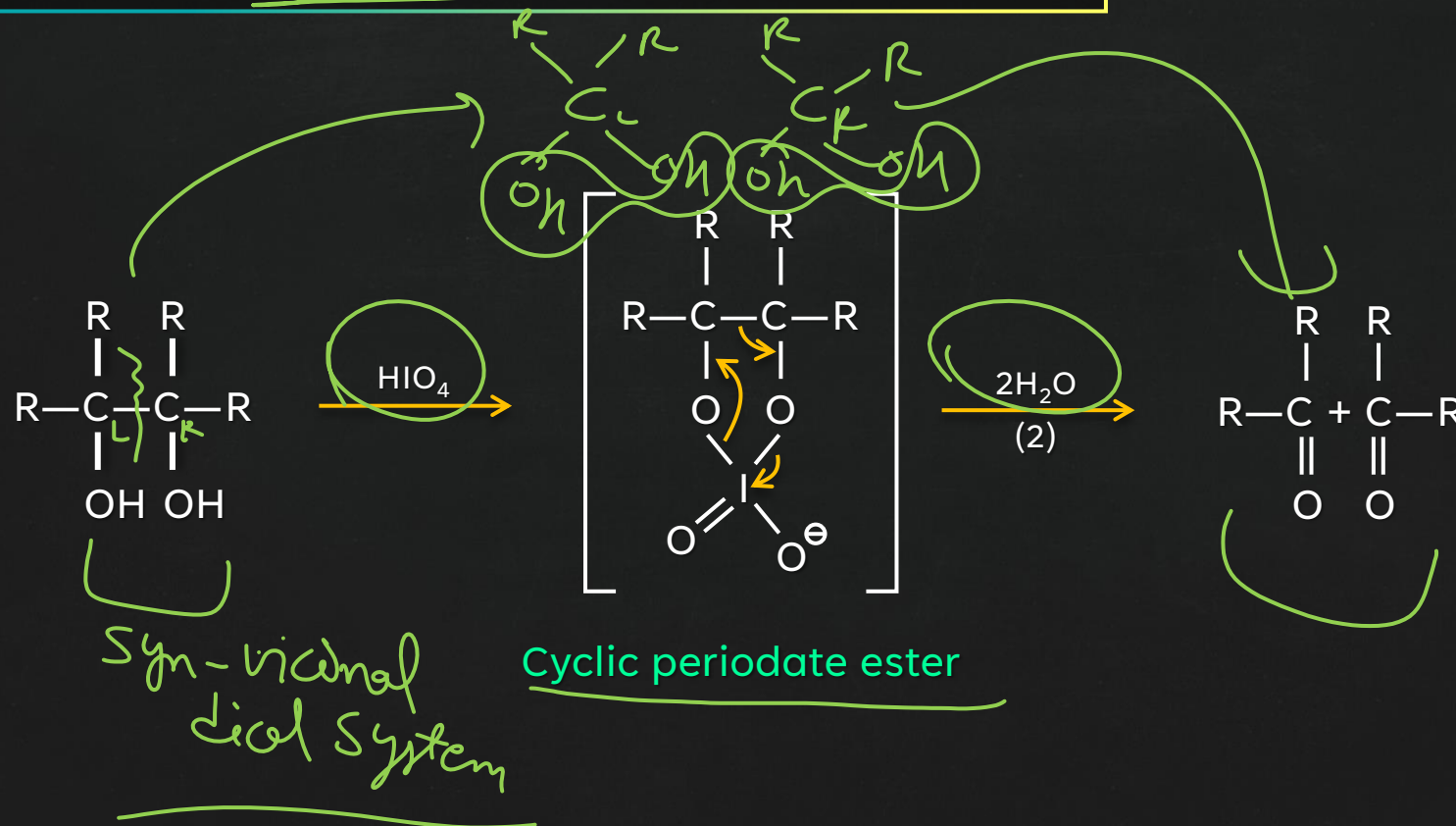


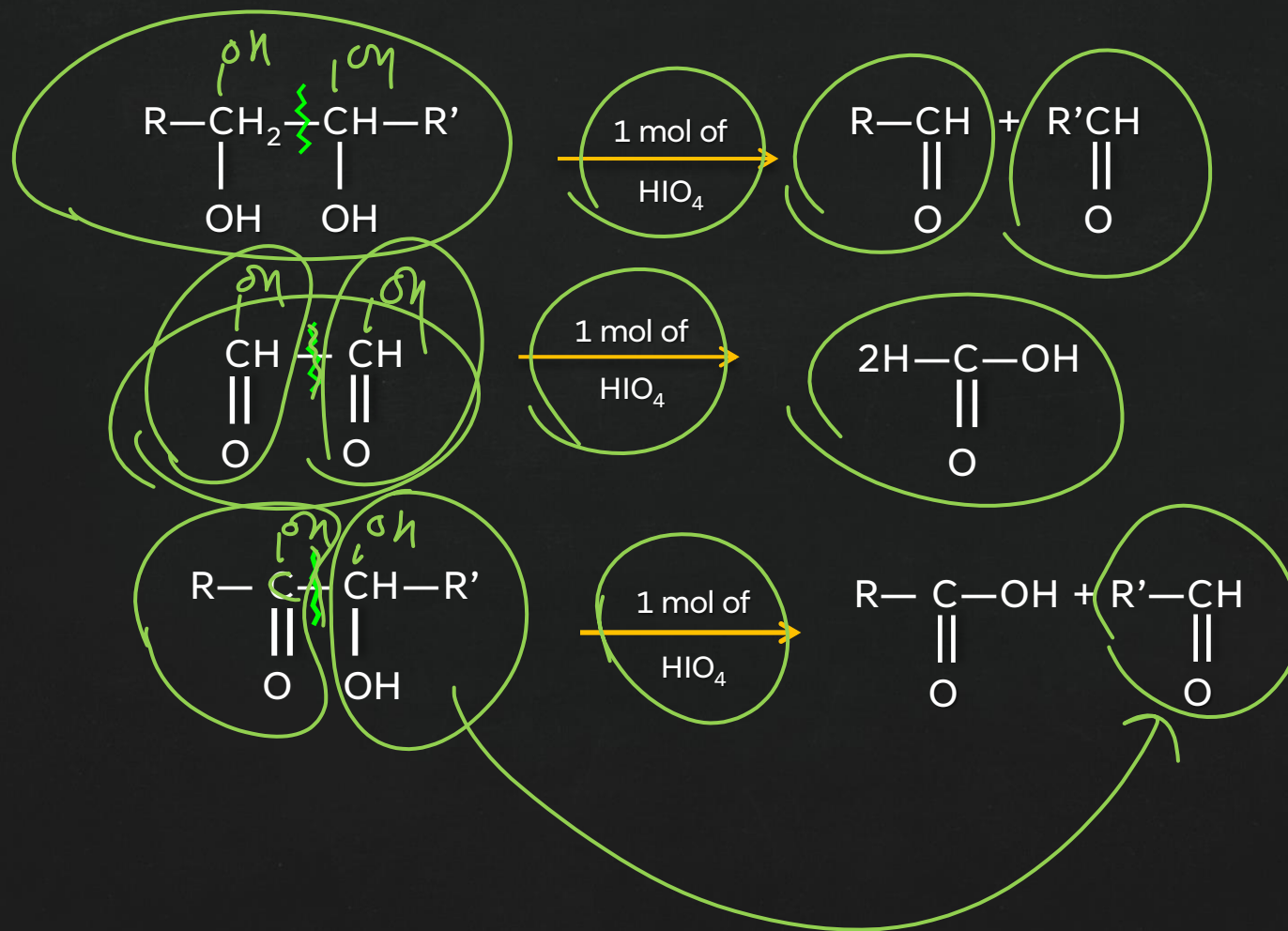
2° Allylic alcohol



Reagent - (i) $\text{HIO}_4 \cdot 2\text{H}_2\text{O}$ (ii) NaIO_4 (iii) $\text{Pb}(\text{OAc})_4$

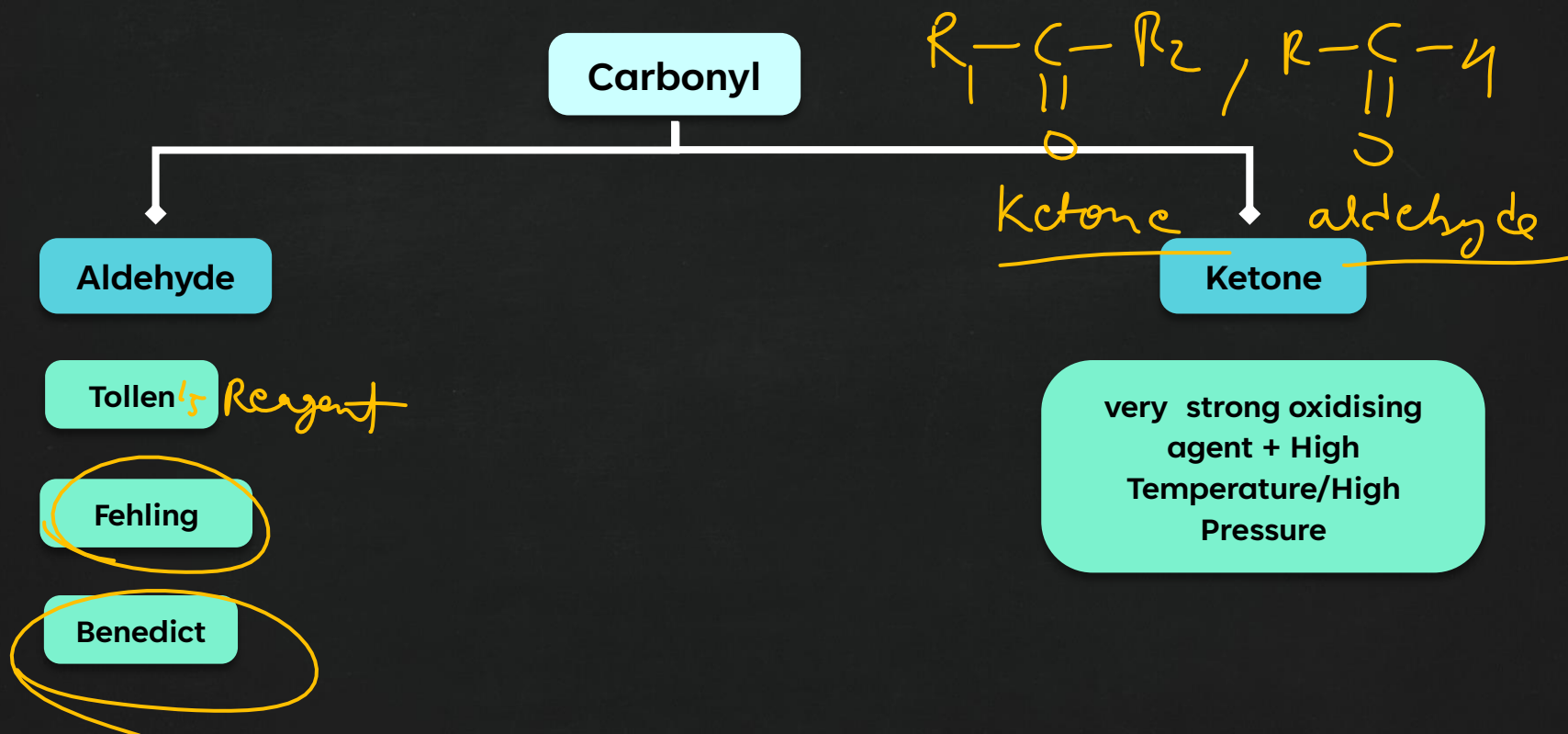
Function - Oxidation of Cis 1, 2-diol





Remarks

- ❑ In cyclic diols only cis-vicinal diols are oxidised. Trans isomers are not oxidised.



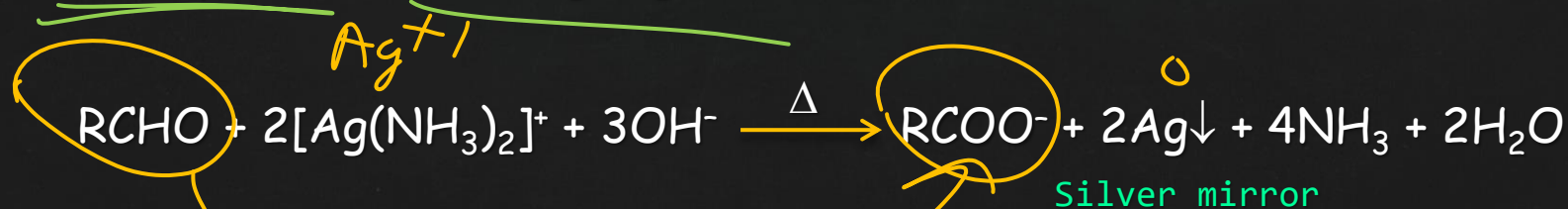
Reagent - Tollens' reagent

Tollens' reagent [ammoniacal AgNO_3 ; $\text{Ag}(\text{NH}_3)_2^+$]

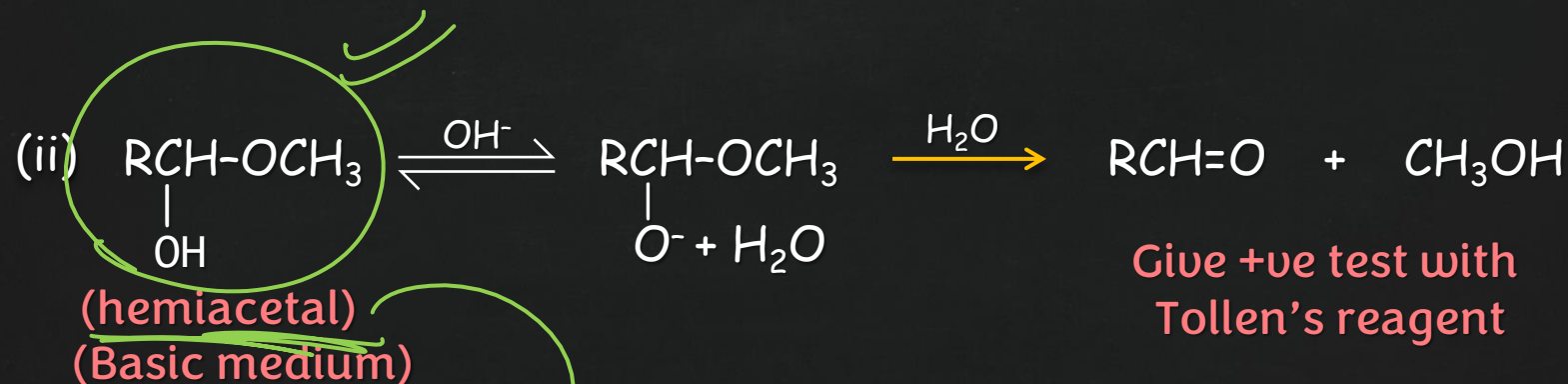
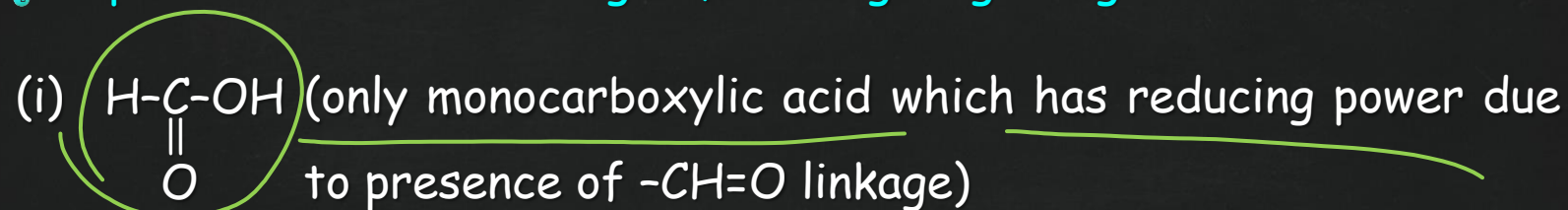


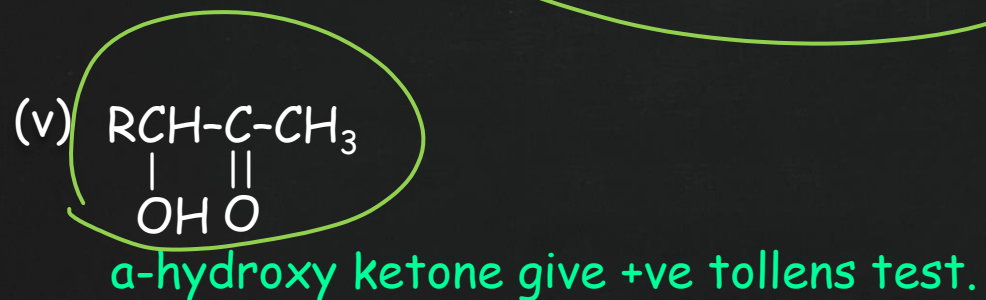
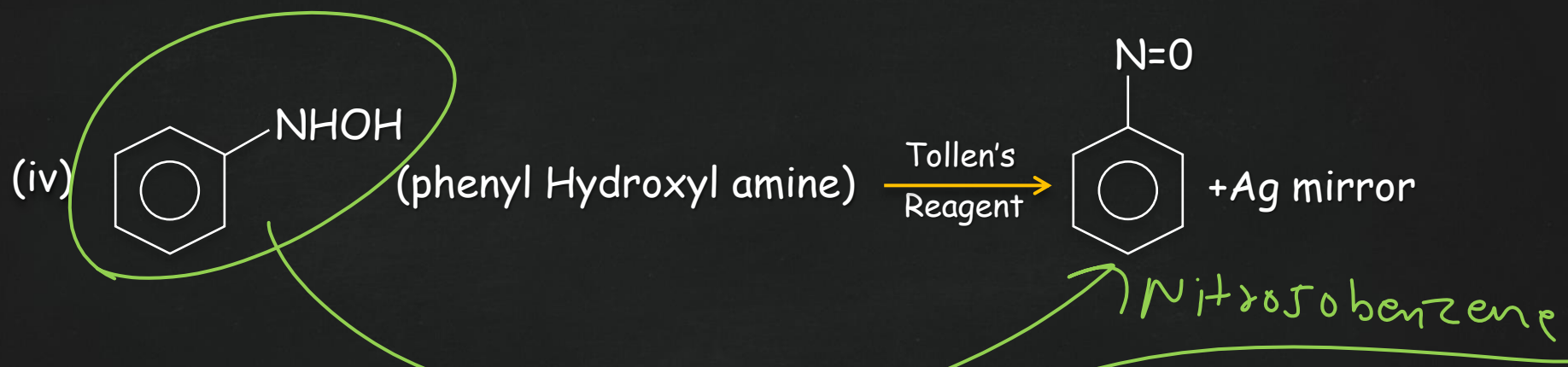
Colorless solution

Aldehydes reduce Tollens' reagent to Ag and appears in the form of silver mirror is called **silver-mirror test**. It is given by all aldehydes and reducing sugars.



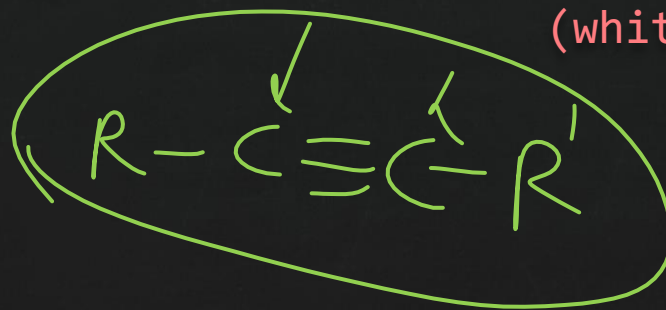
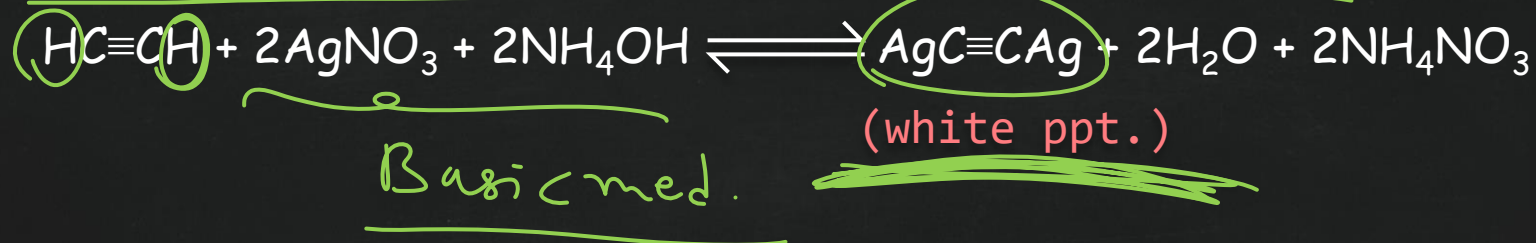
 In presence of Tollens' reagent, following reagents give silver mirror test.





Note

Terminal alkyne $RC\equiv CH$; $HC\equiv CH$ reacts with tollen's reagent.



**Example**

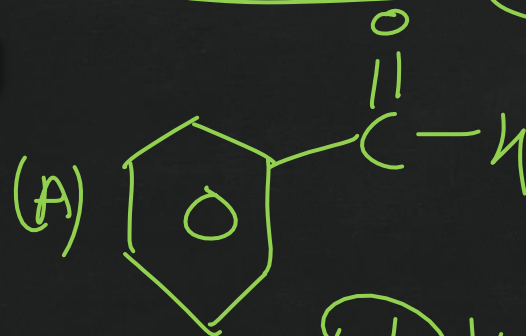
Which will give silver mirror test with Tollens reagent :

(A) $\text{C}_6\text{H}_5\text{CHO}$

(B) $\text{CH}_3\text{-CHO}$

(C) HCOOH

(D) All of these

Solution

Aromatic Aldehyde

(+)ve tollen's test

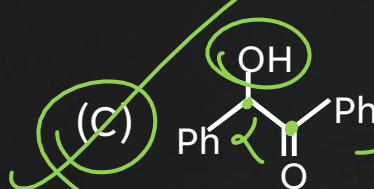
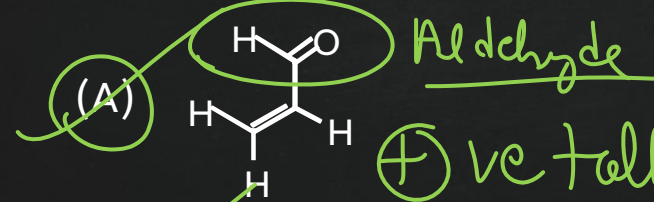
Ans. (D)



Example

[JEE-Advance 2016]

Positive Tollen's test is observed for



→ (-)ve Tollen's test

Solution

Hydroxy ketone

Ans. (A, B, C)

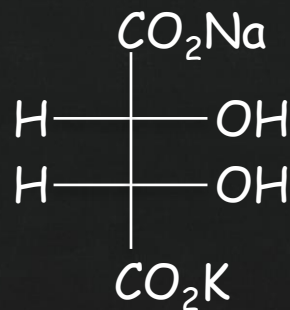
Reagent - Fehling's solution



It is a mixture of Fehling A and Fehling B solution.

Fehling A $\Rightarrow$ Aqueous solution of CuSO_4

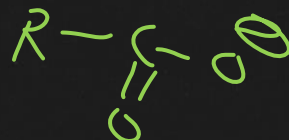
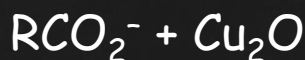
Fehling B $\Rightarrow$ Sodium potassium tartarate (Rochelle salt)



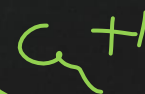
(Rochelle salt)



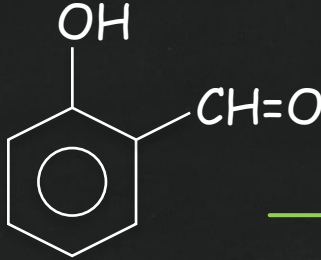
$\xrightarrow[\text{Fehling solution}]{\text{Cu}^{2+} \text{ (tartarate)}}$



(red precipitate)

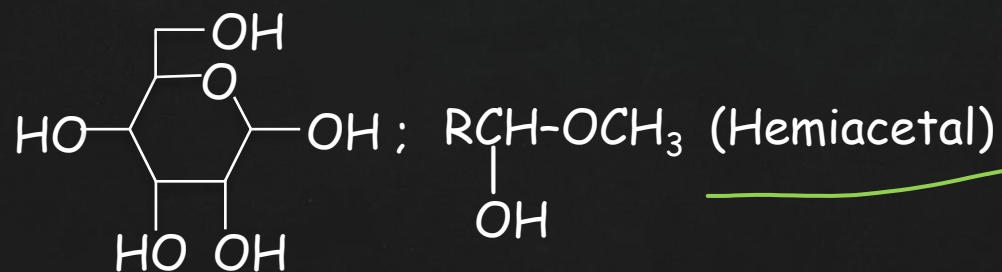


Points to Remember

1. Aromatic aldehyde, PhCH=O ;  gives -ve Fehling test
but +ve tollens test.

So oxidising power Tollens reagent > Fehling solution.

2. HCO_2H ; RCH(OH)-C(=O)-R (α -hydroxy ketone) ;



Also give, +ve fehling test.

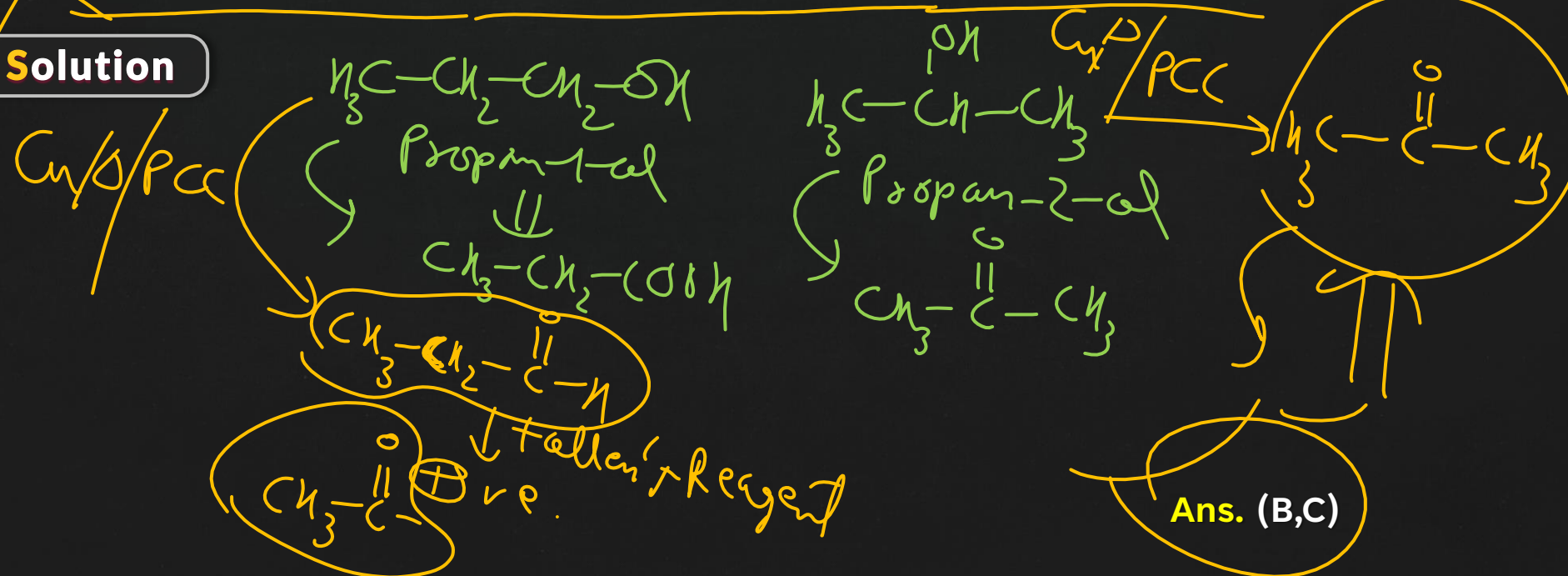


Example

Propan-1-ol and Propan-2-ol can be best distinguished by

- (A) Oxidation with alkaline KMnO_4 followed by reaction with H_2O
- (B) Oxidation with PCC followed by reaction with Tollen's reagent
- (C) Oxidation by heating with copper followed by reaction with iodoform test
- (D) Reaction with cone. H_2SO_4 followed by reaction with Fehling solution.

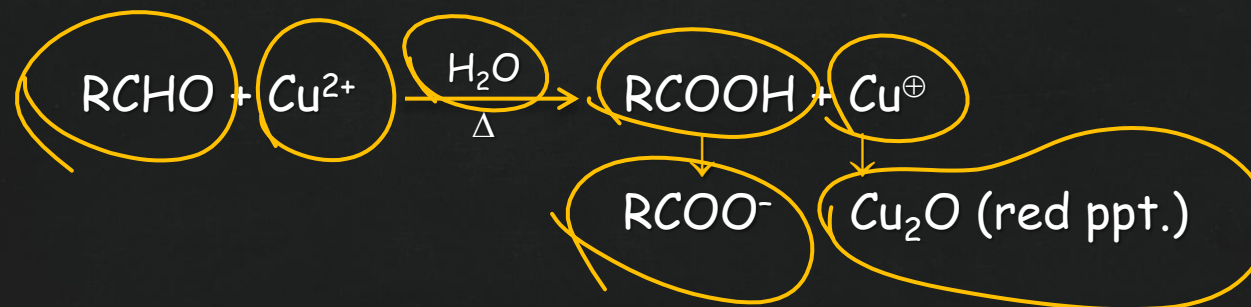
Solution



Reagent – Benedict's solution

Benedict's solution $\rightarrow$

Sodium citrate + NaOH + NaHCO₃ + CuSO₄





Carbohydrates which has aldehyde functional group (not aromatic aldehyde) or having alpha hydroxy ketone gives positive benedict test it is in mild basic medium.

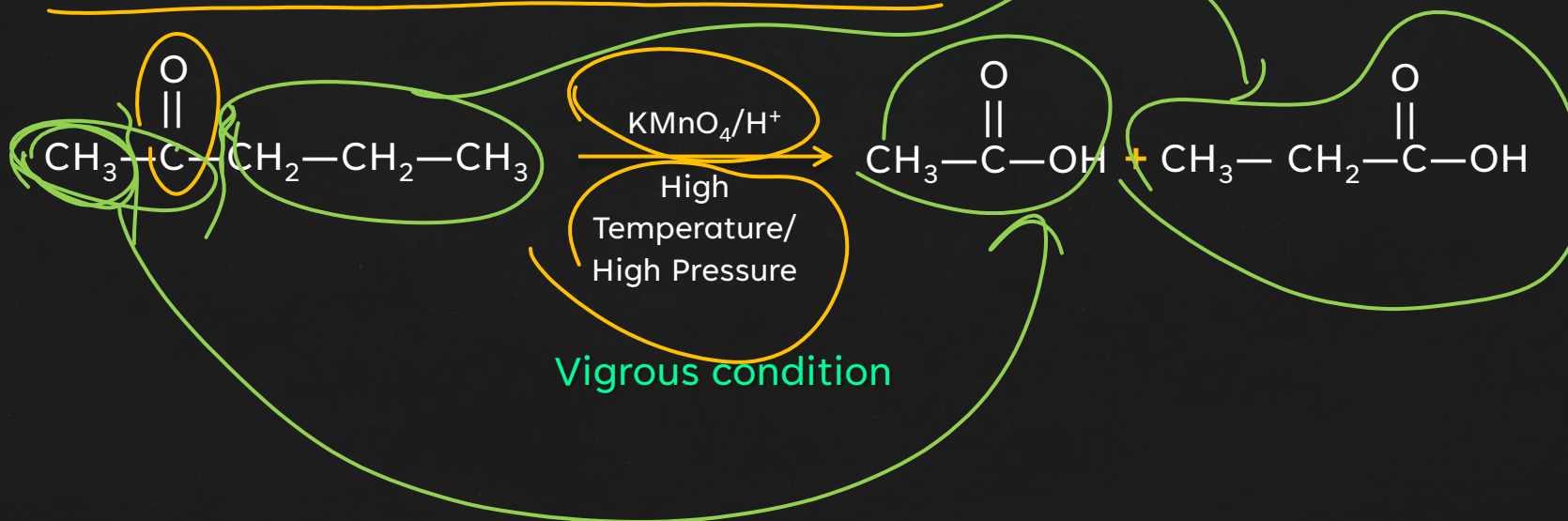
Reagent - KMnO_4/H^+ + High Temperature/High Pressure

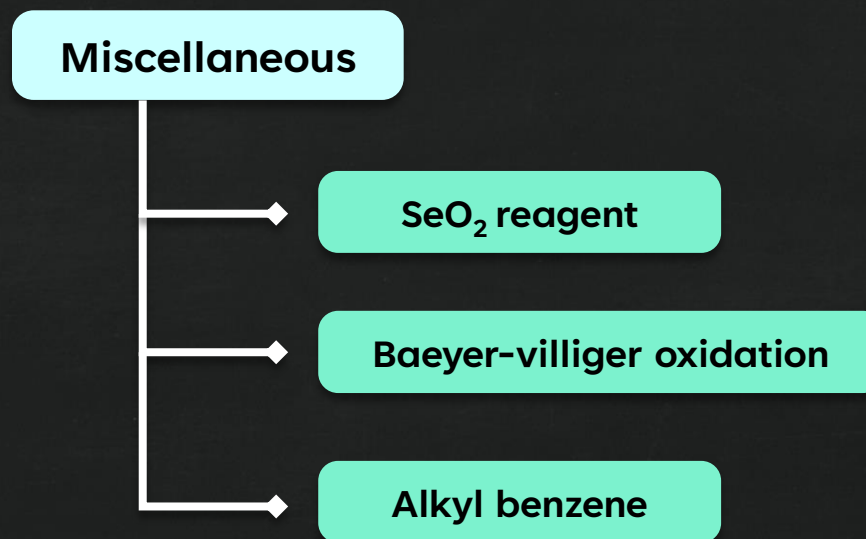
Function - Oxidation of ketone

Stereochemistry -

 Ketones can be oxidized from their enolic form at high temperature with very strong oxidizing agent.

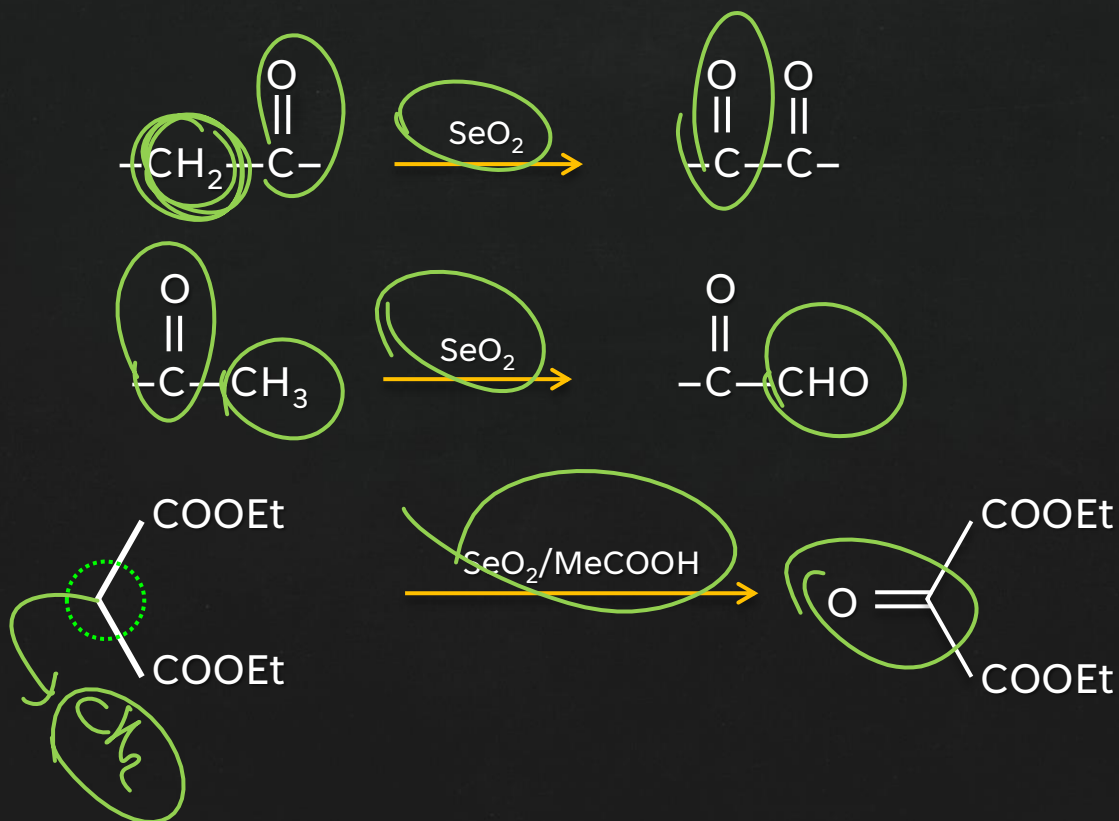
 In case of mixed or unsymmetrical ketones the $>\text{C}=\text{O}$ group remains with the smaller alkyl group. (**Popoff's rule**)





Reagent - SeO_2

Function - SeO_2 in acetic acid or acetic anhydride at $100-140^\circ\text{C}$, oxidises active methyl and active methylene group to aldehyde Or ketone, respectively, e.g.,

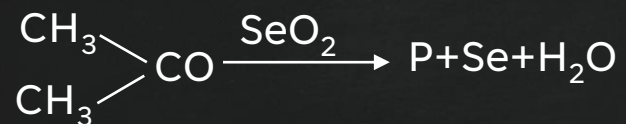




Example

[IIT 1995]

In the reaction, P is



(A) CH_3COCHO

(B) $\text{CH}_3\text{COOCH}_3$

(C) $\text{CH}_3\text{COCH}_2\text{OH}$

(D) None

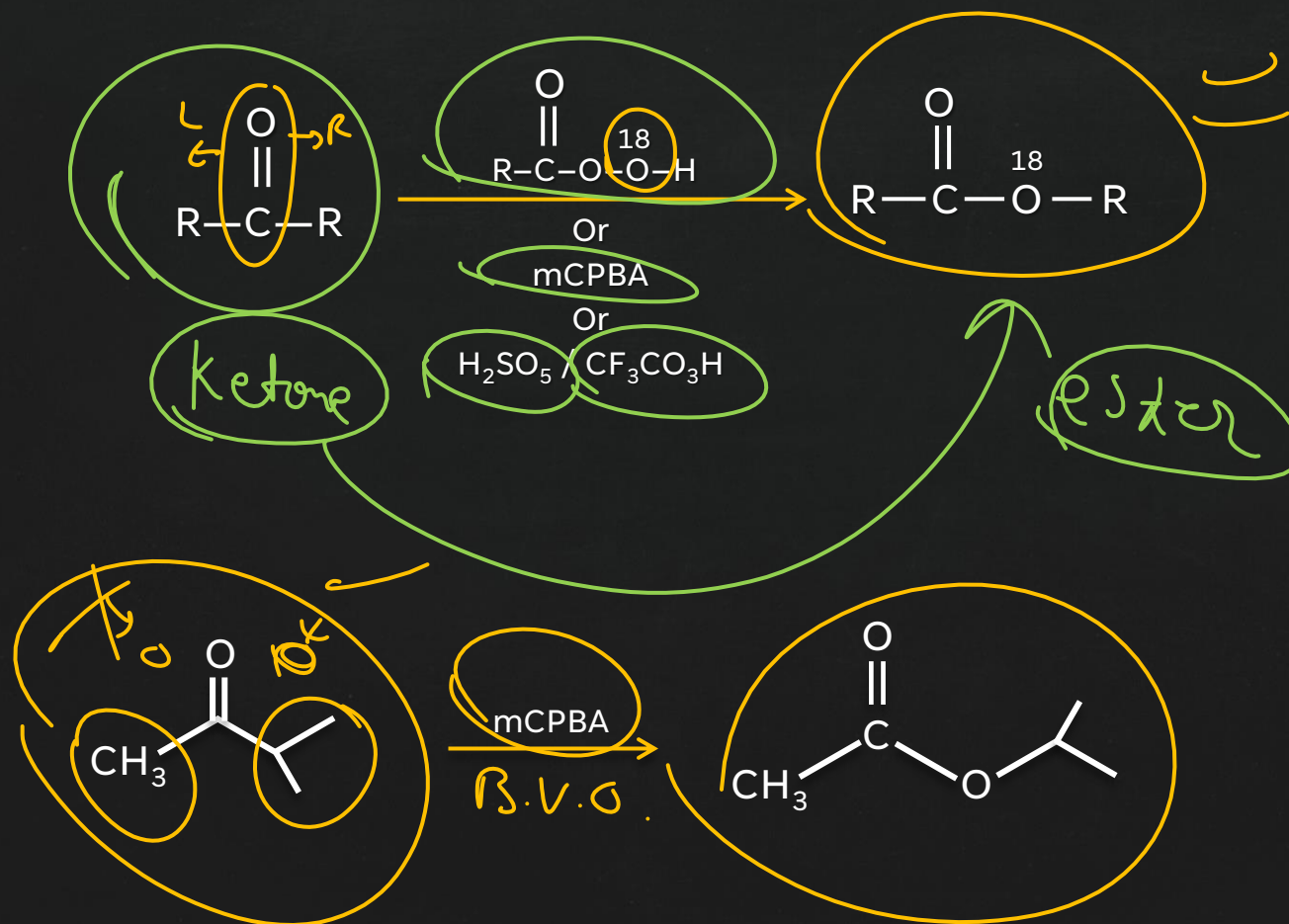
Solution



Ans. (A)

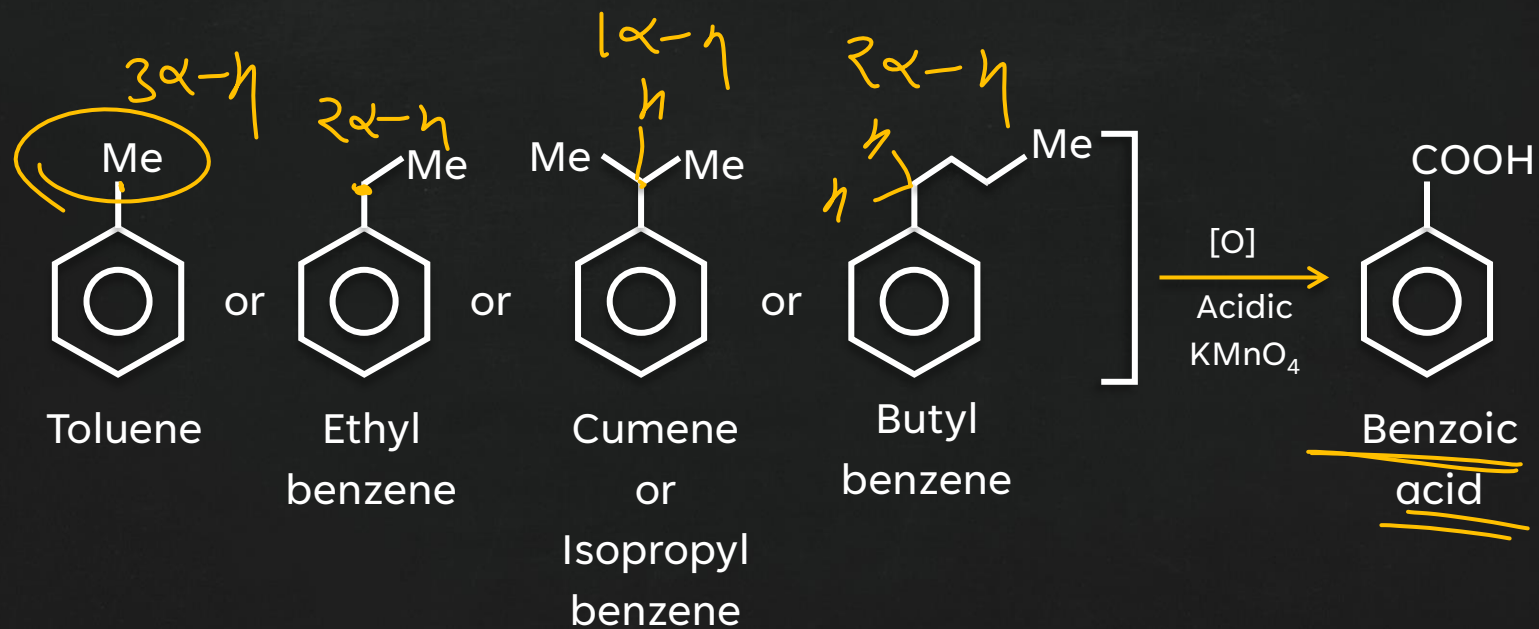
Reagent - Per Acid (Baeyer-villiger oxidation)

Function - Formation of ester from Carbonyl group



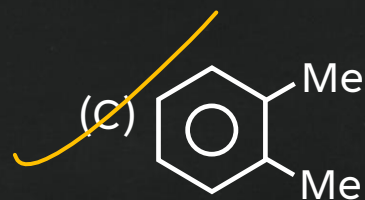
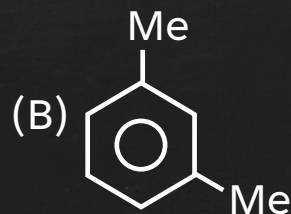
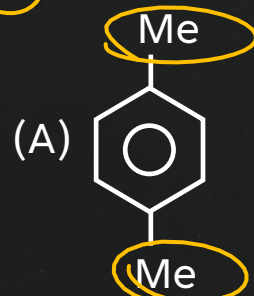
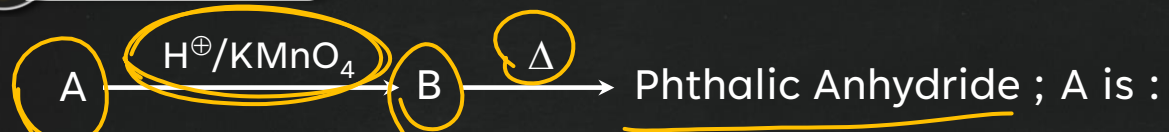
Reagent - KMnO_4/H^+ , $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$

Function - Side Chain Oxidation of Benzene and its derivative

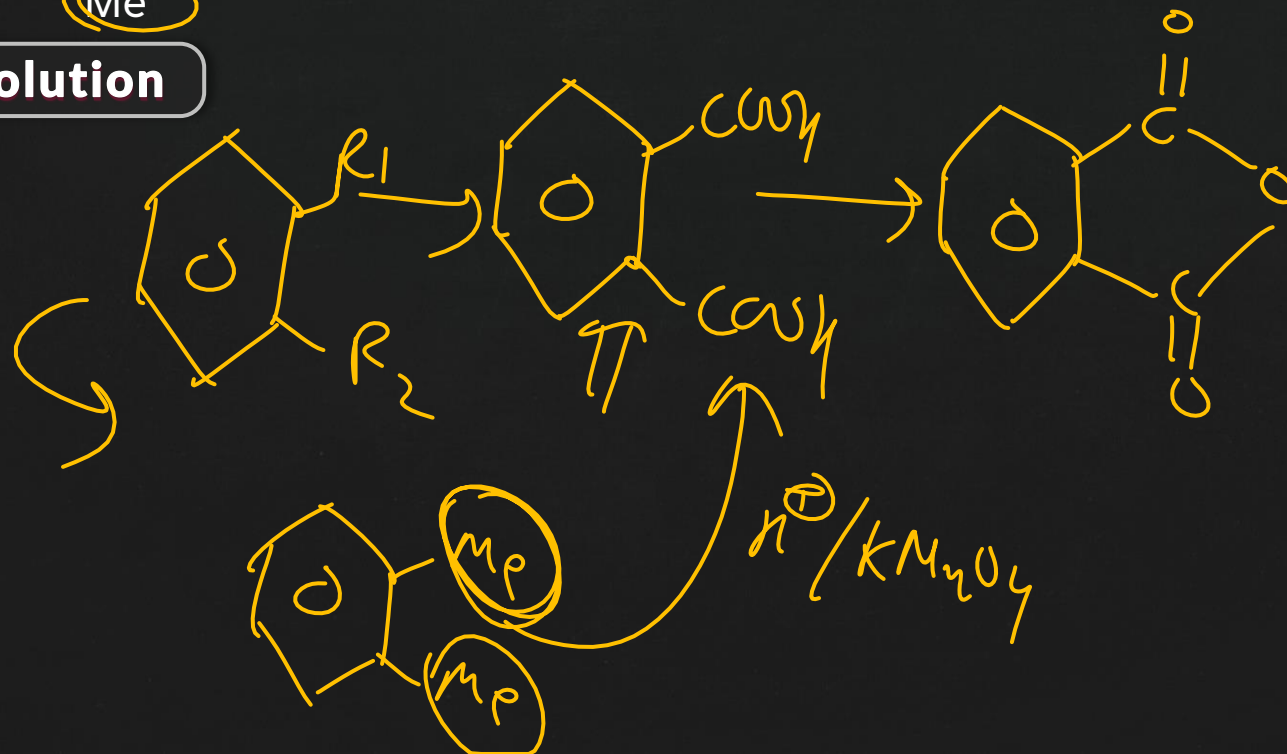




Example



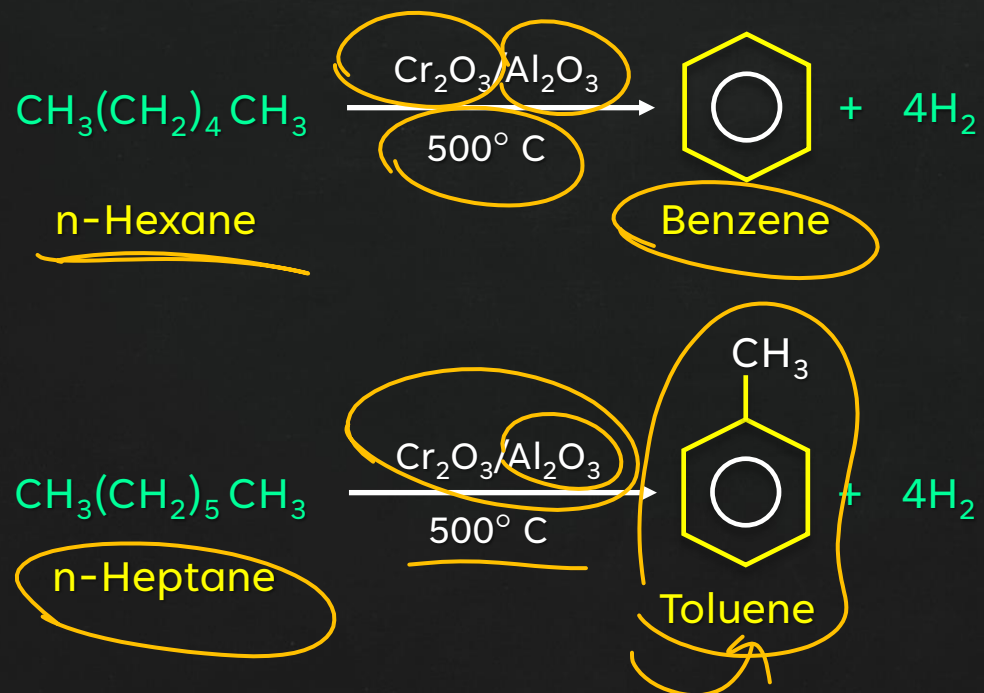
Solution



Ans. (C)

Reagent - $\text{Cr}_2\text{O}_3/\text{Al}_2\text{O}_3/\text{Mo}_2\text{O}_3$, High T,P

Function - Aromatization of Alkane

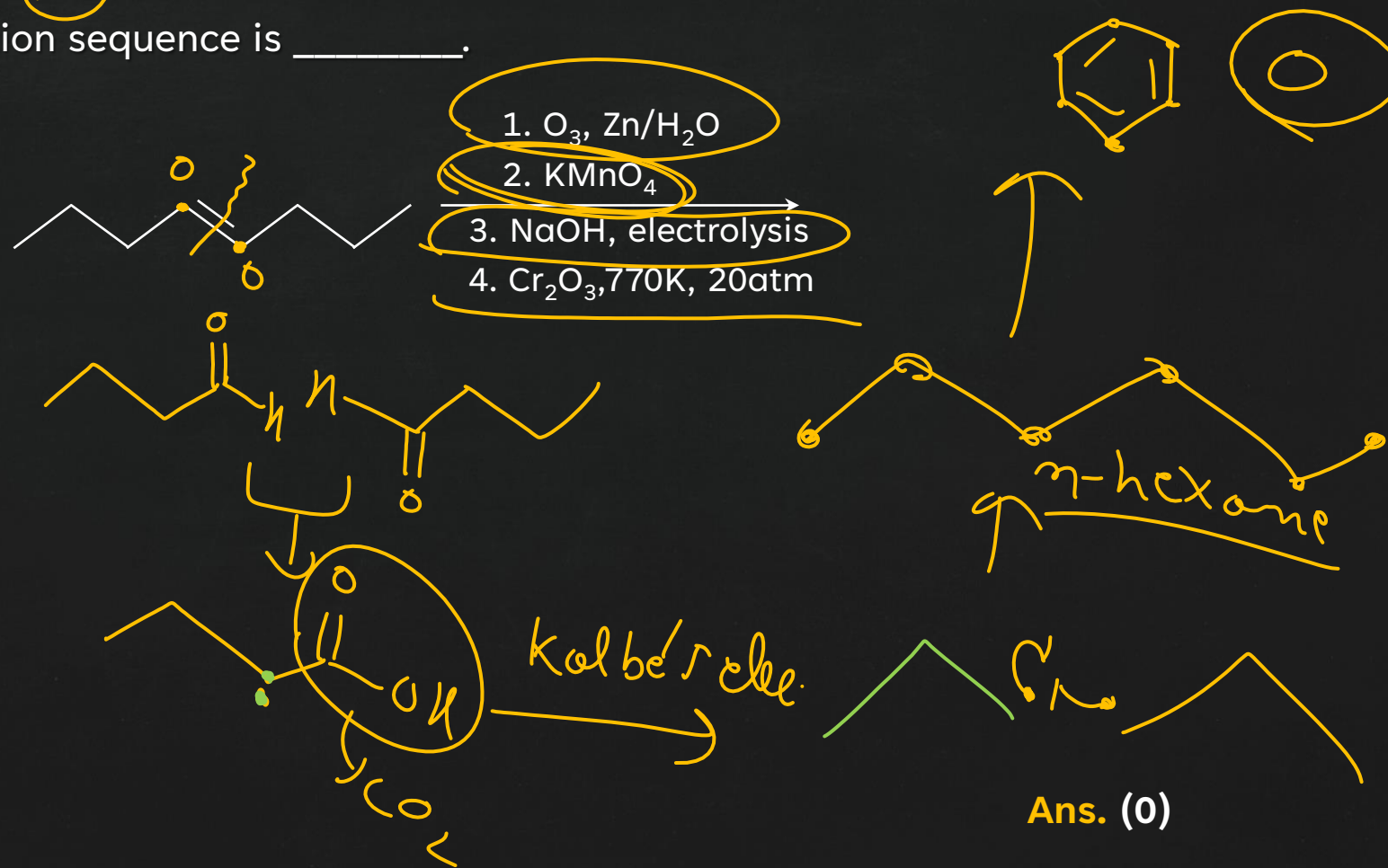




Example

[JEE-Advance 2022]

The number of $-\text{CH}_2-$ (methylene) groups in the product formed from the following reaction sequence is _____.

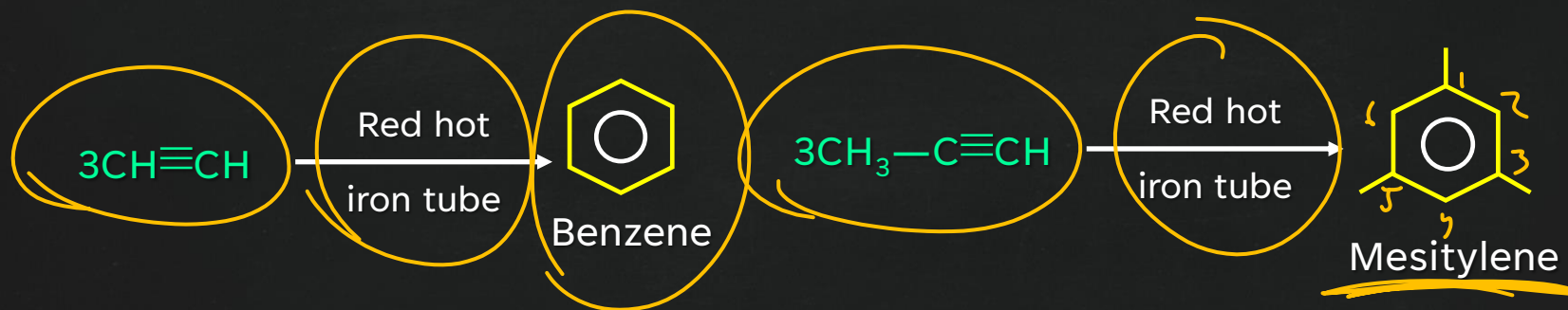


Solution

Ans. (0)

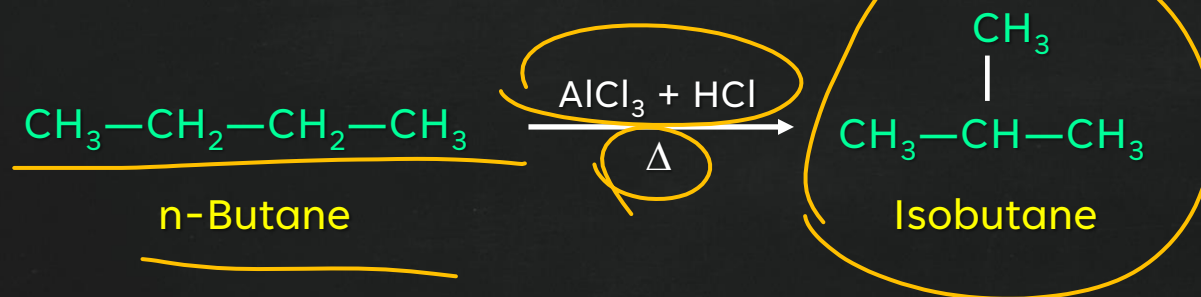
Reagent - Red hot Fe Tube

Function - Cyclic polymerization of Alkyne



Reagent - $\text{AlCl}_3 + \text{HCl}$

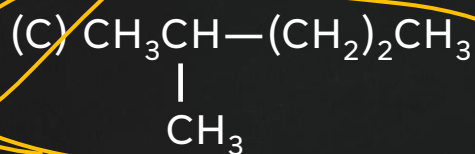
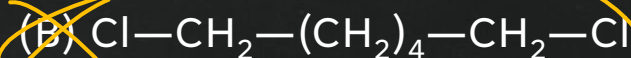
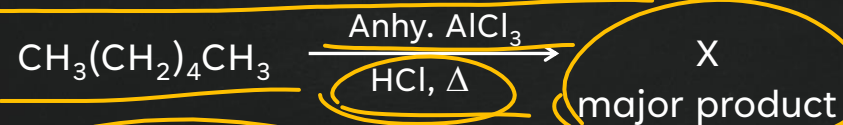
Function - Isomerisation of Alkane



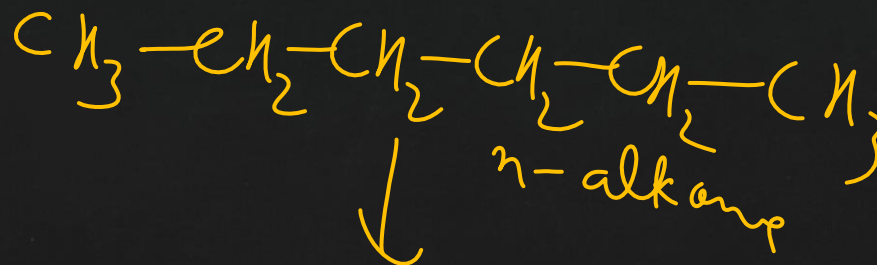


Example

In the following reaction 'X' is



Solution



Branch-Alkane

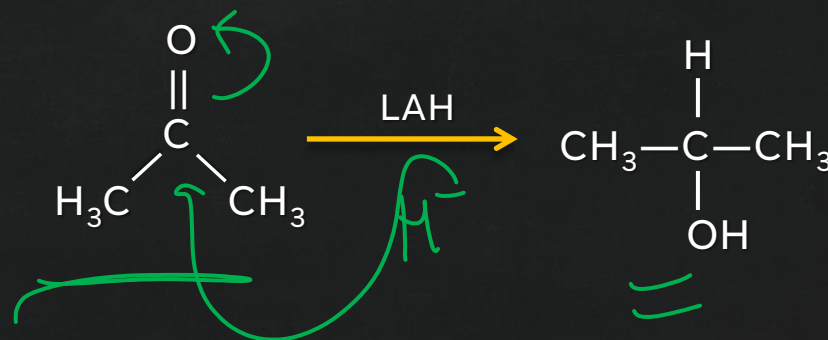
Ans. (C)

REDUCING REAGENTS

Reagent - Lithium Aluminium Hydride (LiAlH_4) $\text{LAH} \rightarrow$ 

Function - Reduction of carbonyl compounds, imines, carboxylic acids and its derivatives, conjugated double bond.

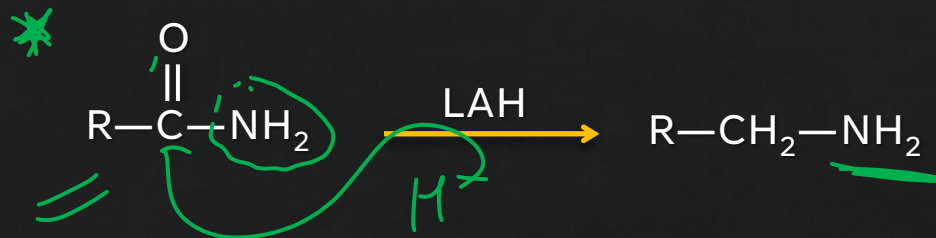
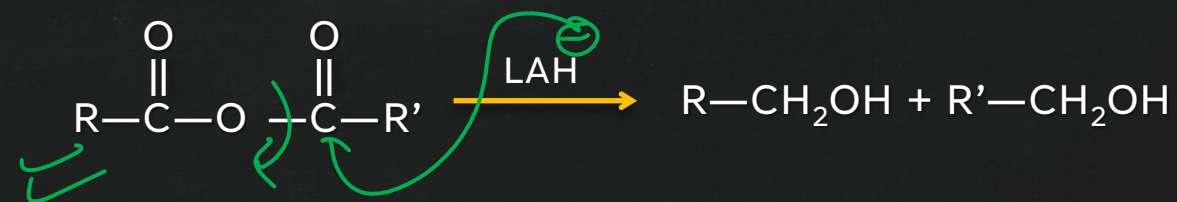
Stereochemistry - Enantiomers and Diastereomers



It is sensitive to protic solvent and therefore used in ether.

Substrate	Product
✓ RCOOR	RCH_2OH
✓ RCOOH	RCH_2OH
✓ RCOCl	RCH_2OH
✗ RNO_2	RNH_2
✗ RCN	RCH_2NH_2
✓ RCHO	RCH_2OH
✓ RCOR	RCHOHR
✗ $\text{RC} \equiv \text{CR}$	No Reaction
✗ $\text{RCH}=\text{CHR}$	No Reaction

Example



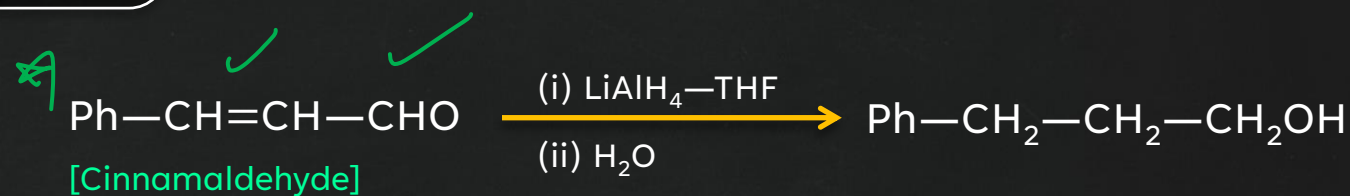
...

Example

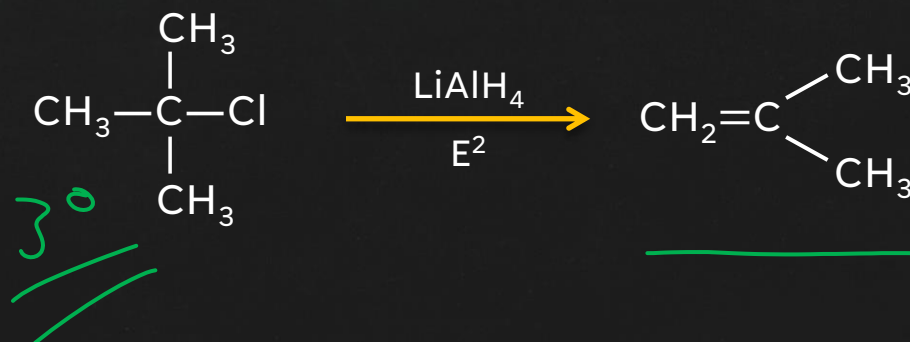
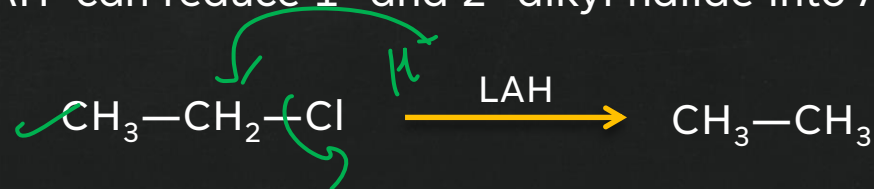
...

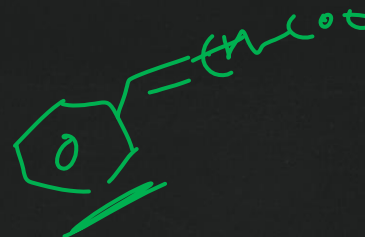


 Key Point

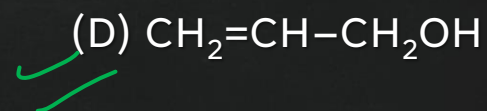
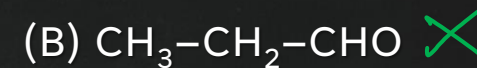
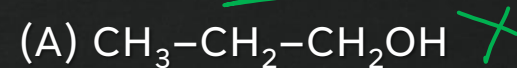


 LAH can reduce 1° and 2° alkyl halide into Alkane but not 3°



**Example**

When $\text{CH}_2=\text{CH}-\text{COOH}$ is reduced with LiAlH_4 , the compound obtained will be

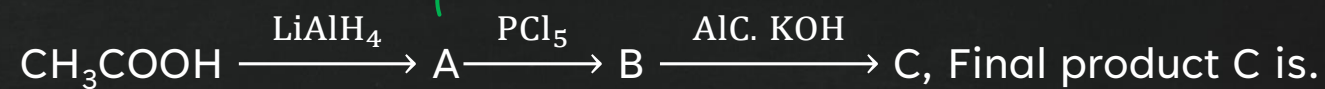
**Solution**

✓
Ans. (D)

[JEE 2014]

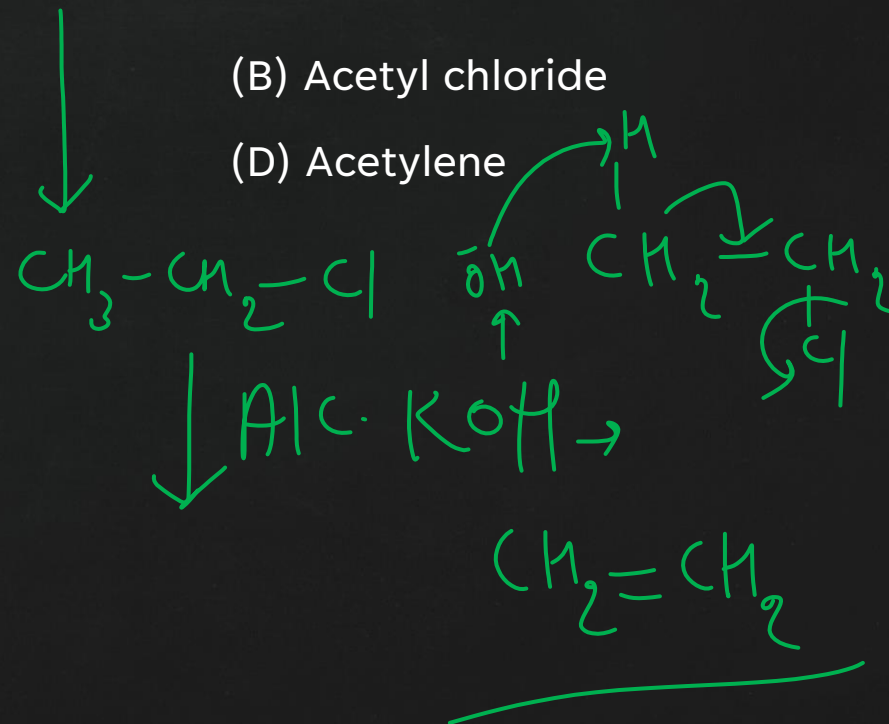
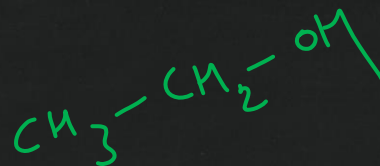
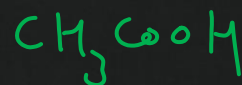
Example

In the reaction,



- (A) Ethylene
(B) Acetyl chloride
(C) Acetaldehyde
(D) Acetylene

Solution



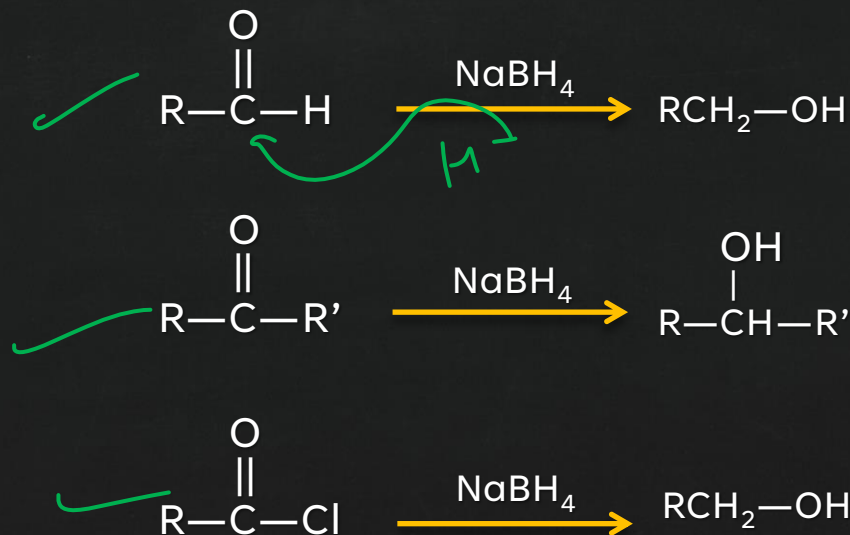
Ans. (A)

Reagent - Sodium borohydride [NaBH_4 in $\text{C}_2\text{H}_5\text{OH}$ or Ether]

Function - Reduction of carbonyl compounds, imines and Acid Chloride

Weak reducing agent as compared to LAH

Stereochemistry - Enantiomers and Diastereomers



NOTE

NaBH_4 can also reduce imine group into amine.

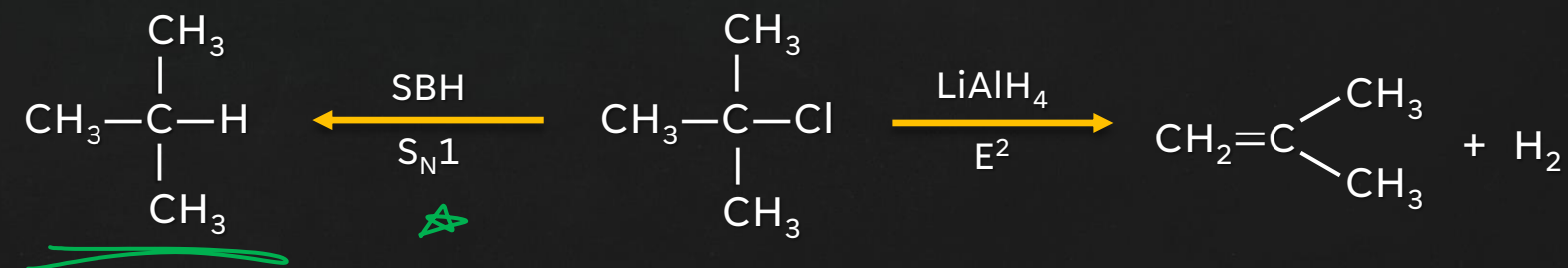
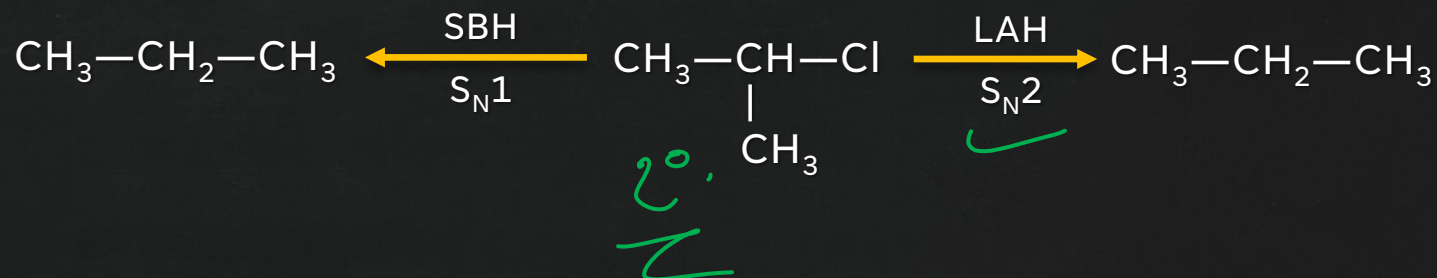
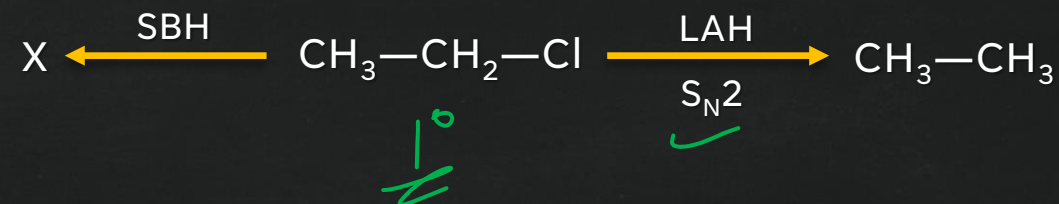
LAH SBH

more
powerful
↓
weak
↓
fast H^-

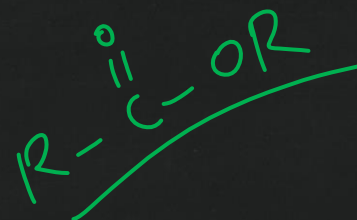
strong H^-



It can reduce 1° and 2° alkyl halide into Alkane but not 3°

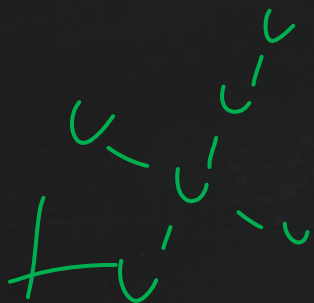


NOTE



LiBH_4 reduces Esters into Alcohols (Selective Reduction)

~~4~~



Example

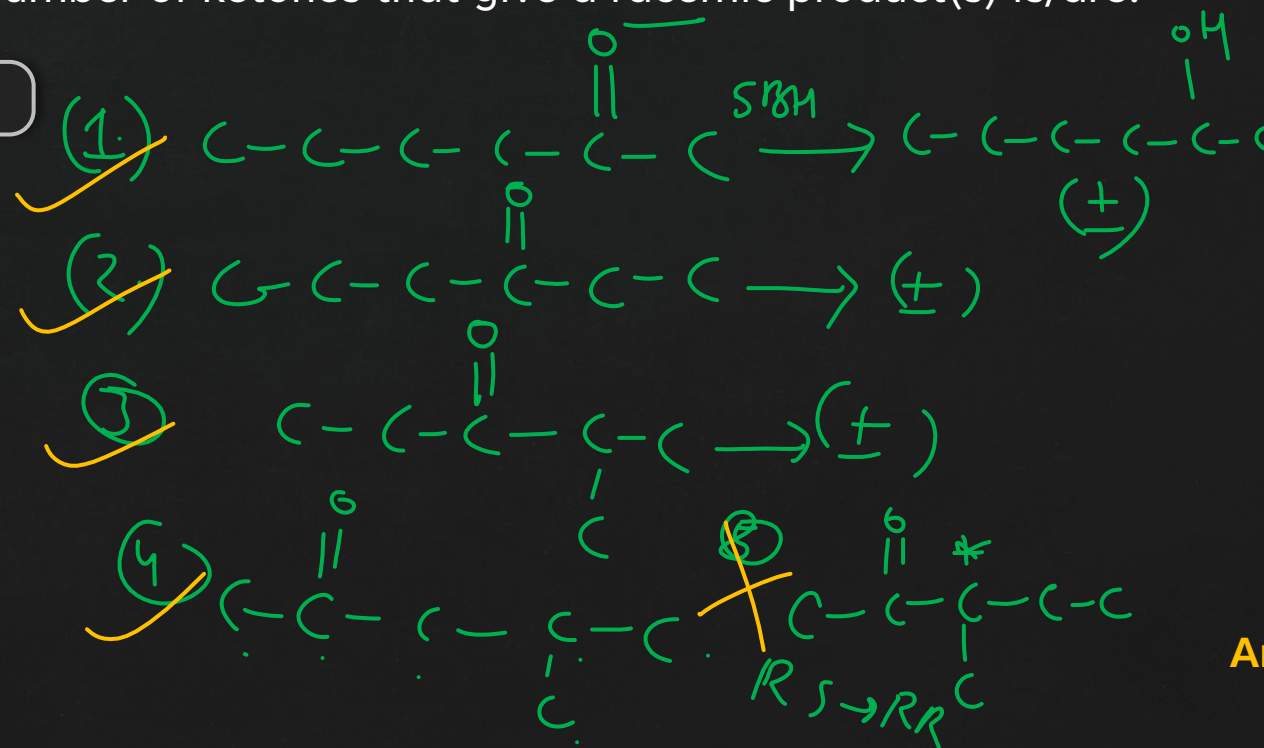


[JEE 2014]

Consider all possible isomeric ketones including stereoisomers of MW = 100,
All these isomers are independently reacted with $NaBH_4$ (NOTE : stereoisomers
are also reacted separately).

The total number of ketones that give a racemic product(s) is/are.

Solution

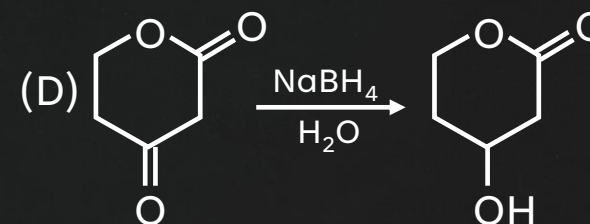
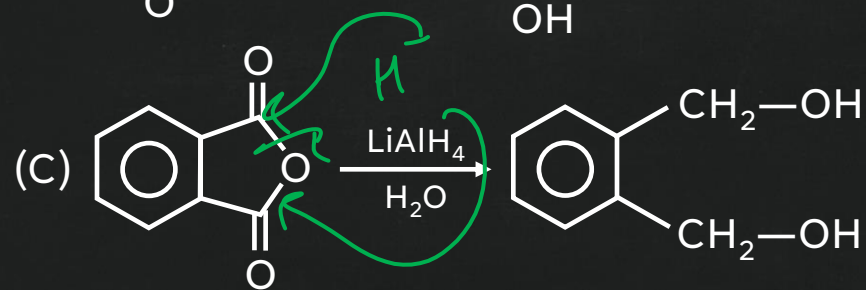
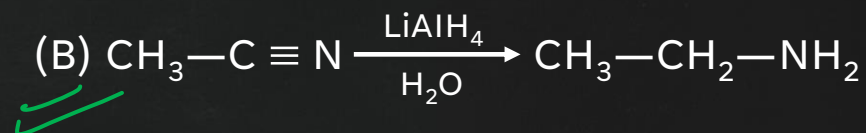
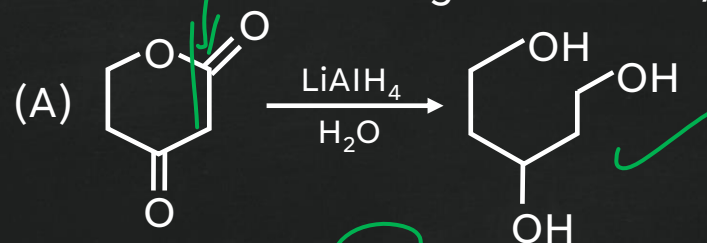


Ans. (5)



Example

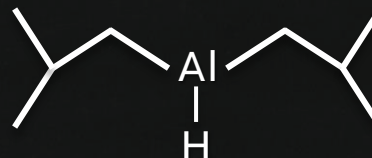
Which of the following reactions is/ are correct?



Solution

Ans. (A,B,C,D)

Reagent - Diisobutyl Aluminium Hydride [DIBAL-H / Inert solvent]



DIBAL-H is used at -78°C .

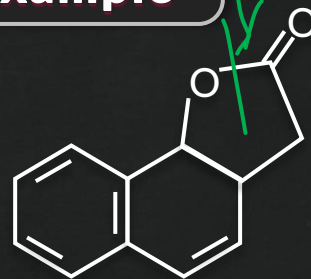
Function - Partial Reduction of Esters, Cyanides

Stereochemistry - Enantiomers and Diastereomers

Reagent	DIBAL-H/ (-78°C)	DIBAL-H/ (-78°C)
Reactant	Ester ✓	Cyanide ✓
Product	Aldehyde	Aldehyde



Example



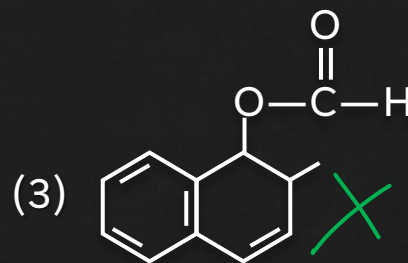
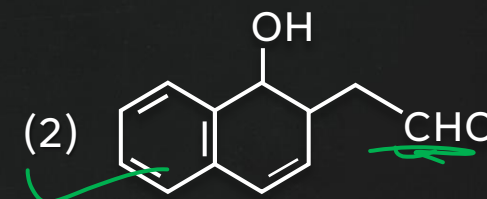
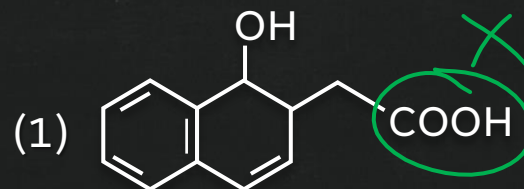
(i) DIBAL-H, Toluene, -78°C

(ii) $\text{H}_3\text{O}^{\oplus}$

"P"

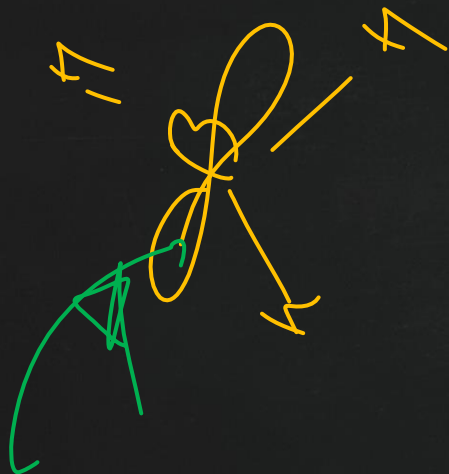
(Major Product)

The product "P" in the above reaction is :



Solution

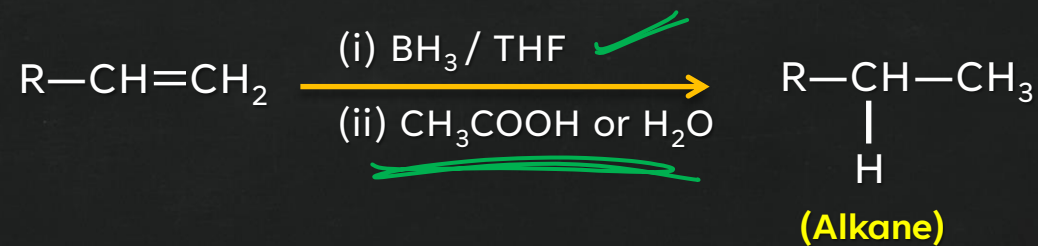
DIBAL-H at R.T. $\rightarrow$ (complete Reduction) Ans. (2)



Reagent - B_2H_6 or BH_3/THF in CH_3COOH or H_2O

Function - Reduction of Nucleophilic Functional group

Stereochemistry - **Syn** Addition



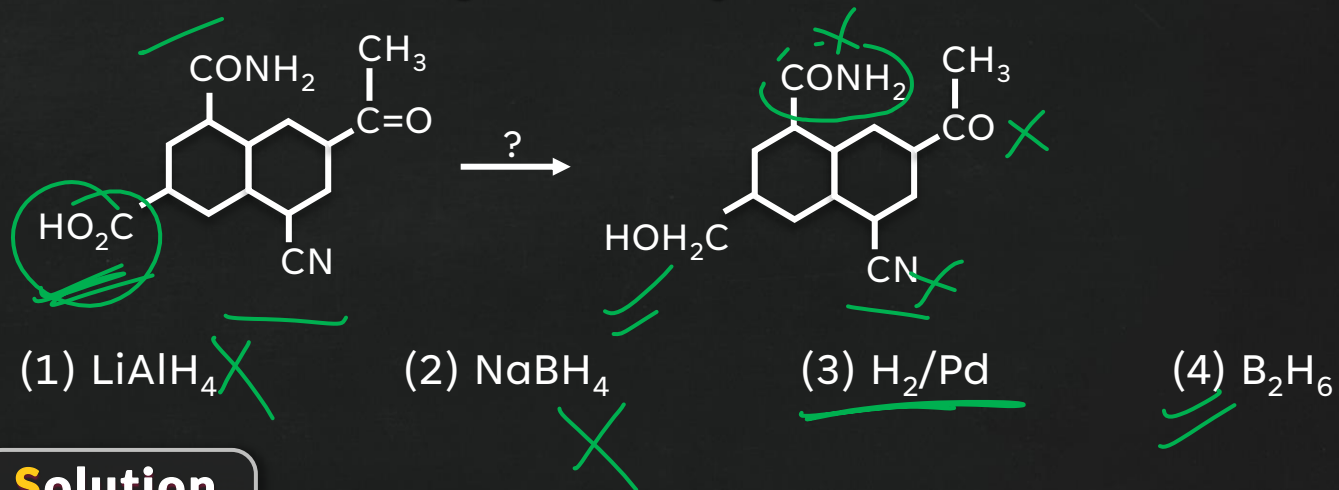
NOTE

[Clayden]

Diborane is a highly chemoselective reagent for the **reduction of carboxylic acids** in the presence of other reducible functional groups such as esters and even ketones.

 Example

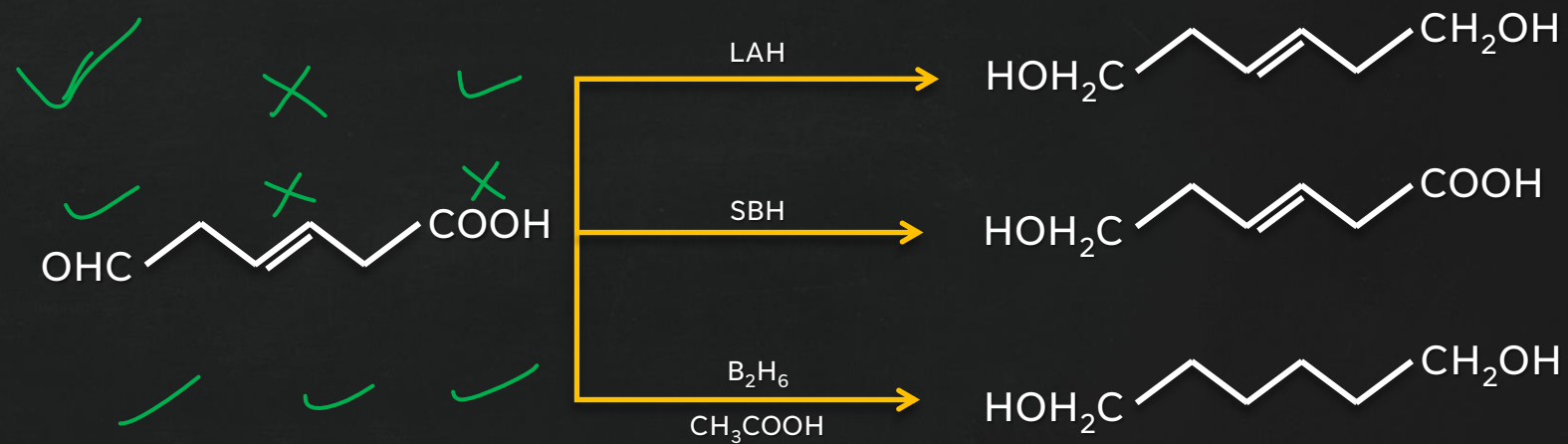
The most suitable reagent for the given conversion is :



Solution

Ans. (4)

Example



Reagent - H_2/Ni , Pt, Pd, Raney nickel, Palladium on charcoal (Pd/C),

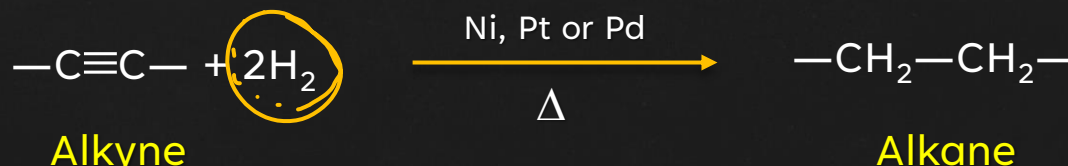
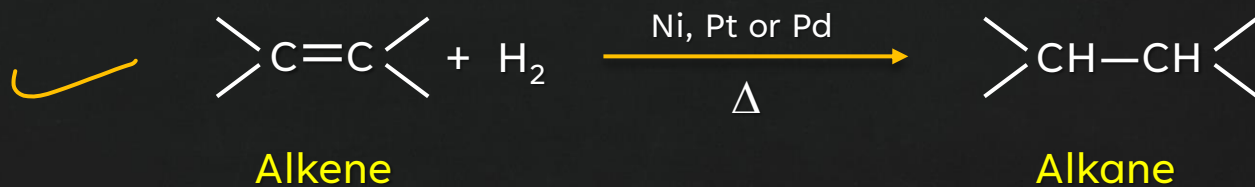
PtO_2 (Adam's Catalyst) and $[(\text{Ph}_3\text{P})_3\text{RhCl}] + \text{H}_2$

Function - Hydrogenation

Stereochemistry - **Syn** Addition ✓



Alkenes and alkynes on catalytic hydrogenation gives alkanes



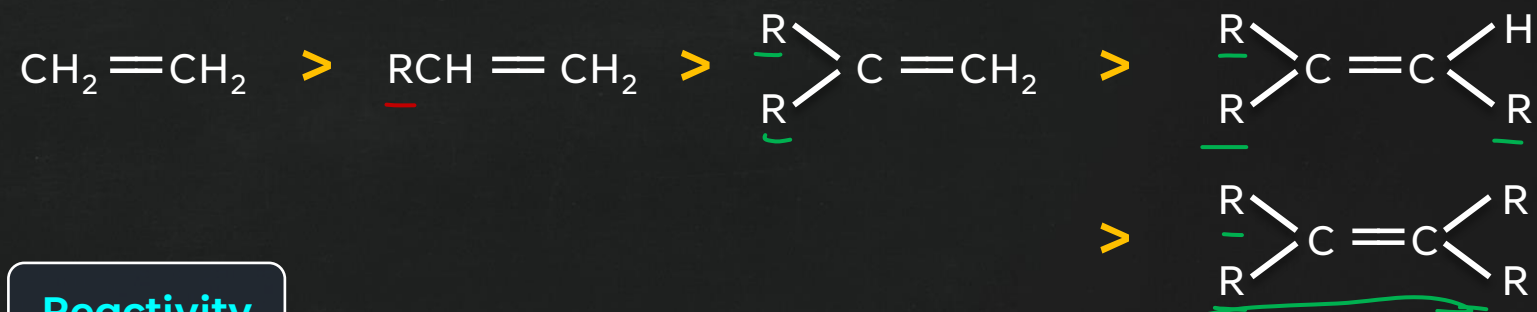
Hydrogenation Products

Substrate	Product
RCOCl	RCH_2OH
RNO_2	RNH_2
$\text{RC}\equiv\text{CR}$	$\text{RCH}_2\text{CH}_2\text{R}$
RCHO	RCH_2OH
RCH=CHR	$\text{RCH}_2\text{CH}_2\text{R}$
RCOR	RCHOHR
RCN	RCH_2NH_2

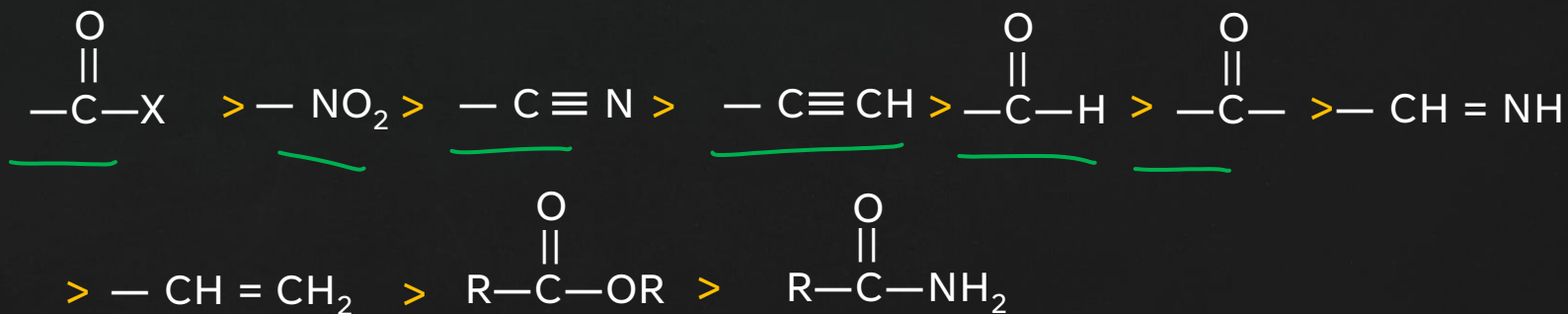


Greater the crowding at double bond carbon more difficult the hydrogenation

Rate of Hydrogenation

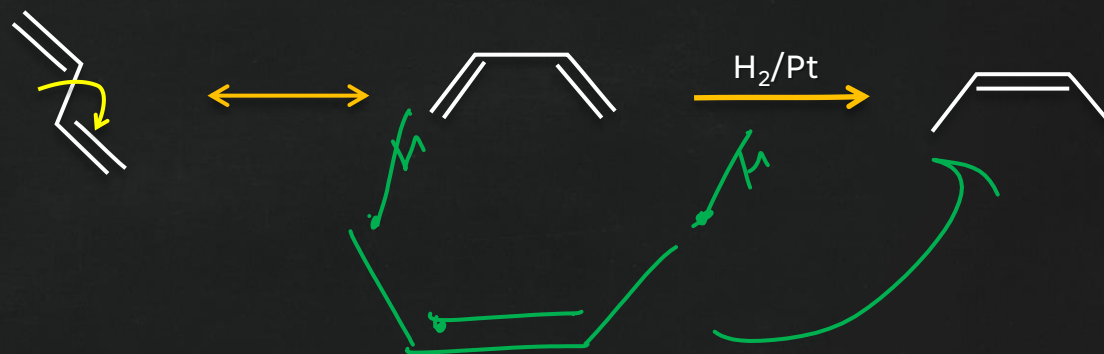


Reactivity



They do not reduce —COOH.

NOTE

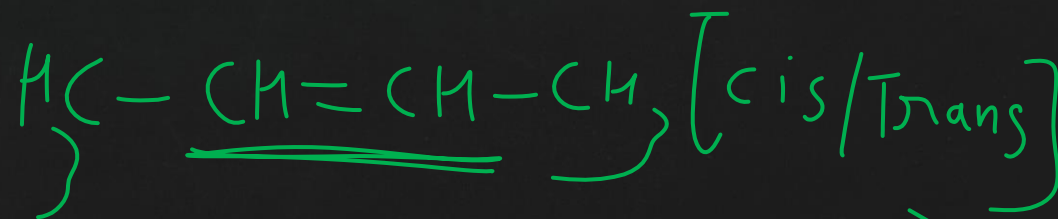
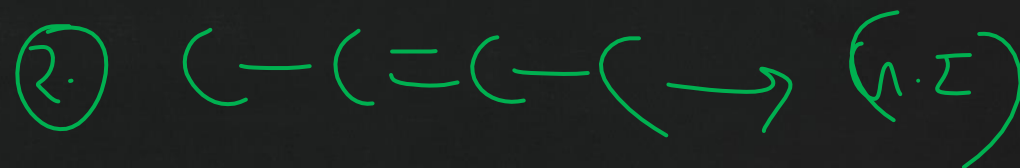
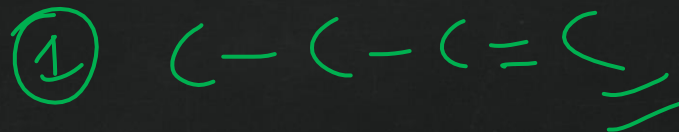




Example

On catalytic reduction with H_2/Pt how many alkenes will give n-butane ?

Solution

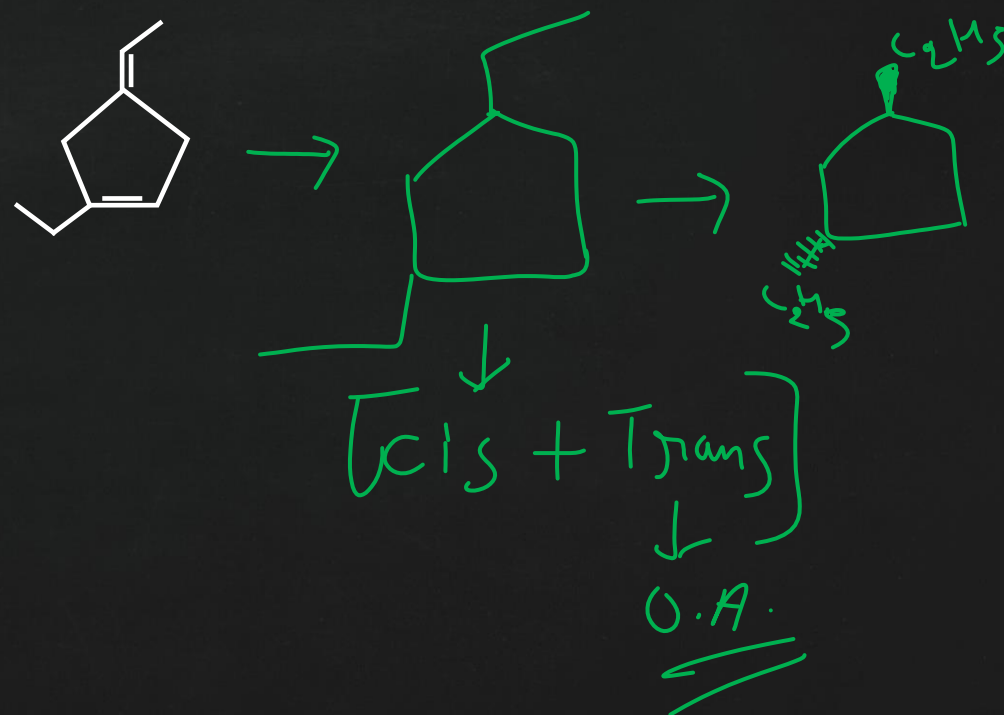


Ans. (3)

 Example

If the following compound is treated with Pd in excess of H_2 gas, how many stereoisomers of the product will be obtained ?

Solution

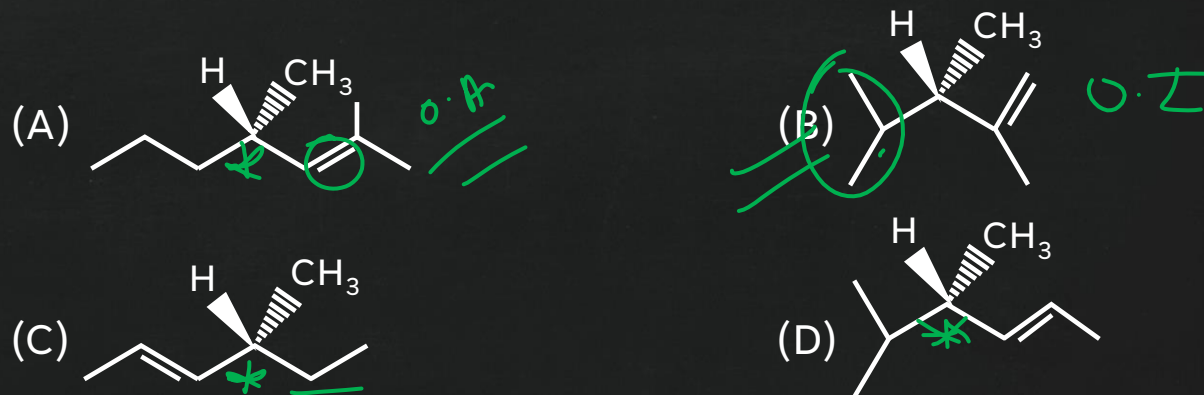


Ans. (3)



Example

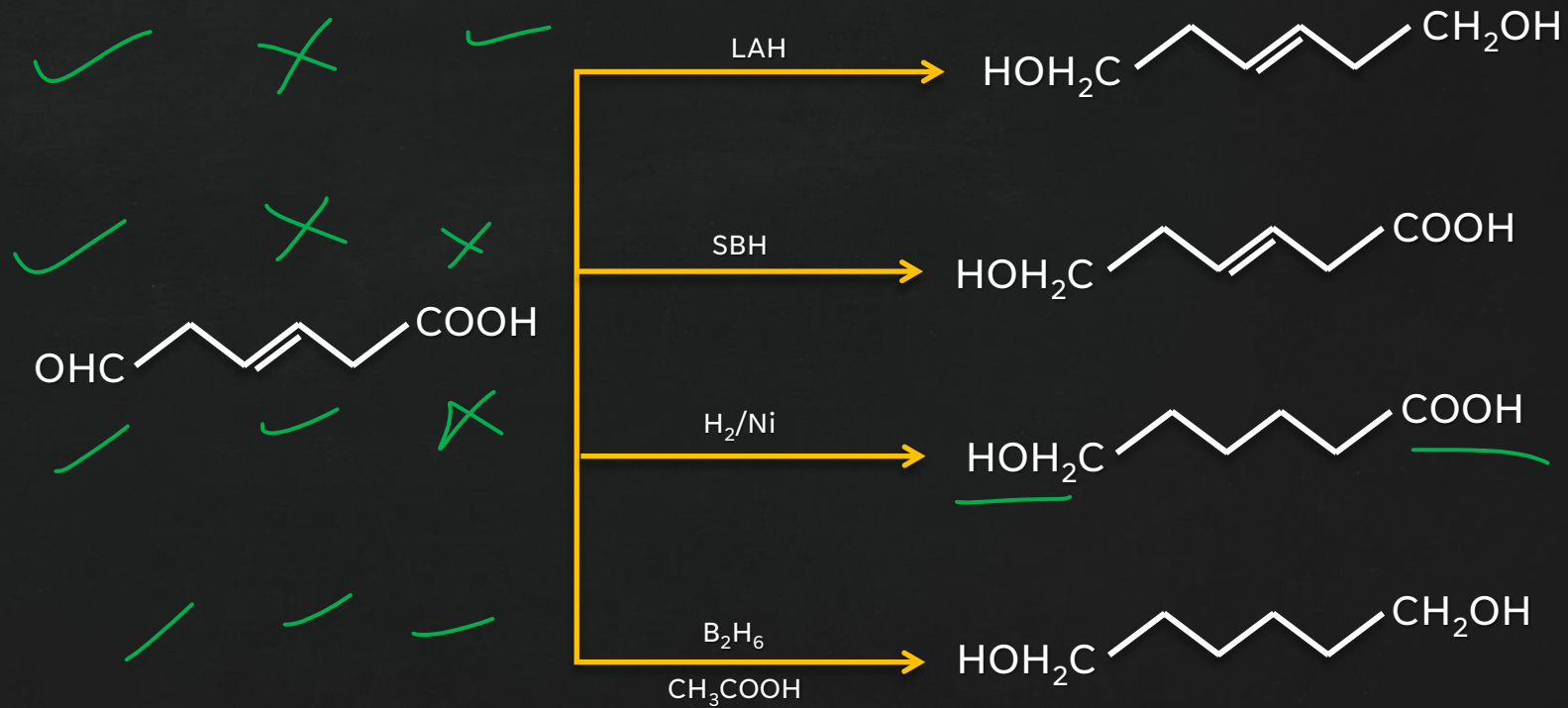
Which of the following compounds produces an optically inactive compound on hydrogenation ?



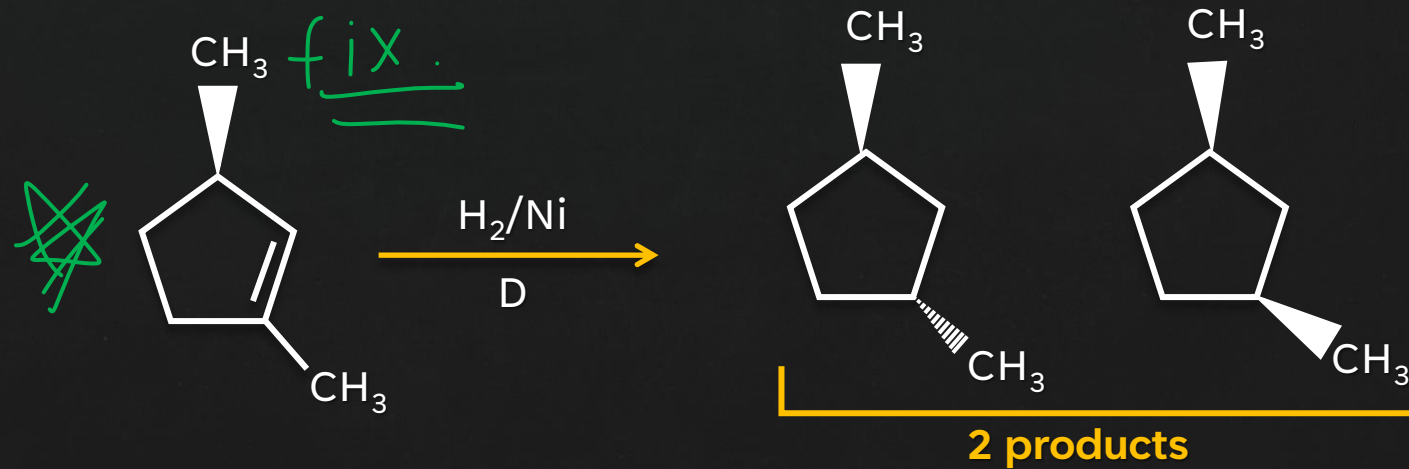
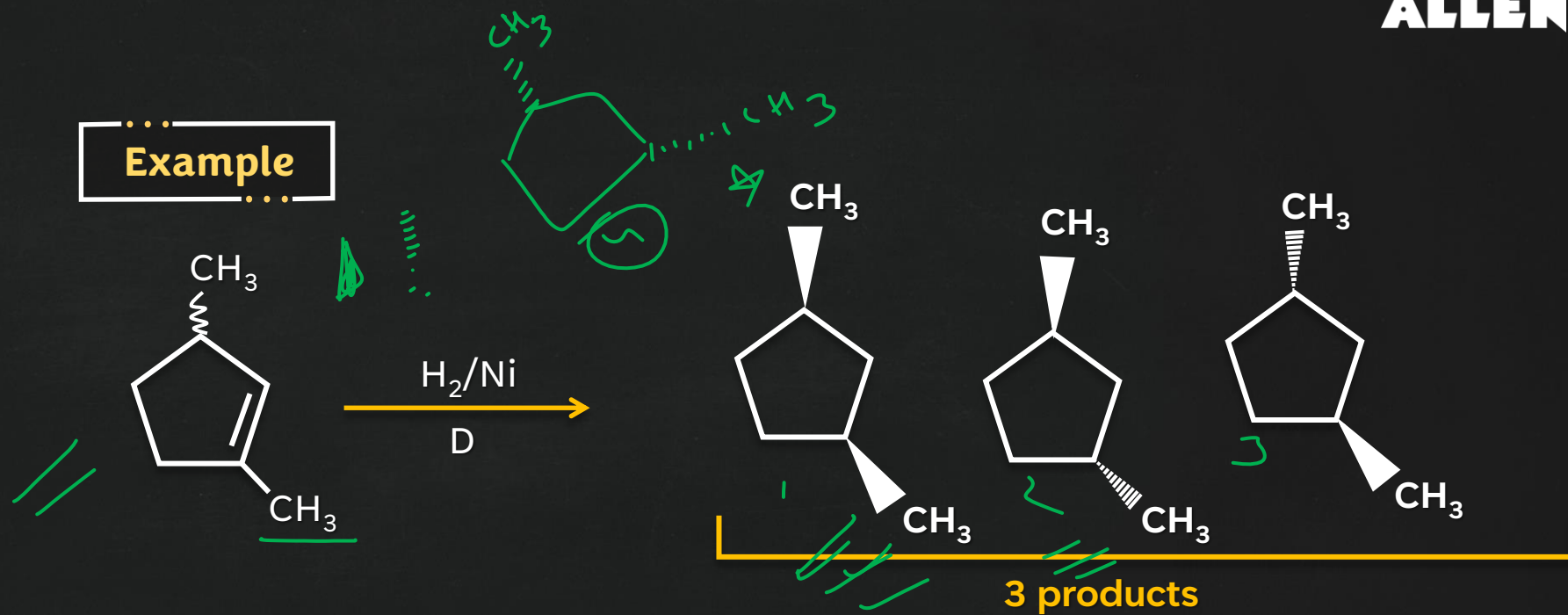
Solution

Ans. (B)

Example

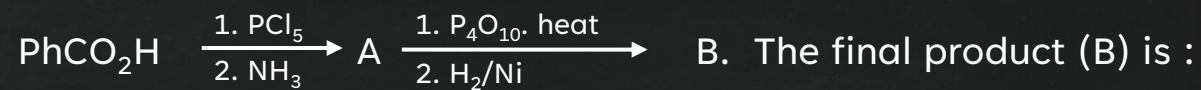


Example



**Example**

Consider the following sequence of reactions:

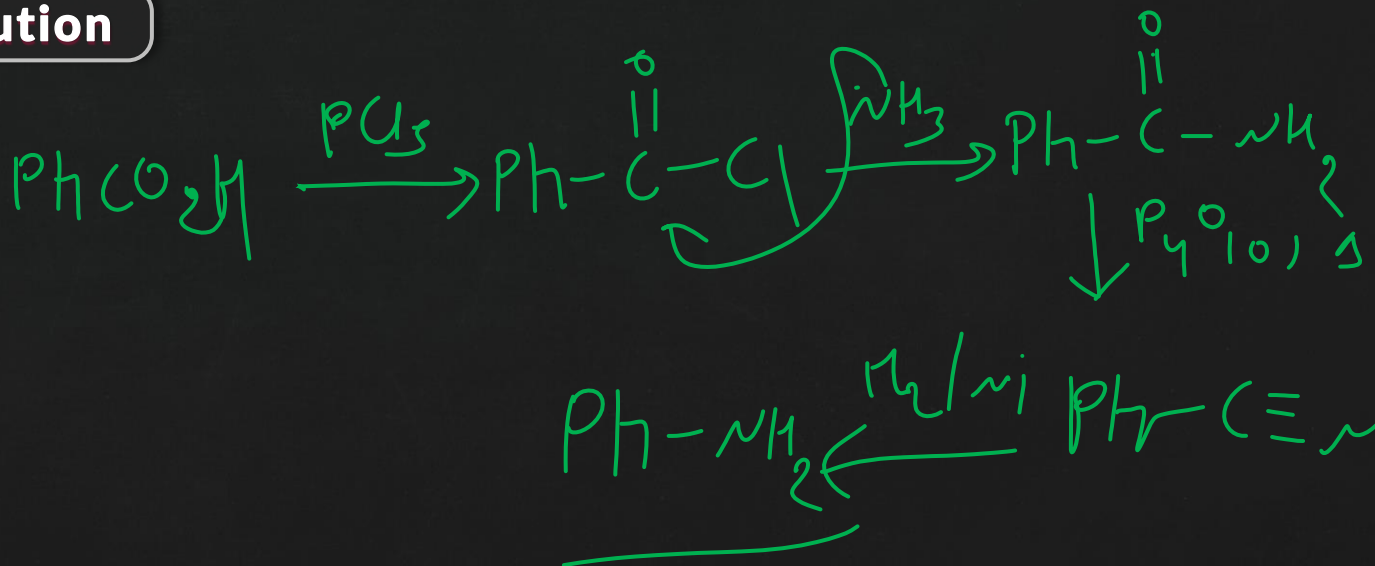


(A) benzonitrile

(B) benzylamine

(C) aniline

(D) benzamide

Solution

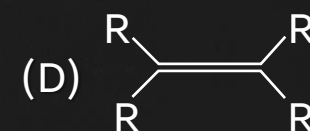
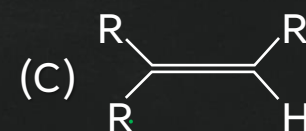
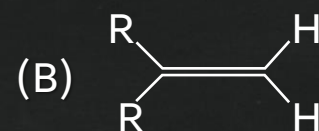
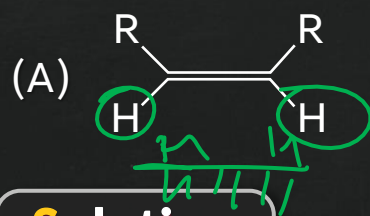
Ans. (B)



Example

[IIT 2000]

Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition



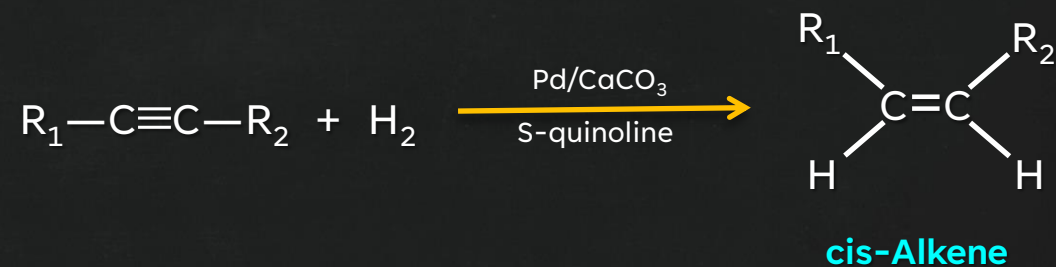
Solution

✓ **Ans. (A)**

Reagent - Lindlar's catalyst ($\text{H}_2/\text{Pd}, \text{CaCO}_3, \text{quinoline}$)

Function - Partial Reduction of Alkyne into Alkene

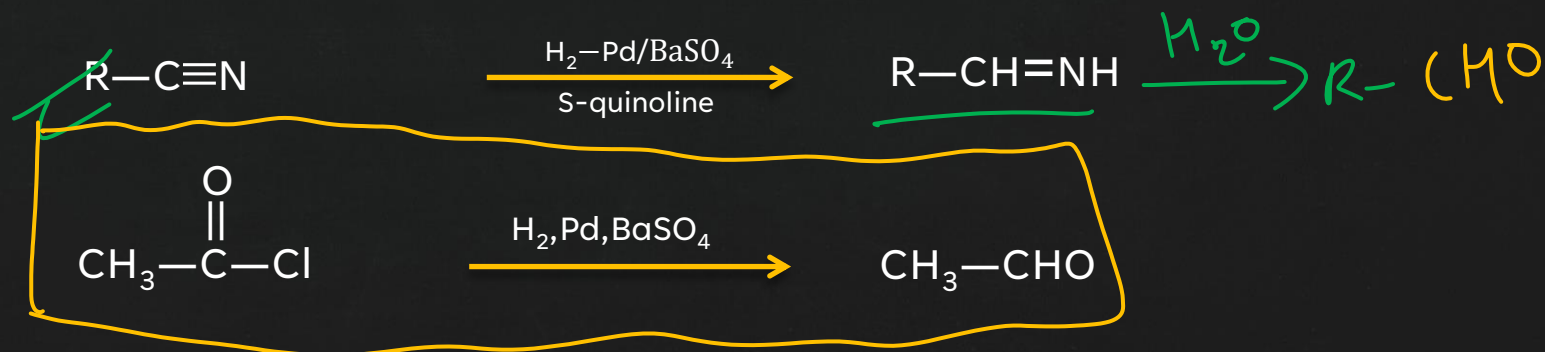
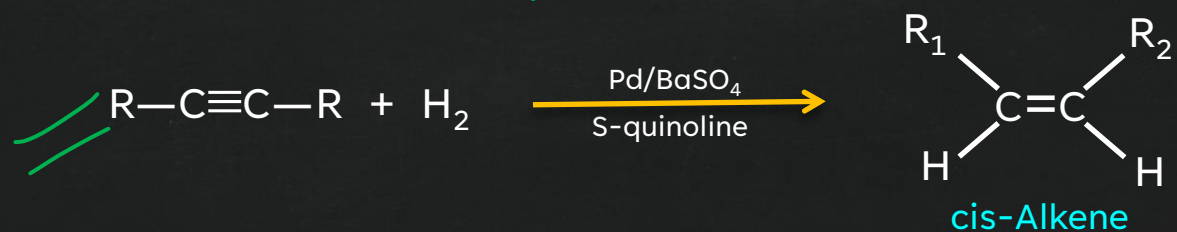
Stereochemistry - **Syn** Addition



Reagent - ($\text{H}_2/\text{Pd}, \text{BaSO}_4, \text{quinoline}$)

Function - Partial Reduction of Alkyne into Alkene

Stereochemistry - **Syn** Addition



NOTE

Hydrogenation of acyl chloride gives aldehyde known as Rosenmund reduction.

**Example**

[IIT 1992]

Hydrogenation of benzoyl chloride in the presence of Pd and BaSO_4 gives :

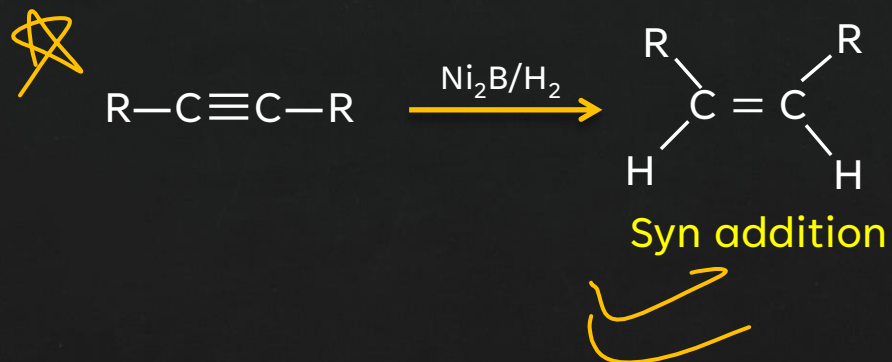
- (A) Benzyl alcohol ~~(B) Benzaldehyde~~
(C) Benzoic acid (D) Phenol

Solution**Ans. (B)**

Reagent - P-2 catalyst ($\text{Ni}_2\text{B}/\text{H}_2$)

Function - Nickel boride Ni_2B (P-2 catalyst) is an excellent alternative catalyst for the conversion of alkyne into alkene.

Stereochemistry - **Syn** Addition



Reagent - Metal + Acid

Function - Reduction of Hydrocarbon Derivatives like Alkyl Halide, Nitro Compound and Cyanide

Stereochemistry - Enantiomers and Diastereomers



✱ Reducing agents may be

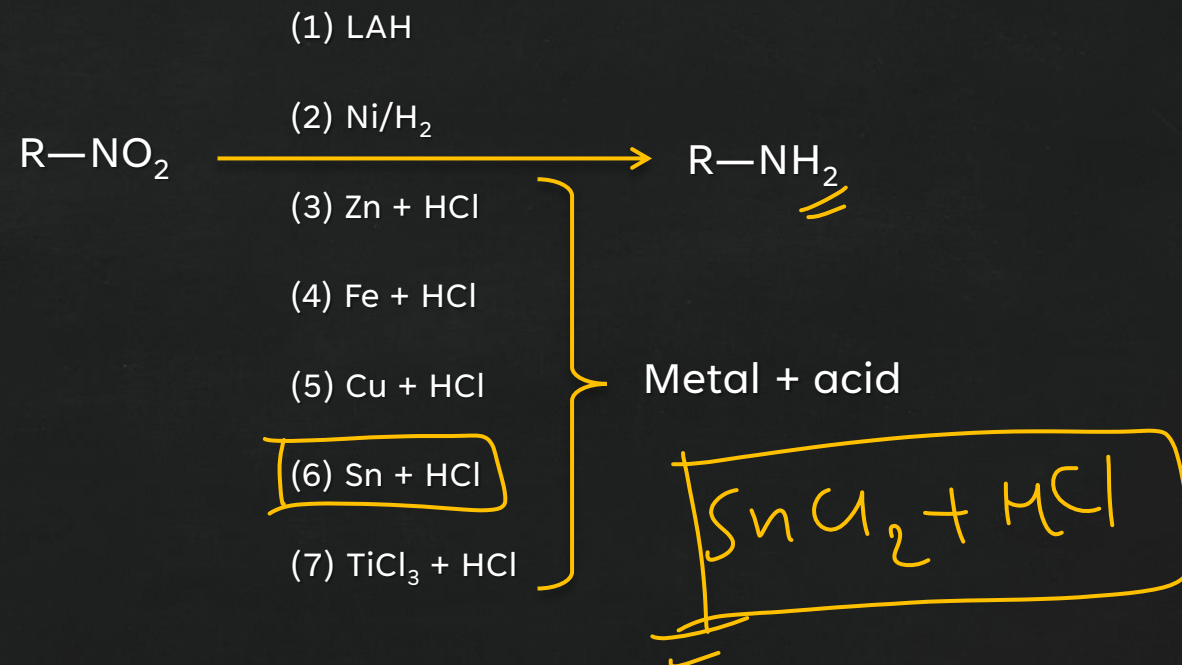
✓ * Zn + HCl

✓ * Sn + HCl

✓ * Fe + HCl

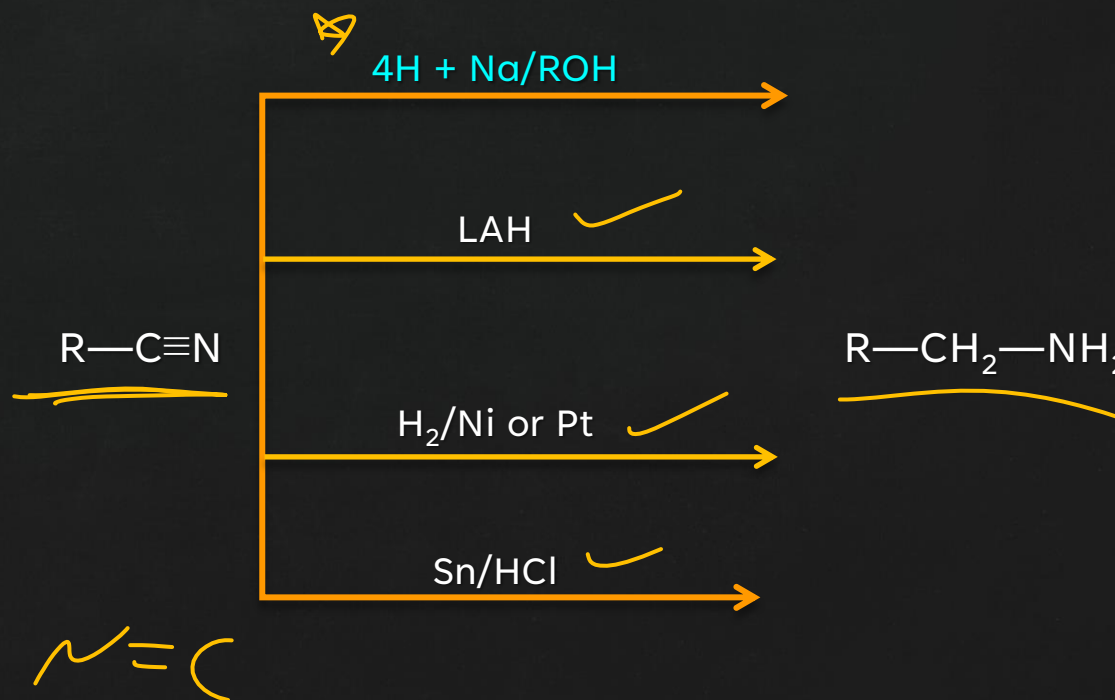
✓ * Zn + CH₃COOH

} Metal + acid

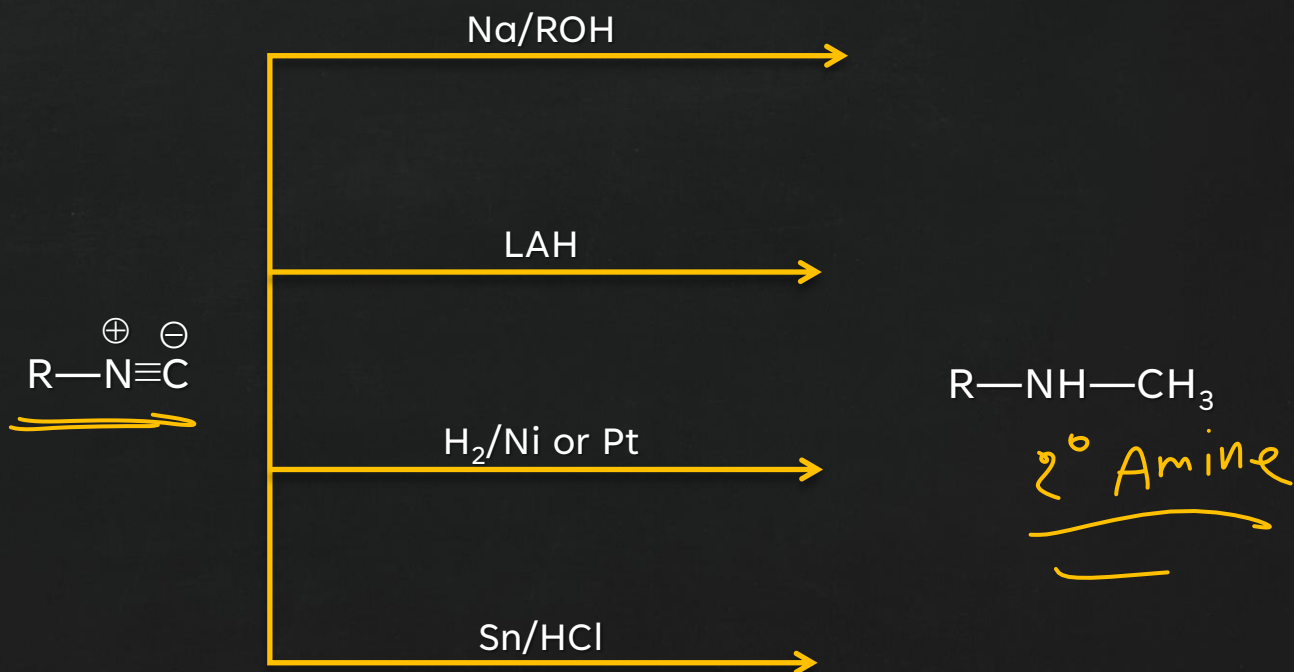


Reagent - (Na/ROH) Mendius reduction (complete reduction)

Function - Alkyl cyanides (nitriles) can be reduced to 1° Alkyl cyanides (nitriles) can be reduced to 1° amine and isocyanides are reduced to 2° amine.

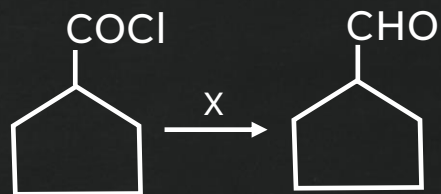


Example





Example



Partial Reduction

. X cannot be

~~(A) $\text{NaBH}_4 / \text{EtOH}$~~

~~(C) Na / EtOH~~

~~(B) $\text{LiAlH}_4 / \text{THF}$~~

(D) $\text{H}_2 / \text{Pd} - \text{BaSO}_4$

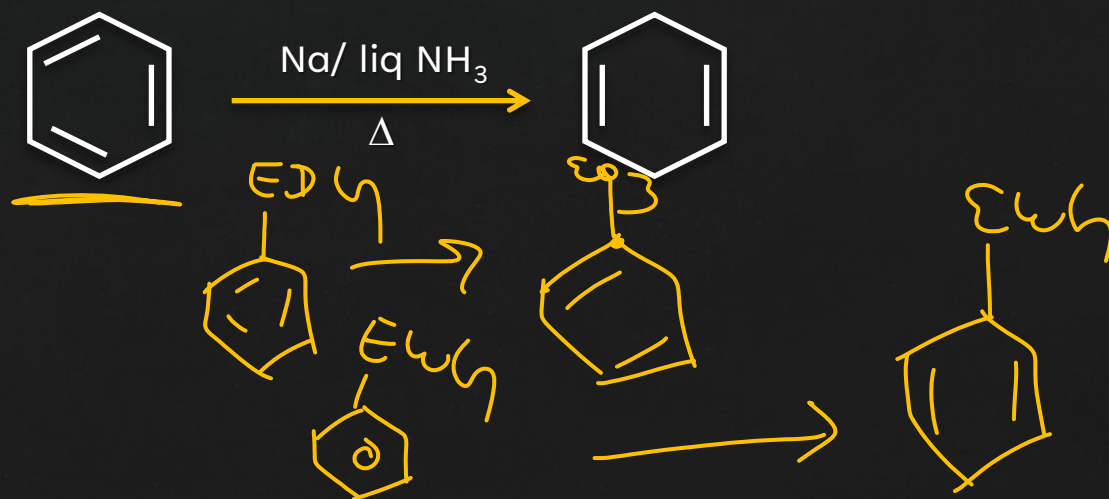
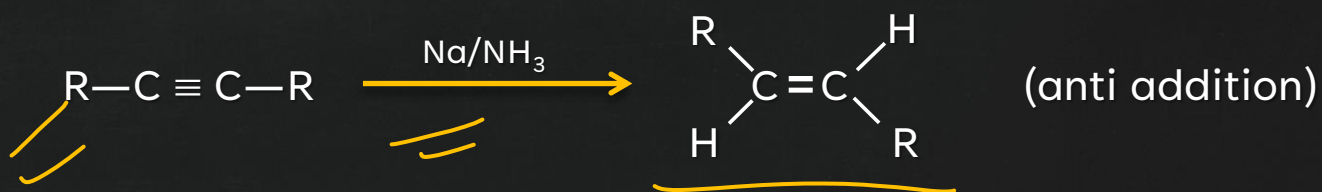
Solution

Ans. (A,B,C)

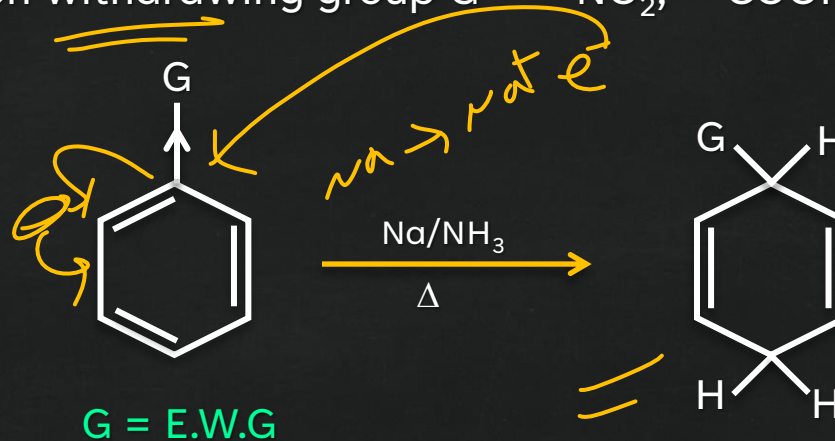
Reagent - Birch reduction [Na or Li/NH₃(liq.) or (ethyl alcohol)]

Function - Partial Reduction of Alkyne and Benzene DERIVATIVES:

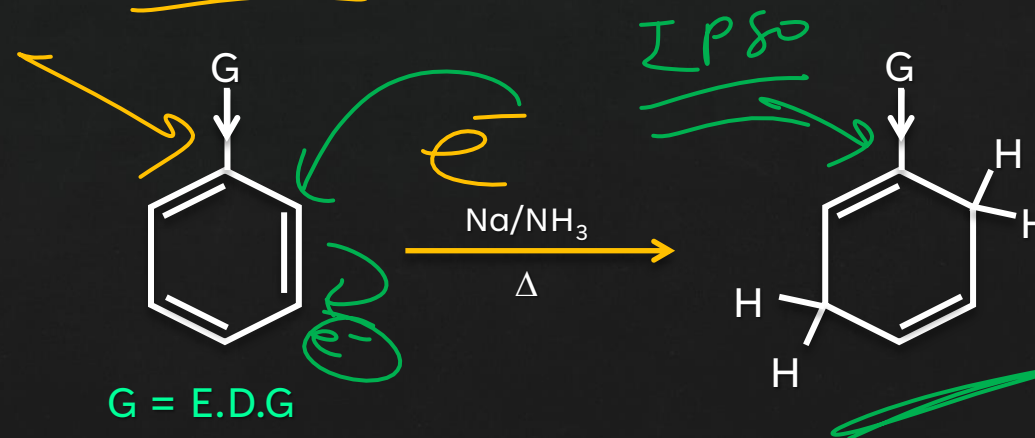
Stereochemistry - Anti-addition.



(electron withdrawing group $G = -\text{NO}_2, -\text{COOH}, -\text{CHO}, -\text{CN}$)



(electron releasing group $-\text{R}, -\text{OR}, -\text{NH}_2$)





Example

[JEE 2014]

The reagent needed for converting is :



(A) H_2 / Lindlar Cat. ~~/~~

~~/~~ (B) Cat. hydrogenation

(C) LiAlH_4 ~~/~~

(D) Li / NH_3

Solution

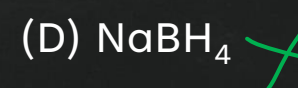
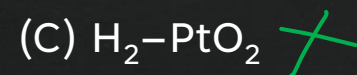
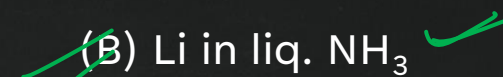
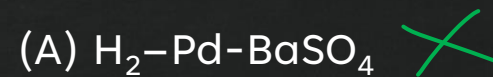
Ans. (D)



Example

[IIT 2004]

2-hexyne can be converted into trans-2-hexene by the action of:



Solution

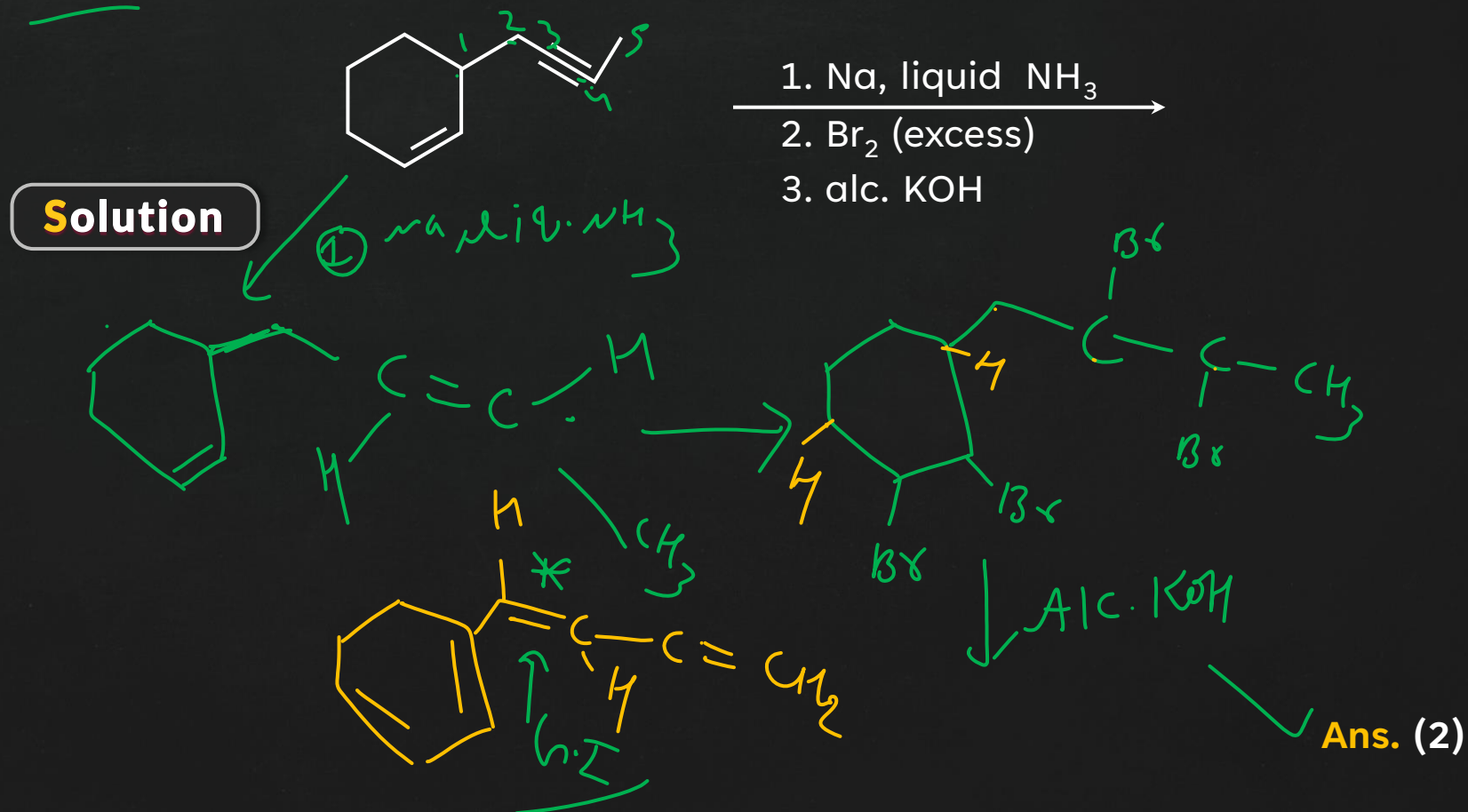


Ans. (B)

Example

[JEE-Advance 2022]

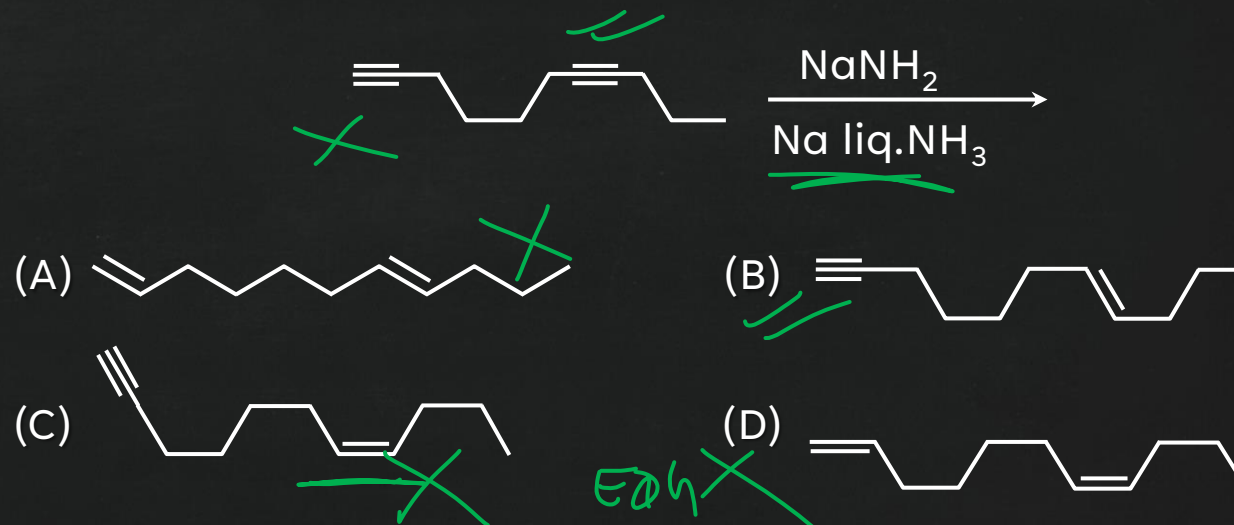
The number of isomeric tetraenes (NOT containing sp-hybridized carbon atoms) that can be formed from the following reaction sequence is _____.



 Example

[JEE-Advance 2021]

The major product formed in the following reaction is



Solution



✓ **Ans. (B)**

Reagent - $[\text{Na}/\text{C}_2\text{H}_5\text{OH}]$ (Bouveault-Blanc reduction)

Function - Reduction of Hydrocarbon derivatives like Carbonyl, cyanide, ester

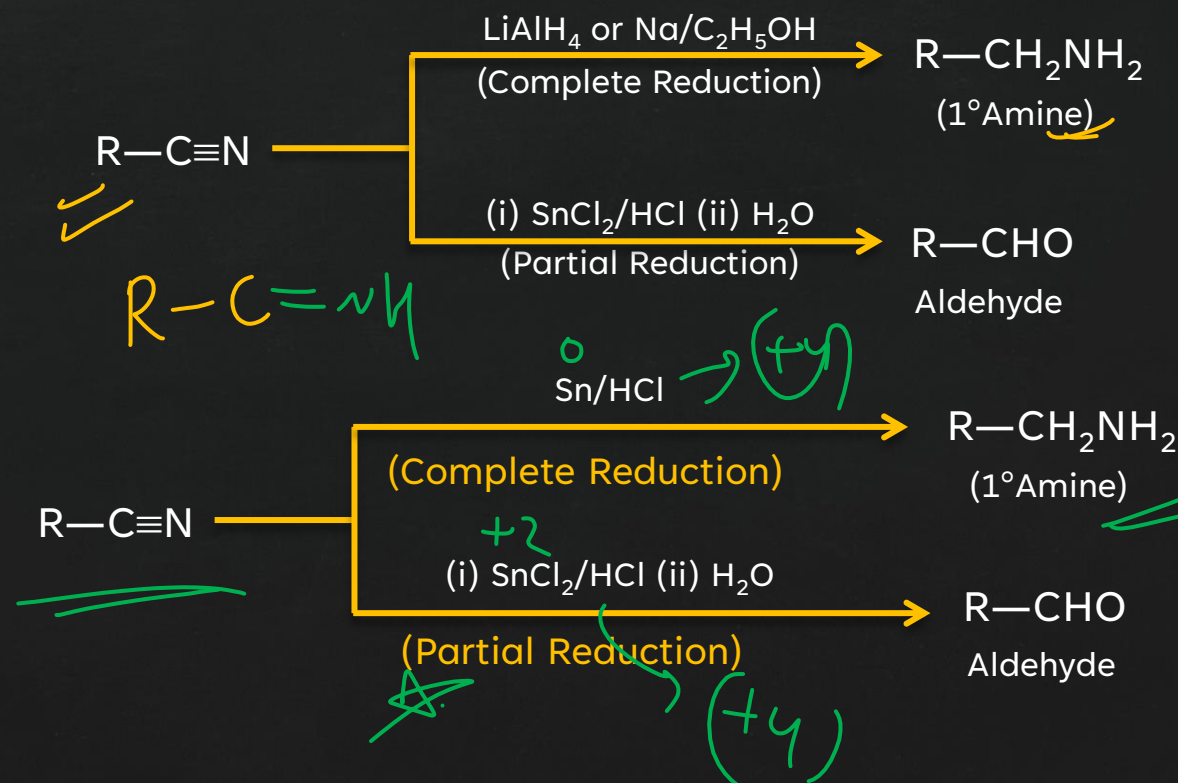
Stereochemistry - Enantiomers and Diastereomers

Reagent	$\text{Na}/\text{C}_2\text{H}_5\text{OH}$	$\text{Na}/\text{C}_2\text{H}_5\text{OH}$	$\text{Na}/\text{C}_2\text{H}_5\text{OH}$	$\text{Na}/\text{C}_2\text{H}_5\text{OH}$	$\text{Na}/\text{C}_2\text{H}_5\text{OH}$
Reactant	Aldehyde	Ketone	Cyanide	Ester	acid halide
Product	1° alcohol	2° alcohol	1° amine	1° alcohol	1° alcohol

Sn/HCl

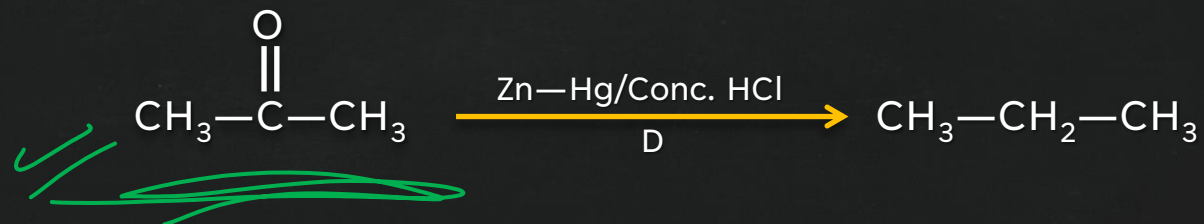
Reagent - Stephen's Reduction $\text{SnCl}_2 / \text{HCl}$ followed by H_3O^+

Function - Partial REDUCTION OF Cyanide



Reagent - Zn-Hg/Conc. HCl (Clemmenson's Reduction)

Function - Aldehyde/Ketone to Alkane

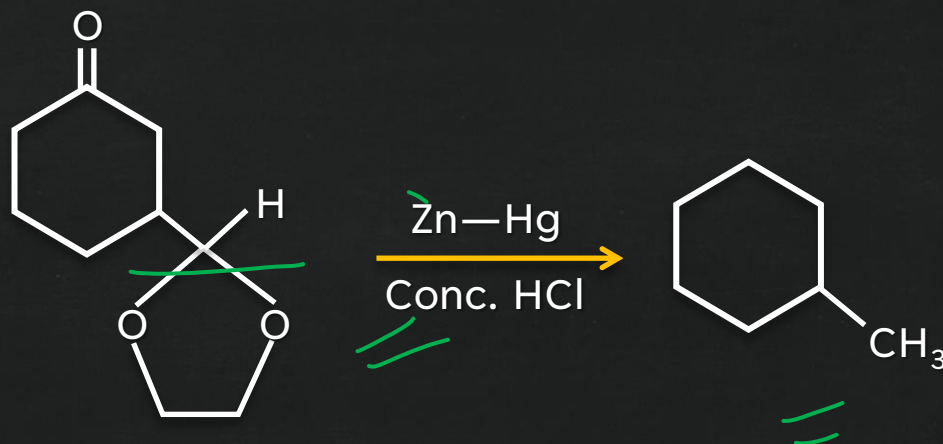


If other acid sensitive group is also attached, then H⁺ will react with that group.



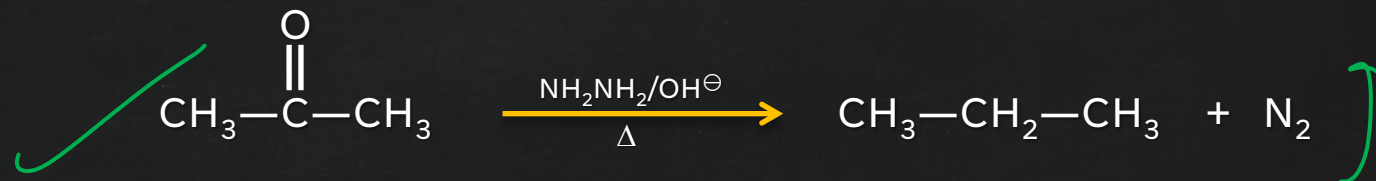
Clemmensen reduction is avoided for compounds which have **acid sensitive** group. [Like: Alcohol, Alkene, Alkyne, Ether, Amine, Amide].

Example



Reagent - $\text{N}_2\text{H}_4/\text{OH}^-/\Delta$ (Wolff-Kischner Reduction) (Using high boiling solvent (ethylene glycol))

Function - Aldehyde/Ketone to Alkane

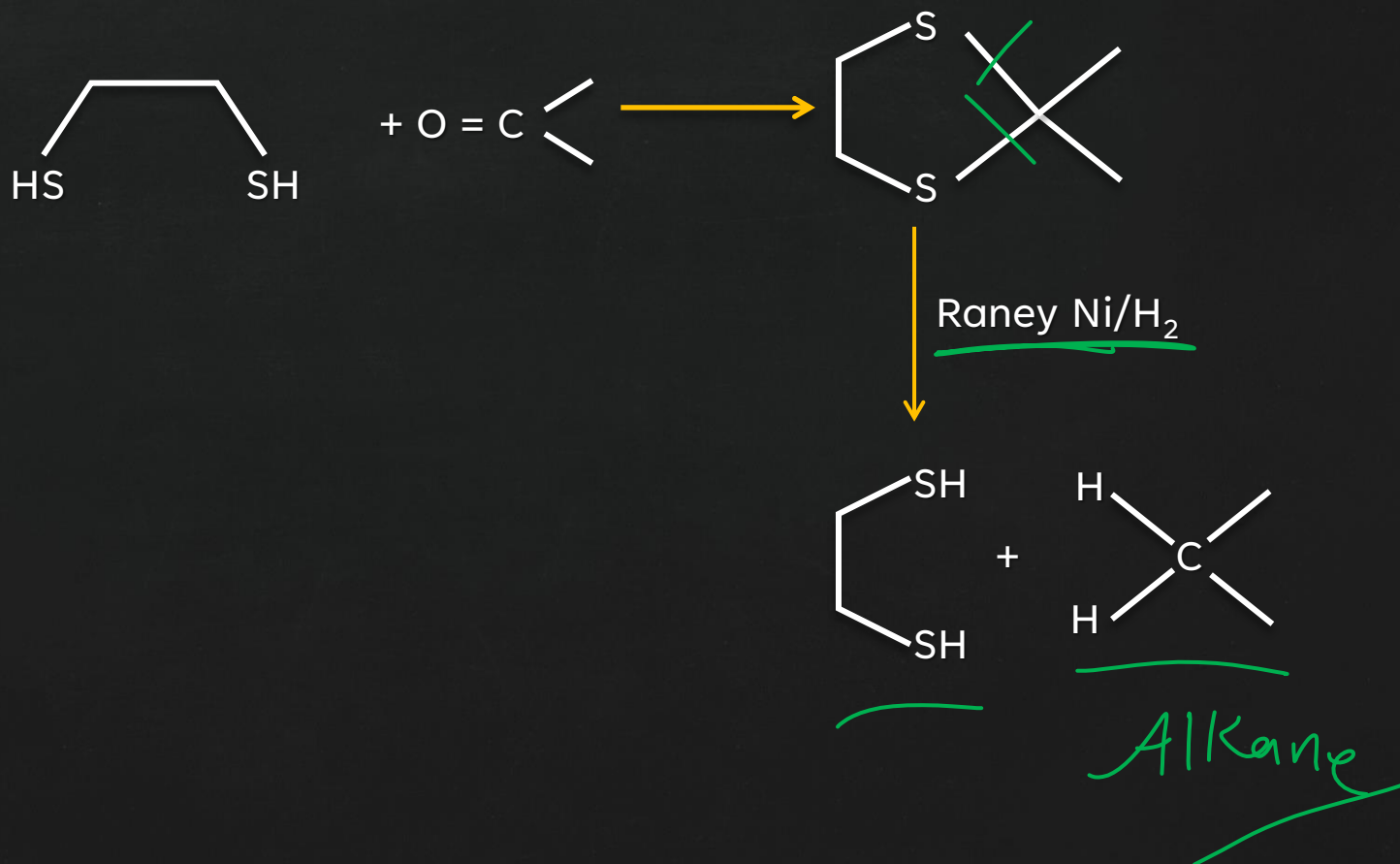


NOTE

Wolff-kishner reduction is avoided for compounds which possess base sensitive groups. [Like : Halogens, Acid halide, Esters, Anhydride, Acid]

Reagent - $\begin{array}{|c} \text{SH} \\ \hline \text{SH} \end{array}$ + Dry HCl, followed By Raney Ni **Mozingo Reaction**
Function - Aldehyde/Ketone to Alkane

Example

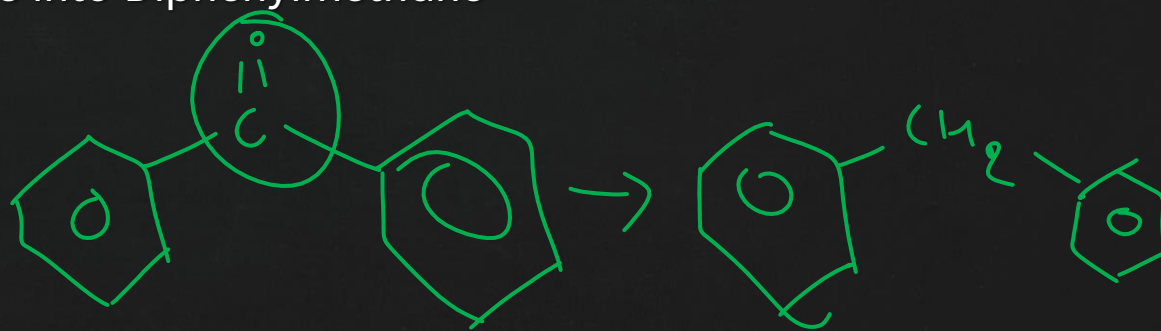


**Example**

[IIT 1995]

Under Wolff Kishner reduction conditions, the conversions which may be brought about is?

- (A) Benzaldehyde into Benzyl alcohol ✗
- (B) Cyclohexanol into Cyclohexane ✗
- (C) Cyclohexanone into Cyclohexanol ✗
- (D) Benzophenone into Diphenylmethane

Solution**Ans. (D)**



Example

[IIT 2000]

The appropriate reagent for the following transformation:



(A) Zn(Hg), HCl ✗

(B) $\text{NH}_2\text{NH}_2, \text{OH}^\ominus$ ✓

(C) H_2/Ni ✗

(D) NaBH_4 ✗

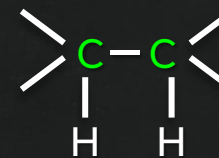
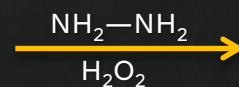
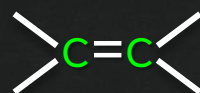
Solution

Ans. (B)

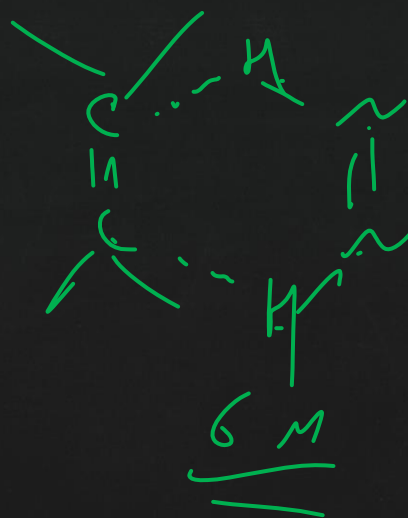
Reagent - $\text{N}_2\text{H}_4/\text{H}_2\text{O}_2$ Or N_2H_2

Function - Reduction of Alkene into Alkane

Stereochemistry - Syn Addition



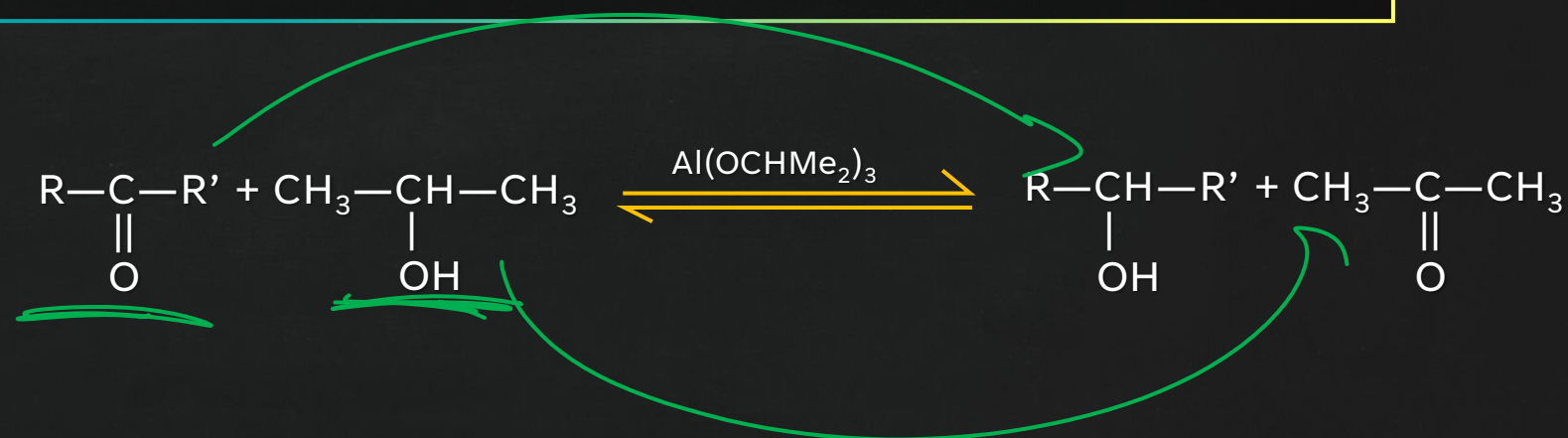
Syn addition



Reagent - MPV (Meerwein - Ponndorff - Verley) Reduction

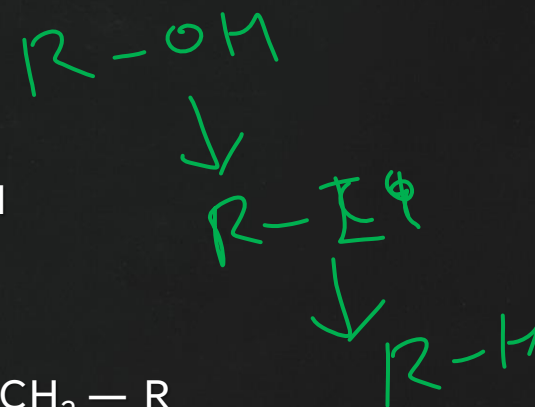
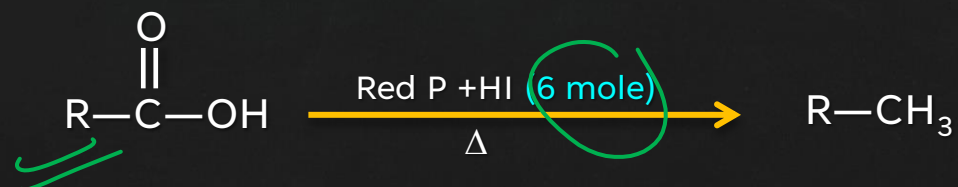
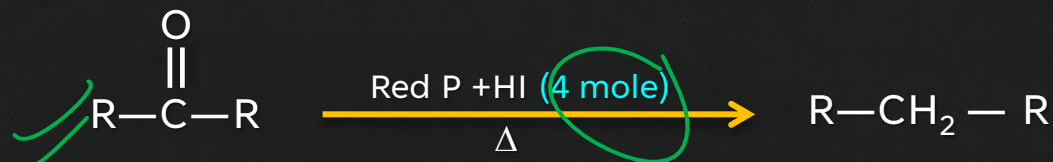
Function - Reduction of Aldehyde and Ketone into Alcohol

Stereochemistry - Enantiomers and Diastereomers



Reagent - Red P + HI

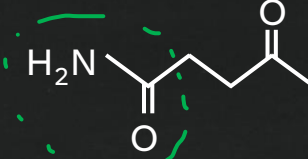
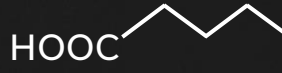
Function - All organic compounds Containing O convert into Alkane
Containing N Convert into Amine

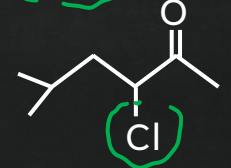
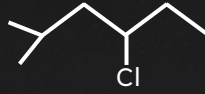


Reactant	Product	Reagent	No. of moles required
$R-I$	$R-H$	Red P + HI	1
$R-OH$	$R-H$		2
$\begin{array}{c} O \\ \\ R-C-H \end{array}$	$R-CH_3-H$		4
$\begin{array}{c} O \\ \\ R-C-R \end{array}$	$R-CH_2-R$		4
$\begin{array}{c} O \\ \\ R-C-OH \end{array}$	$R-CH_3$		6

 Example

Given below are two statements :

Statement I :  under Clemmensen reduction conditions will give  ✓

Statement II :  under Wolff-Kishner reduction condition will give  ✗

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is false but Statement II is true ✗

(2) Both Statement I and Statement II are false ✗

(3) Statement I is true but Statement II is false ✓

(4) Both Statement I and Statement II are true

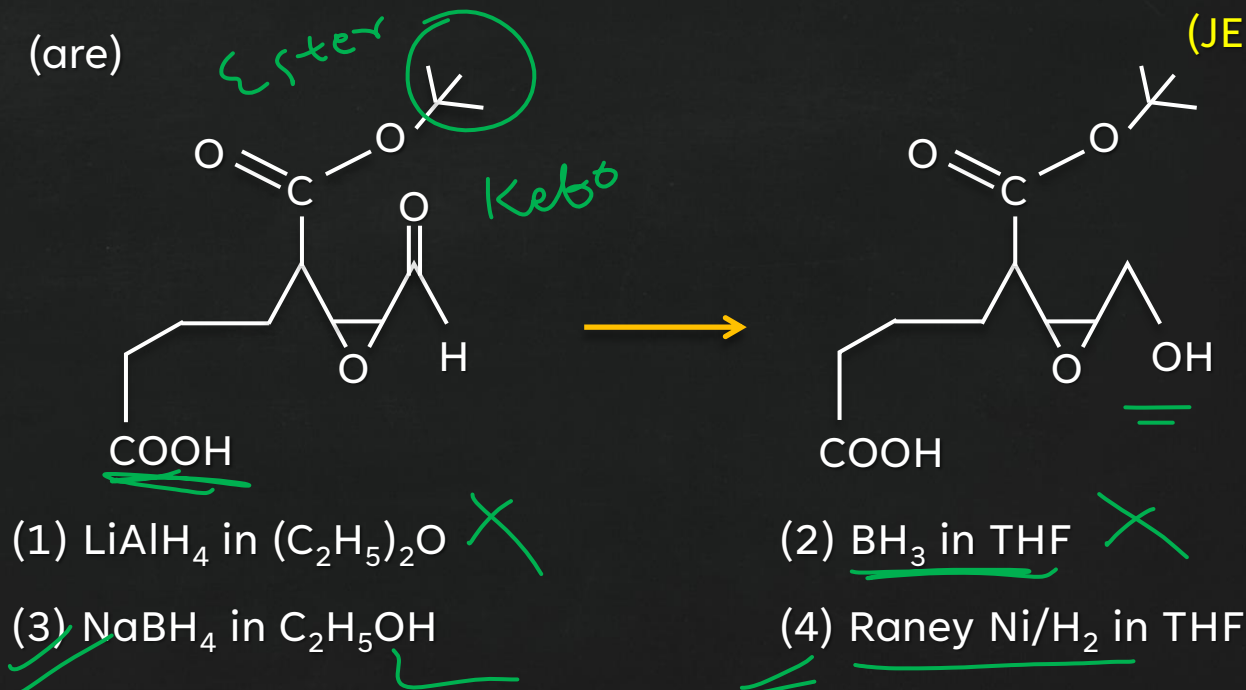
Solution

Ans. (3)



Example

Reagent(s) which can be used to bring about the following transformation is (are) (JEE Advanced 2016)



Solution

✓ **Ans. (3,4)**

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